

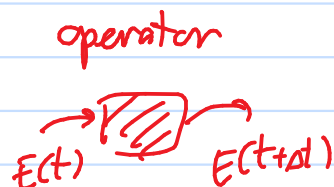
Symmetry Method (Group theory)

Symmetry = invariant property under transformation

- Symmetry is behind physics $\Rightarrow$ conservation

e.g. conservation of energy $E_{ini} = E_{final}$

- $E(t) = E(t + \Delta t) = \hat{T}(\Delta t) E(t)$
= $\xrightarrow{\text{transformation = time translation}}$



- $\frac{dE(t)}{dt} = 0$

More example

$$\frac{d}{dt} = 0$$

conservation of momentum $\Rightarrow$ spatial translation
conservation of angular momentum $\Rightarrow$ rotation
conservation of mass, charge $\Rightarrow$??

- Symmetry is powerful!

Newton vs Conservation of energy
 $\nearrow$ 2nd order ODE $\Rightarrow$ difficult to solve
 $\searrow$ no DE $\Rightarrow$ easier

$\rightarrow$ *require specific situation
i.e. Symmetry.

We will first study the structure of symmetry.

$\Rightarrow$ We will study the discrete symmetry (finite)

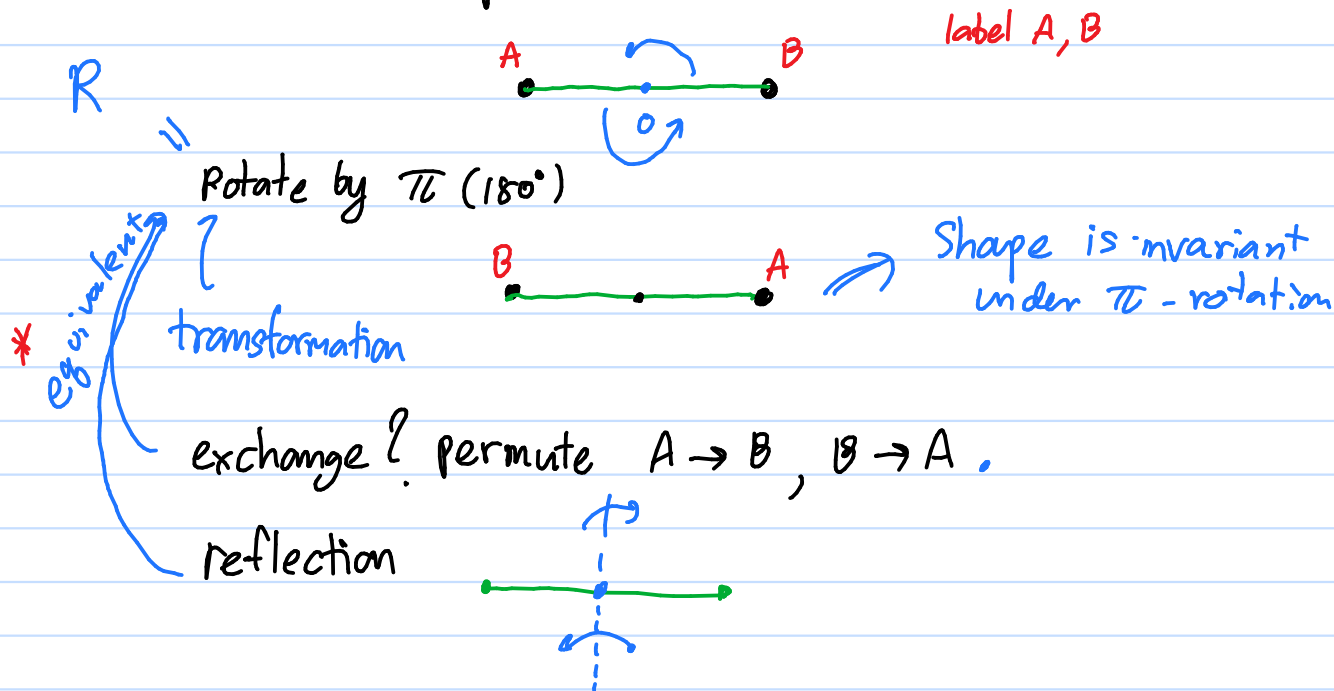
Consider a point

- $\Rightarrow$ trivial symmetry
rotation...

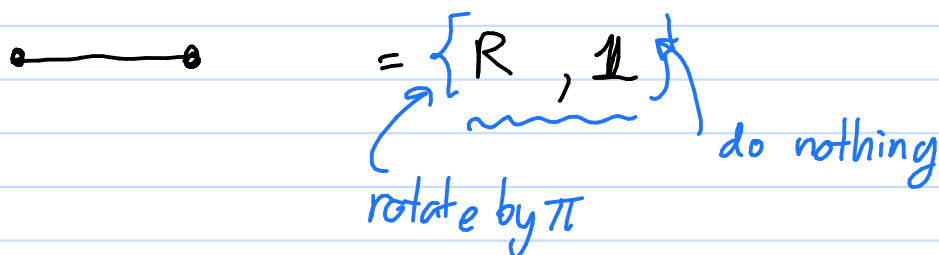
Consider two points connected to each other



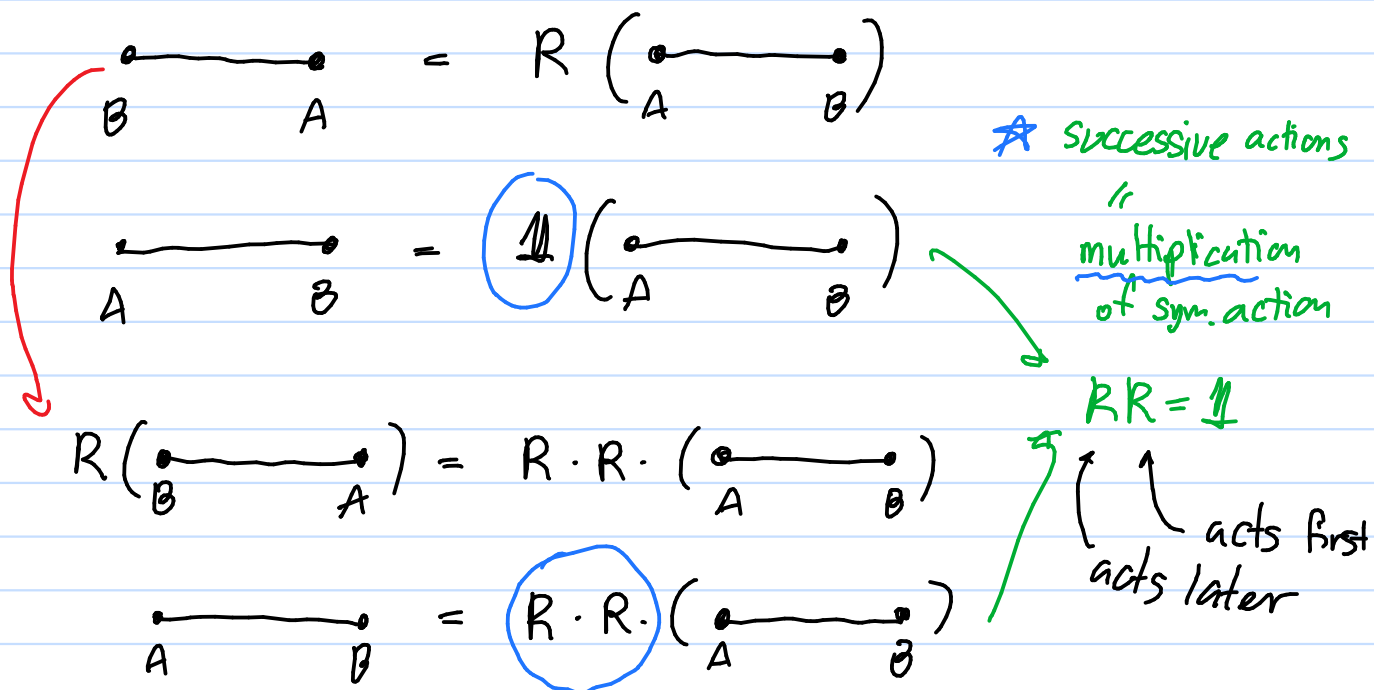
Consider two points connected to each other



Symmetry action = action on object that leaves the shape invariant



We can play with series of actions



$\underbrace{RR = \mathbb{1}}$, $\underbrace{\mathbb{1}R = R}$, $\underbrace{R\mathbb{1} = R}$, $\underbrace{\mathbb{1}\mathbb{1} = \mathbb{1}}$
 multiplication table

	$\mathbb{1}$	R
$\mathbb{1}$	$\mathbb{1}$	R
R	R	$\mathbb{1}$

Side note notice ($\mathbb{Z}_2$)

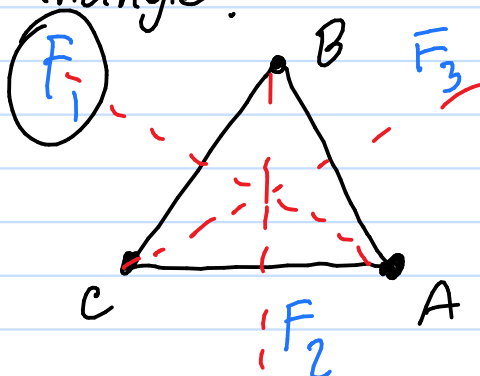
$$(-1)(-1) = 1 \quad -1 \rightarrow R$$

$$(-1)1 = -1 \quad 1 \rightarrow \mathbb{1}$$

$$1(-1) = -1$$

$$11 = 1$$

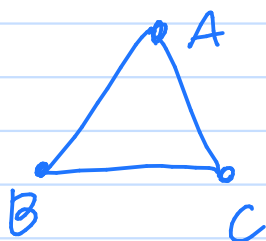
Let's do a triangle!



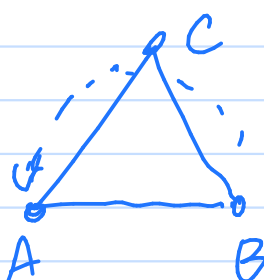
Let's list all symmetry actions!

$$\{\mathbb{1}, R_1, R_2, F_1, F_2, F_3\}$$

Rotate $120^\circ \left(\frac{2\pi}{3}\right)$, Rotate $240^\circ \left(\frac{4\pi}{3}\right)$

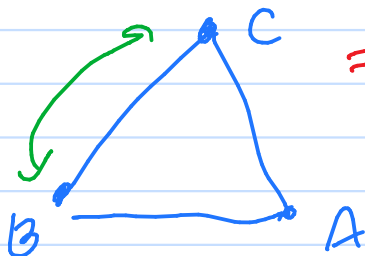


$$= R_1$$

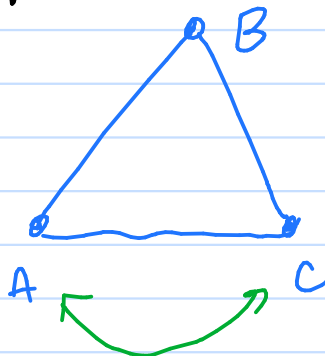


$$= R_2$$

Flip around A axis, Flip around B axis

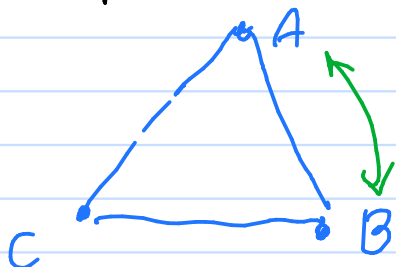


$$= F_1$$



$$= F_2$$

Flip around C axis



$$= F_3$$

Let's do the multiplication table.

	$\mathbb{1}$	R_1	R_2	F_1	F_2	F_3
$\mathbb{1}$	$\mathbb{1}$	R_1	R_2	F_1	F_2	F_3
R_1	R_1	R_2	$\mathbb{1}$	F_3		
R_2	R_2	$\mathbb{1}$	R_1			
F_1	F_1			$\mathbb{1}$		
F_2	F_2			R_2	$\mathbb{1}$	
F_3	F_3					$\mathbb{1}$

$$X\mathbb{1} = X$$

$$\mathbb{1}X = X$$

$$R_1 R_1 = R_2$$

$$R_2 R_1 = \mathbb{1}$$

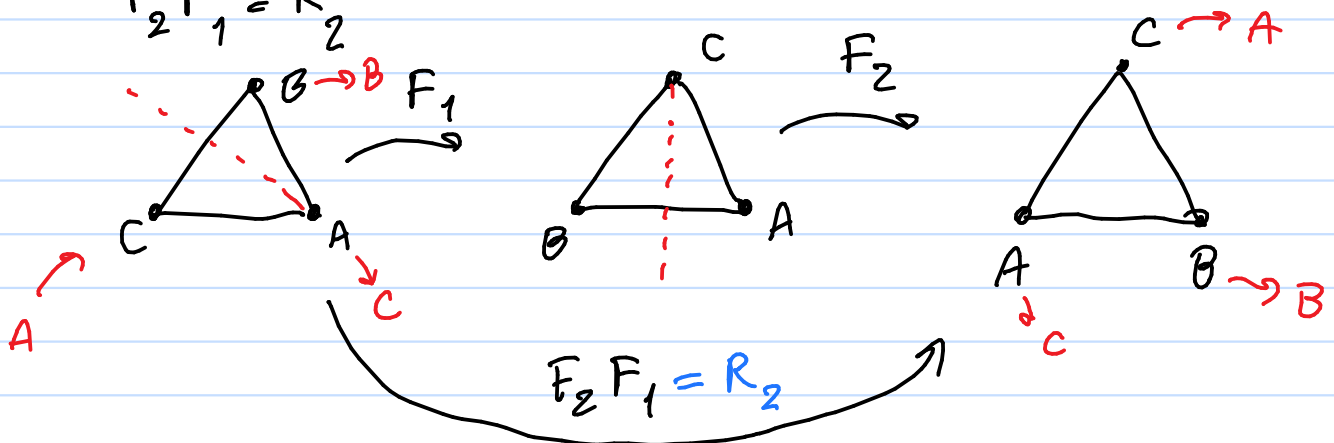
$$R_1 R_2 = \mathbb{1}$$

$$R_2 R_2 = R_1$$

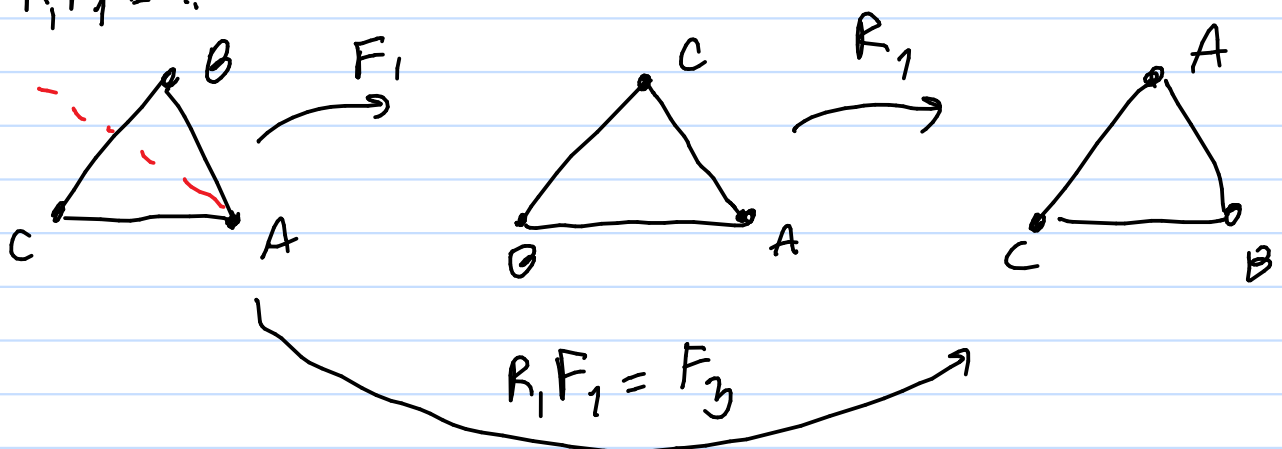
$$F_1 F_1 = F_2 F_2 = F_3 F_3 = \mathbb{1}$$

Fill the table as an exercise!

$$F_2 F_1 = R_2$$



$$R_1 F_1 = ?$$



The heart of Math is generalisation

1). Notice the patterns

2). Define the "object" by definitions

↳ study more examples & properties.

1). They are all described by a set of symmetry actions

$\{1, R\}$, $\{1, R_1, R_2, F_1, F_2, F_3\}$ "completed"

For example $\{1, F_1, F_2, F_3\}$ is not completed $F_2 F_1 = R_2$

2). There is a "multiplication" $\Rightarrow$ symmetry action
- successive action

$$\begin{array}{c} X \cdot Y = Z \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{sym action} \end{array}$$

$$Y \cdot (\text{---}) = \text{---}$$

↑
action on objects

$$\begin{array}{c} \vec{F} \cdot \vec{S} = W \\ \uparrow \quad \uparrow \\ \text{vector} \quad \text{vector} \end{array} \quad \begin{array}{c} \text{Scalar} \\ \Rightarrow \text{inner product / dot product} \end{array}$$

$$\vec{A} \times \vec{B} = \vec{C} \quad (\text{เวกเตอร์คูณเวกเตอร์})$$

3) The multiplication is associative

$$X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z = X \cdot Y \cdot Z$$

$\Rightarrow$ we can check this for symmetry action
(สมมาตร)

★ This is not equal to $X \cdot Y = Y \cdot X$

(This $X \cdot Y = Y \cdot X$ is not necessary true!)

4). The existence of $\mathbb{1}$ = identity

5). Every element has an "inverse"

$$x, x^{-1} \text{ is defined as } x \cdot x^{-1} = x^{-1} \cdot x = \mathbb{1}$$

$$\underline{\text{Ex}} \quad \{\mathbb{1}, R_1, R_2, F_1, F_2, F_3\}$$

$$R_1 \text{ has an inverse} \Rightarrow R_1 \cdot R_2 = \mathbb{1} = R_2 \cdot R_1$$

R_2 has R_1 as an inverse.

F_1 has itself as an inverse, $F_2, F_3 \dots$

All the properties above are the properties of "group"

Let's do this a proper way.

(multiplication)

G = group = a set together with a binary operator

$(G, \cdot)$ = group.
set $\nearrow$ multiplication

$(G, \star), (G, \times)$

$(G, +)$

binary operator: $\boxed{} \cdot \boxed{} = \boxed{}$
input $\nearrow$ output

with the following properties: $\nearrow$ is an element of

1). Closure, $\forall g_1, g_2 \in G$ then $g_1 \cdot g_2 \in G$
 $\nwarrow$ for all

2). Associativity, $g_1(g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$

3). Identity, $\mathbb{1}$, $\forall g \in G$, $g \cdot \mathbb{1} = \mathbb{1} \cdot g = g$

4). Inverse, $\forall g \in G$, $g^{-1} \cdot g = g \cdot g^{-1} = \mathbb{1}$

Examples of group

the normal multi.
number $\times$ number = number

1). $\mathbb{Z}_2$ group $\{1, -1\}$ with a usual multiplication.

Closure: $1 \times -1 = -1$, $1 \times 1 = 1$, $-1 \times -1 = 1$, ...

Associativity: $x(yz) = (xy)z$ $\checkmark$ (follow from the usual multiplication)

Identity: 1

Inverse: $1 \rightarrow 1^{-1} = 1$, $-1^{-1} = -1$

Parity group.

$$(\{1, -1\}, \cdot) = \mathbb{Z}_2$$

2). The trivial group: $\{1\}$ with a usual multi.

Closure $\checkmark$ Asso $\checkmark$

Identity $\checkmark$ Inverse $\checkmark$

3). Let's consider $\{1, -1, i, -i\}$; $i = \sqrt{-1}$
with a usual multi.

Closure: $\checkmark$ $i \cdot (-i) = 1$, $i \cdot i = -1$, $i \cdot (-1) = -i$, ...

Associativity: $\checkmark$ from a multi..

Identity: 1

Inverse: $i \rightarrow -i$, $-i \rightarrow i$

4). The real number set can also form a group.

$(\mathbb{R}, +) = \text{group.}$ $\Rightarrow (\mathbb{Q}, +) = \text{group.}$

Closure: $x+y=z$ $\checkmark$, Associativity: $(x+y)+z = x+(y+z)$

Identity: $x+0 = 0+x = x$, $0 = \text{identity}$

Inverse: $x: x+(-x) = 0$

5). Can we use the usual multi on the $\mathbb{R}$?

Closure: $\checkmark$, Asso: $\checkmark$

Identity: 1: $1 \cdot x = x$, Inverse: $x \rightarrow \frac{1}{x}$

0 has no inverse

$\frac{1}{0}$

Group: $(\mathbb{R} - \{0\}, \cdot)$

6). A set of all 3D vectors $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ with a usual addition (vector)

Closure: $\checkmark$ , Associativity: $\checkmark$

Identity: $\vec{0}$: $\vec{v} + \vec{0} = \vec{v}$, Inverse: $\vec{v}$: $-\vec{v}$

7). A set of all rotations of 2D space

with successive operations as multiplication

(ex. $\ast R(60^\circ)R(30^\circ) = R(90^\circ)$)

Closure: $\checkmark$, Associativity: $\checkmark$ the same with sym. action.

Identity: $\mathbb{1} = \text{do nothing} = R(0^\circ)$

Inverse: $R(\theta)$: $R(-\theta)$

8). We can write 7) as a 2×2 matrix

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$\uparrow$ rotation

The multiplication is matrix multiplication:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

rotation

The multiplication is matrix multiplication:

$$\begin{pmatrix} \cos(60^\circ) & -\sin(60^\circ) \\ \sin(60^\circ) & \cos(60^\circ) \end{pmatrix} \begin{pmatrix} \cos(30^\circ) & -\sin(30^\circ) \\ \sin(30^\circ) & \cos(30^\circ) \end{pmatrix}$$

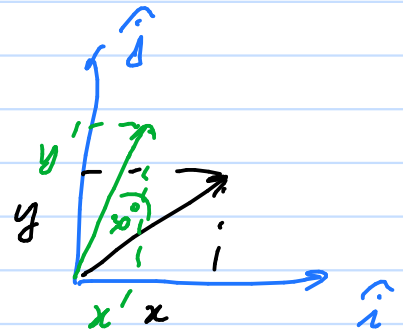
$$= \text{still rotation matrix} = \begin{pmatrix} \text{closure?} \end{pmatrix}$$

Motivation for choosing matrix multiplication

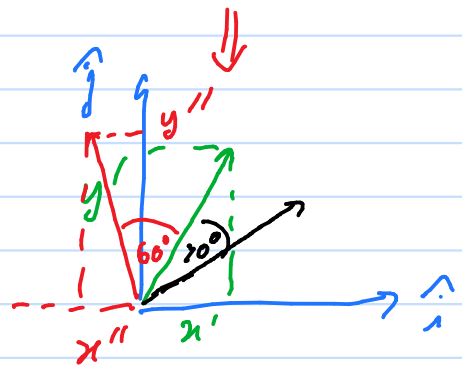
$R(30^\circ) \sim$ action of rotating by 30°

$R(60^\circ) \sim$ action of $\sim 60^\circ$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$



$$\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$



$$\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

* This is why we need matrix multi..

$$\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix} \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

We claim that the set of all rotation matrices (2D) with matrix multiplication is a group.

$$\left\{ \begin{pmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{pmatrix} \right\}, \theta \in [0, 2\pi) \Rightarrow \text{There are } \infty \text{ number of elements.}$$

$$* c\theta = \cos\theta$$

$$s\theta = \sin\theta$$

1). Closure

$$\underbrace{\begin{pmatrix} c\theta_2 & -s\theta_2 \\ s\theta_2 & c\theta_2 \end{pmatrix}}_{R(\theta_2)} \underbrace{\begin{pmatrix} c\theta_1 & -s\theta_1 \\ s\theta_1 & c\theta_1 \end{pmatrix}}_{R(\theta_1)} = \begin{pmatrix} \overbrace{c\theta_2 c\theta_1 - s\theta_2 s\theta_1}^{\cos(\theta_1 + \theta_2)} & \overbrace{-c\theta_2 s\theta_1 - s\theta_2 c\theta_1}^{-\sin(\theta_1 + \theta_2)} \\ \underbrace{s\theta_2 c\theta_1 + c\theta_2 s\theta_1}_{\sin(\theta_1 + \theta_2)} & \underbrace{-s\theta_2 s\theta_1 + c\theta_2 c\theta_1}_{\cos(\theta_1 + \theta_2)} \end{pmatrix}$$

$$\cos(a+b) = \cos a \cos b - \sin a \sin b$$

$$\sin(a+b) = \sin a \cos b + \sin b \cos a$$

$$= \begin{pmatrix} c(\theta_1 + \theta_2) & -s(\theta_1 + \theta_2) \\ s(\theta_1 + \theta_2) & c(\theta_1 + \theta_2) \end{pmatrix}$$

still in the group

2). Associativity ✓

$$(\tilde{A}\tilde{B})\tilde{C} = \tilde{A}(\tilde{B}\tilde{C})$$

follow from matrix mult.

3). Identity.

$$\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \text{identity matrix.}$$

4). Inverse

$$(-\theta \approx 2\pi - \theta)$$

$$\begin{pmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{pmatrix} \rightarrow \begin{pmatrix} c(-\theta) & -s(-\theta) \\ s(-\theta) & c(-\theta) \end{pmatrix}$$

This group is called $SO(2)$

Special Orthogonal group of dim 2.

9) How about we collect all possible 2×2 matrices?

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, a, b, c, d \in \mathbb{R}$$

1). Closure. ✓ 2). Asso ✓

$$(2 \times 2)(2 \times 2) \rightarrow (2 \times 2)$$

3). Identity. $\mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, 4). Inverse.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

This is not a group.

↳ some matrices are non-invertible

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

(det = 0)

★ If we choose only invertible matrices

⇒ This is a group. (det ≠ 0)

1). Closure , 2) Asso ✓

$$(\text{inv}) \cdot (\text{inv}) = \text{inv}$$

3). Identity = $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, 4) Inverse ✓

⇒ This group is called $GL(2, \mathbb{R})$

⇒ "matrix group"

↑
2x2 component

General Linear group of dim 2 over $\mathbb{R}$

"General Linear"

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \Rightarrow \quad \begin{matrix} * \\ \text{basis transformation} \\ \text{coordinate transformation} \end{matrix}$$

$$x' = ax + by$$

$$y' = cx + dy$$

linear equations

"power of $x, y = 1$ "

linear transformation

Scaling

$$x' = ax$$

rotation is a part of this

$$(a = \cos \theta, b = -\sin \theta, c = \sin \theta, d = \cos \theta)$$

linear operator:

$$\vec{x}' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \vec{x} \quad \begin{matrix} \nearrow \begin{pmatrix} x' \\ y' \end{pmatrix} \\ \nearrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ \nearrow \begin{pmatrix} x \\ y \end{pmatrix} \end{matrix}$$

operator

vector



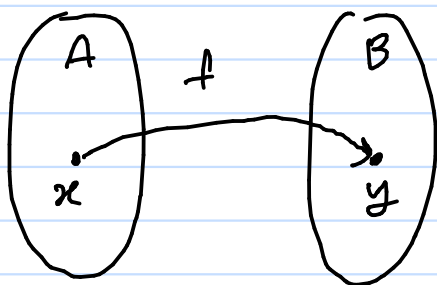
vector (on the same space)

Maps

$\Rightarrow$ connection between sets

Set A, B , a map f denoted by

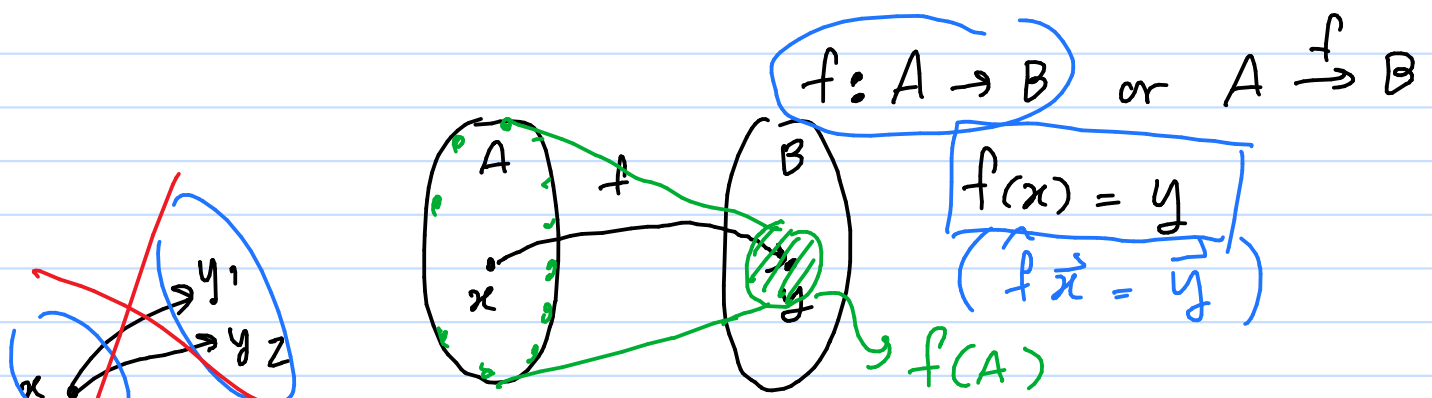
$$f: A \rightarrow B \quad \text{or} \quad A \xrightarrow{f} B$$



$$f(x) = y$$

$$(\hat{f} \vec{x} = \vec{y})$$

* Each element (x) in A maps to only one element in B

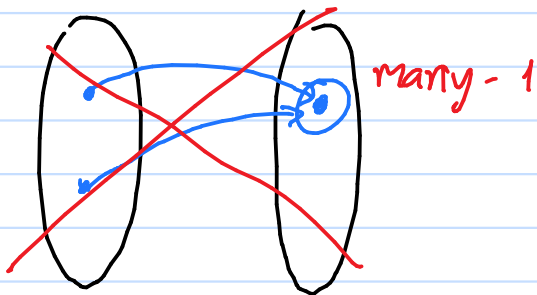


* Each element (x) in A maps to only one element in B

A = domain, B = codomain / target space

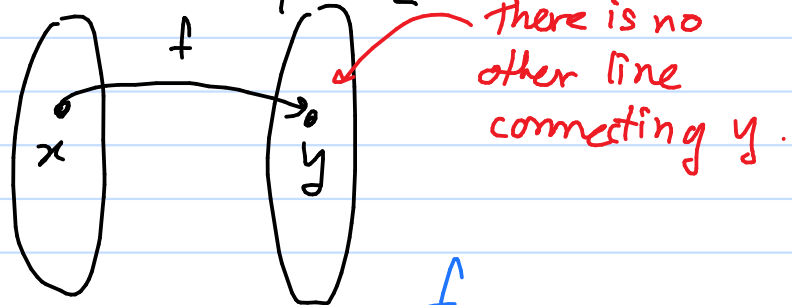
$f(A)$ = range

* injective map (1-1 map)



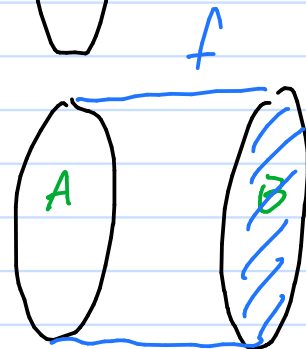
If $f(x_1) = f(x_2)$

then $x_1 = x_2$



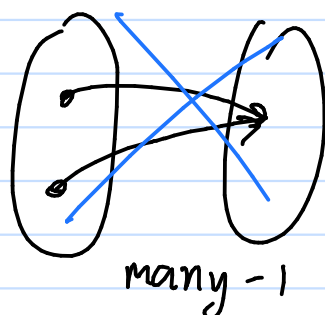
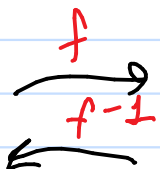
* surjective (onto map)

$f(A) = B$
 range $\swarrow$ $\nwarrow$ codomain

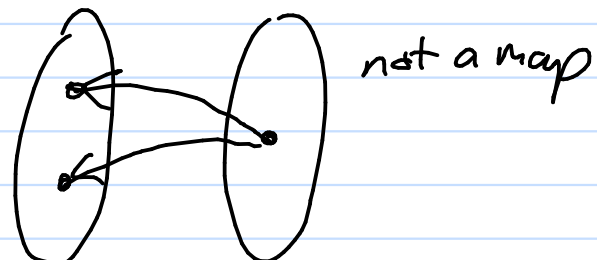


* bijective = injective + surjective (1-1 correspondance)

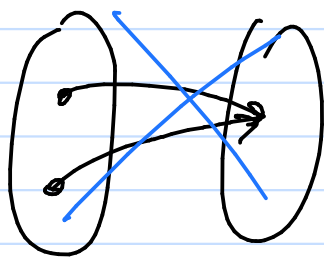
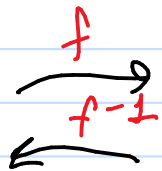
inverse map? $\Rightarrow$ bijective (1-1 correspondance)



$\Rightarrow$

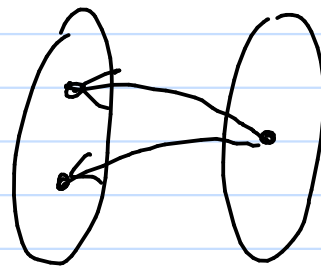


inverse map? $\Rightarrow$ bijective (1-1 correspondence)



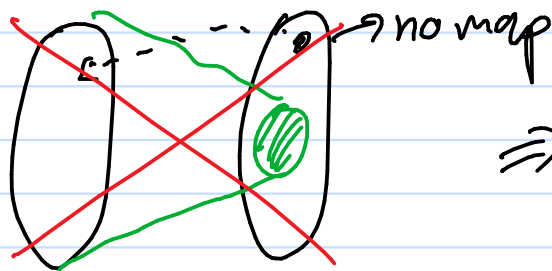
many - 1

$\Rightarrow$



not a map

Since all elements in Θ must have a mapping to it



$\Rightarrow$ bijective (1-1 correspondence)

not surjective (onto)

Now let's get back to group.

$\Rightarrow \{1, -1\} \cong \{I, R\} \rightarrow$ sym action group.

$\uparrow \mathbb{Z}_2$

* Some groups are equivalent (related)

Can we identify their connections?

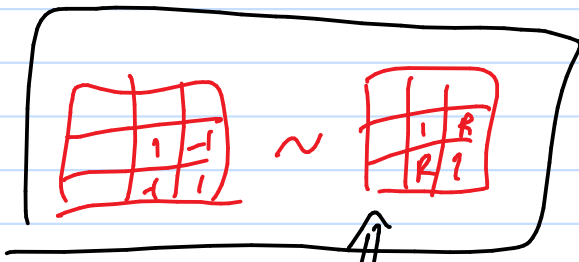
$\Rightarrow$ we need to have similar multiplication table (or multiplication structure)

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & R \\ R & I \end{pmatrix}$$

$$R(\theta_1) R(\theta_2) = R(\theta_1 + \theta_2)$$

$$\begin{pmatrix} \end{pmatrix} \begin{pmatrix} \end{pmatrix} = \begin{pmatrix} \end{pmatrix}$$

$\Rightarrow$ We need to have similar multiplication table (or multiplication structure)



$1 \rightarrow 1$
 $R \rightarrow -1$

$$R(\theta_1) R(\theta_2) = R(\theta_1 + \theta_2)$$

$$\begin{pmatrix} & \end{pmatrix} \begin{pmatrix} & \end{pmatrix} = \begin{pmatrix} & \end{pmatrix}$$

Symmetry action
of rotation in 2D
 $\{ R(\theta) \}$

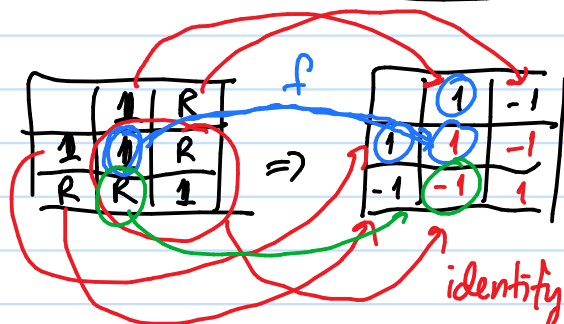
Rotation matrix

$$\left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \right\}$$

Mapping

$$R(\theta) \rightarrow \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$f(R(\theta)) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



Define mapping
 f

identify the equivalence

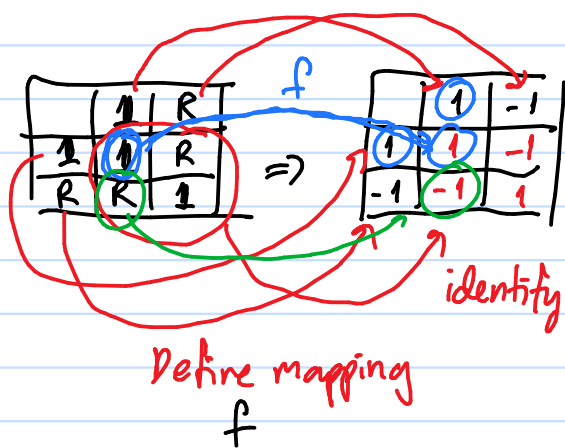
$$11 = 1 \quad \boxed{1 \cdot 1 = 1}$$

$$f(1) \cdot f(1) = f(11) \quad \checkmark$$

$$\underbrace{f(R)}_{-1} \cdot \underbrace{f(1)}_1 = \underbrace{f(R1)}_{-1} \quad \checkmark$$

$$f(1) \cdot f(R) = f(1R)$$

$$f(R) \cdot f(R) = f(RR)$$



$$1 \cdot 1 = 1$$

$$1 \cdot 1 = 1$$

$$f(1) \cdot f(1) = f(1 \cdot 1) \checkmark$$

$$f(R) \cdot f(1) = f(R \cdot 1) \checkmark$$

$$f(1) \cdot f(R) = f(1 \cdot R)$$

$$f(R) \cdot f(R) = f(R \cdot R)$$

Start from basic again

$G = \{1, R\}$ = symmetry action of $\bullet \longrightarrow \bullet$

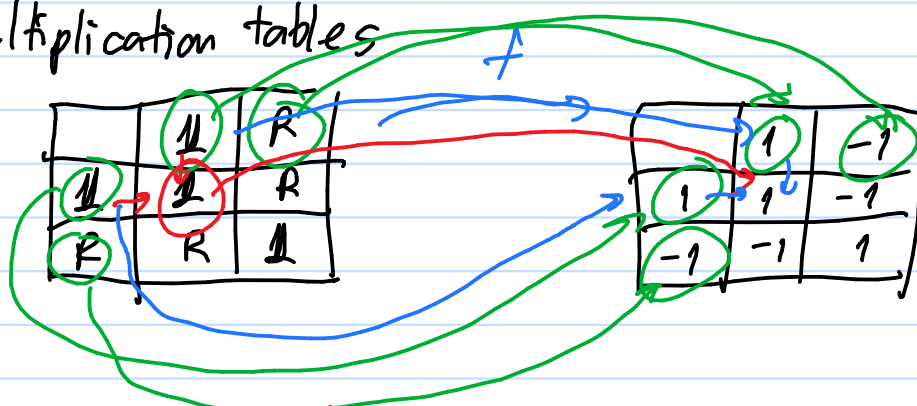
$\mathbb{Z}_2 = \{1, -1\}$ = parity group.

We can write a mapping $f: G \rightarrow \mathbb{Z}_2$ such that

$$f(1) = 1, f(R) = -1.$$

we want this mapping to show equivalence of 2

multiplication tables



1) we look at the first table $1 \cdot 1 = 1$

$$\text{multiply then map} = f(1 \cdot 1)$$

2) we look at the second table : map then multiply
 $= f(1) \cdot f(1)$

check this for every element

$\checkmark$
 $\Downarrow$
 equivalent!

Let's go back to the rotation group

Symmetry action
of rotation in 2D

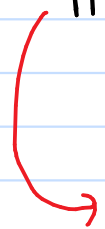
$$\{ R(\theta) \}$$

Rotation matrix

$$\left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \right\}$$

Mapping

$$R(\theta) \rightarrow \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

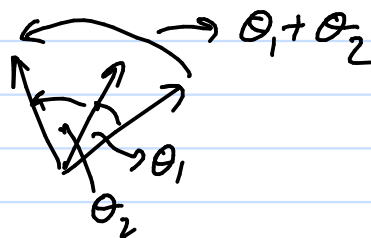


$$f(R(\theta)) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

We can use the equivalence condition

for every θ_1, θ_2 : $\underbrace{f(R(\theta_1))f(R(\theta_2))}_{\text{map then multiply}} = \underbrace{f(R(\theta_1)R(\theta_2))}_{\text{multiply then map}}$

Since $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$



We can check that

$$\underbrace{f(R(\theta_1))f(R(\theta_2))}_{\text{LHS}} = \begin{pmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{pmatrix} \begin{pmatrix} \cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{pmatrix}$$

LHS

$$= \begin{pmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{pmatrix}$$



$$\underbrace{f(R(\theta_1)R(\theta_2))}_{\text{RHS}} = f(R(\theta_1 + \theta_2)) = \begin{pmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{pmatrix}$$

RHS

Homomorphism : Let $(G, \star)$ & $(H, \bullet)$ be 2 groups.

↑
similar
↑
shape

Homomorphism is a map $f: G \rightarrow H$

such that $f(g_1) \bullet f(g_2) = f(g_1 \star g_2)$, $\forall g_1, g_2$

then we say that group G & group H are homomorphic

(★) If the map is 1-1 correspondence $\Rightarrow$ isomorphism
(bijective) $\hookrightarrow$ equivalence

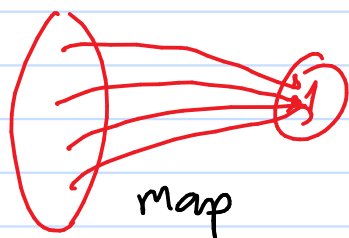
Example

- $\{1, R\}$ & $\mathbb{Z}_2 = \{1, -1\}$ are isomorphic
- Sym act. of rotation on 2D plane & rotation matrix on 2D plane are isomorphic

Let's consider group $G = \{1, R_1, R_2, F_1, F_2, F_3\}$

and consider the trivial group $\{1\}$

Can we find a homomorphism for these 2 groups?



$$f(x) = 1, x \in G$$

this is the only choice!

$$f(R_1) \cdot f(R_1) \stackrel{?}{=} f(\underbrace{R_1 R_1}_{R_2}) \quad \checkmark$$

$$1 \cdot 1 = 1 \quad \text{This is always true!}$$

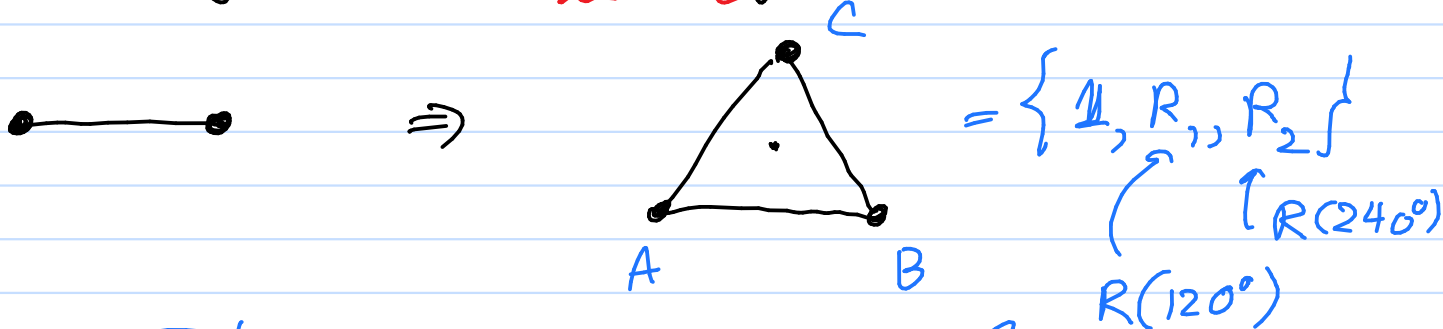
$\therefore$ The map from any group to the trivial group

is always a homomorphism

isomorphic $\sim$ equivalence (iso = equal)

Let's consider some more groups.

Symmetry action of rotating 3 objects $\equiv \mathbb{Z}_3$



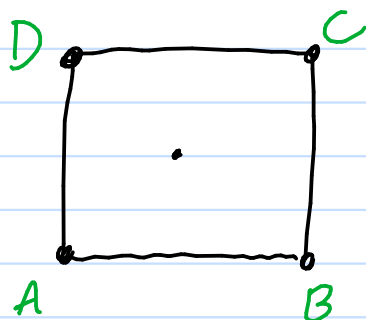
→

	1	R_1	R_2
1	1	R_1	R_2
R_1	R_1	R_2	1
R_2	R_2	1	R_1

Is this a group?

- 1 Closure ✓
- 2 Assoc ✓
- 3 Identity ✓
- 4 Inverse ✓ $R_1^{-1} = R_2$

How about 4 objects



Sym. act. of rotating 4 objects.

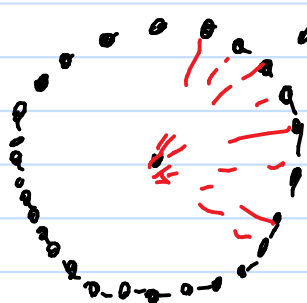
$$= \{1, R(90^\circ), R(180^\circ), R(270^\circ)\}$$

Is this a group ✓

$$= \mathbb{Z}_4$$



n objects



⇒ Sym. act. of rotating n objects

$$= \{1, R_1, \dots, R_{n-1}\} = \mathbb{Z}_n$$

= cyclic group of order n $\swarrow$ Zyklic.

Notice that a cyclic group can be written as a "power" of a single element.

$$\hookrightarrow g \cdot g = g^2, \quad \underbrace{g \cdot g \cdots g}_n = g^n$$

generator.

$$\text{Ex. } \mathbb{Z}_3 = \{ \mathbb{1}, R(120^\circ), R(240^\circ) \} = \{ R^3, R, R^2 \}$$

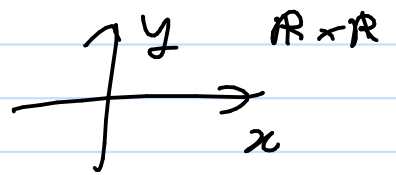
$\uparrow$
 $\equiv R$

$$\mathbb{Z}_4 = \{ R^4, R, R^2, R^3 \}, \quad \mathbb{Z}_n = \{ R^n, R, \dots, R^{n-1} \}$$

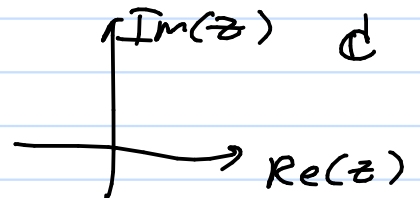
We can extend $n \rightarrow \infty$; $\mathbb{Z}_\infty \sim$ sym action of rotation in 2D plane.

Let's find the isomorphism of $\mathbb{Z}_3 = \{ R^3, R, R^2 \}$

Option 1) $R = \begin{pmatrix} \cos 120^\circ & -\sin 120^\circ \\ \sin 120^\circ & \cos 120^\circ \end{pmatrix} \checkmark$



Option 2) $R = e^{i \frac{2\pi}{3}}$



$$\mathbb{Z}_3 = \{ \mathbb{1}, R_1, R_2 \}, \quad G = \{ 1, e^{\frac{2\pi i}{3}}, e^{\frac{4\pi i}{3}} \}$$

We can check that the map

$$f(\mathbb{1}) = 1, \quad f(R_1) = e^{\frac{2\pi i}{3}}, \quad f(R_2) = e^{\frac{4\pi i}{3}}$$

This is homomorphism (isomorphism), Exercise: Check this.

Now let's check the multiplication table

	$\mathbb{1}$	R_1	R_2
$\mathbb{1}$	$\mathbb{1}$	R_1	R_2
R_1	R_1	R_2	$\mathbb{1}$
R_2	R_2	$\mathbb{1}$	R_1

$$f(1) = 1, \quad f(R_1) = e^{\frac{2\pi i}{3}}, \quad f(R_2) = e^{\frac{4\pi i}{3}}$$

Now let's check the multiplication table

	1	R_1	R_2
1	1	R_1	R_2
R_1	R_1	R_2	1
R_2	R_2	1	R_1

$$f(1) \cdot f(1) = f(11) \quad \checkmark$$

$$1 \cdot 1 = f(1) = 1 \quad \checkmark$$

obvious!

$$x = R_1, R_2; \quad f(x) \cdot f(1) = f(x1)$$

$$f(x) \cdot 1 = f(x) = f(x) \quad \checkmark$$

true for
 R_1, R_2

$$f(1) f(x) = f(1x) \quad \checkmark$$

$$f(R_1) \cdot f(R_1) = f(R_1 R_1)$$

$$e^{\frac{2\pi i}{3}} \cdot e^{\frac{2\pi i}{3}} = f(R_2) = e^{\frac{4\pi i}{3}}$$

$$e^{\frac{4\pi i}{3}} = e^{\frac{4\pi i}{3}} \quad \checkmark$$

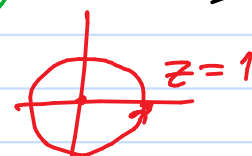
$$f(R_1) \cdot f(R_2) = f(R_1 R_2)$$

$$e^{\frac{2\pi i}{3}} \cdot e^{\frac{4\pi i}{3}} = f(1)$$

$$e^{\frac{6\pi i}{3}} = 1 \quad \checkmark$$

$$e^{2\pi i} = 1$$

$$f(R_2) \cdot f(R_1) = f(R_2 R_1) \quad \checkmark$$



$$f(R_2) f(R_2) = f(R_2 R_2)$$

$$e^{\frac{4\pi i}{3}} e^{\frac{4\pi i}{3}} = f(R_1) = e^{\frac{2\pi i}{3}} \quad \checkmark$$

$$e^{\frac{8\pi i}{3}} = e^{\frac{6\pi i}{3} + \frac{2\pi i}{3}} = (e^{\frac{6\pi i}{3}}) e^{\frac{2\pi i}{3}}$$

$\xrightarrow{2\pi i}$

Can we find more groups that are isomorphic with $\mathbb{Z}_3$?

→ There are a lot more, for example

$$G = \{1, e^{\frac{2\pi i}{3}}, e^{\frac{4\pi i}{3}}\}, \quad G' = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ab & 0 \\ 0 & 1 \end{pmatrix}, \quad f(a) = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

→ There are a lot more, for example

$$G = \{1, e^{\frac{2\pi i}{3}}, e^{\frac{4\pi i}{3}}\}, \quad G' = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ab & 0 \\ 0 & 1 \end{pmatrix}, \quad f(a) = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

We can show that G' & G are isomorphic $f: G \rightarrow G'$

Isomorphism $\Rightarrow f(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f(e^{\frac{2\pi i}{3}}) = \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix}$

$$f(e^{\frac{4\pi i}{3}}) = \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix}.$$

$$e^{\frac{6\pi i}{3}} = e^{2\pi i} = 1$$

$$f(e^{\frac{2\pi i}{3}}) f(e^{\frac{4\pi i}{3}}) = f(e^{\frac{2\pi i}{3}} e^{\frac{4\pi i}{3}}) = f(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{\frac{2\pi i}{3}} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \checkmark$$

We can check all elements in the mul. table

Or simply check

→ homomorphism.

$$f(a)f(b) = f(ab) \quad \text{since } f(a) = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ab & 0 \\ 0 & 1 \end{pmatrix}$$

$$G'' = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 0 \end{pmatrix} \right\}$$

$$G''' = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & e^{\frac{2\pi i}{3}} \end{pmatrix}, \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & e^{\frac{4\pi i}{3}} \end{pmatrix} \right\} \checkmark$$

We have a nontrivial example as well

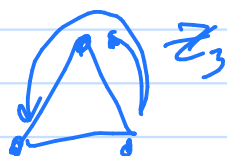
Isomorphism between $\mathbb{Z}_3 = \{1, R_1, R_2\}$ &

$$H = \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \right\}$$

$$f(1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, f(R_1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, f(R_2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Exercise Check that this map is isomorphism!

★ Notice that we often choose the numbers as a group and use homomorphism condition to relate with an abstract group (sym. act.) $\Rightarrow$ It is easier to manipulate!
 $\Rightarrow$ We call this numbers group "representation"



$$\sim \underset{1 \times 1}{*} \{1, e^{\frac{2\pi i}{3}}, e^{\frac{4\pi i}{3}}\} = \text{representation of } \mathbb{Z}_3$$

$$\sim \underset{2 \times 2}{*} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix} \right\} = \text{another representation}$$

"The numbers as a group" is a matrix group = general linear group

mapping

★ Homomorphism between a group and $GL(n, \mathbb{R})$ is called "representation"

Let's get back to basic of group theory.

- For a finite group, the number of elements = order

Ex $\mathbb{Z}_3$ has order 3, $\mathbb{Z}_2$ has order 2

a sym. act. of 3 objects = $\{1, R_1, R_2, F_1, F_2, F_3\}$

order = 6

- Some groups are commutative:

Commutative: $xy = yx$, $\forall x, y$

$\mathbb{Z}_3 = \{1, R_1, R_2\}$

	1	R_1	R_2
1	1	R_1	R_2
R_1	R_1	1	R_2
R_2	R_2	R_1	1

$R_1 R_2 = 1$
 $R_2 R_1 = 1$ ✓

This is called "Abelian group" (Abel)

How about a sym. act. of 3 objects = $\{1, R_1, R_2, F_1, F_2, F_3\}$?

	1	R_1	R_2	F_1	F_2	F_3
1	1	R_1	R_2	F_1	F_2	F_3
R_1	R_1	R_2	1	F_3		
R_2	R_2	1	R_1			
F_1	F_1			1		
F_2	F_2			R_2	1	
F_3	F_3					1

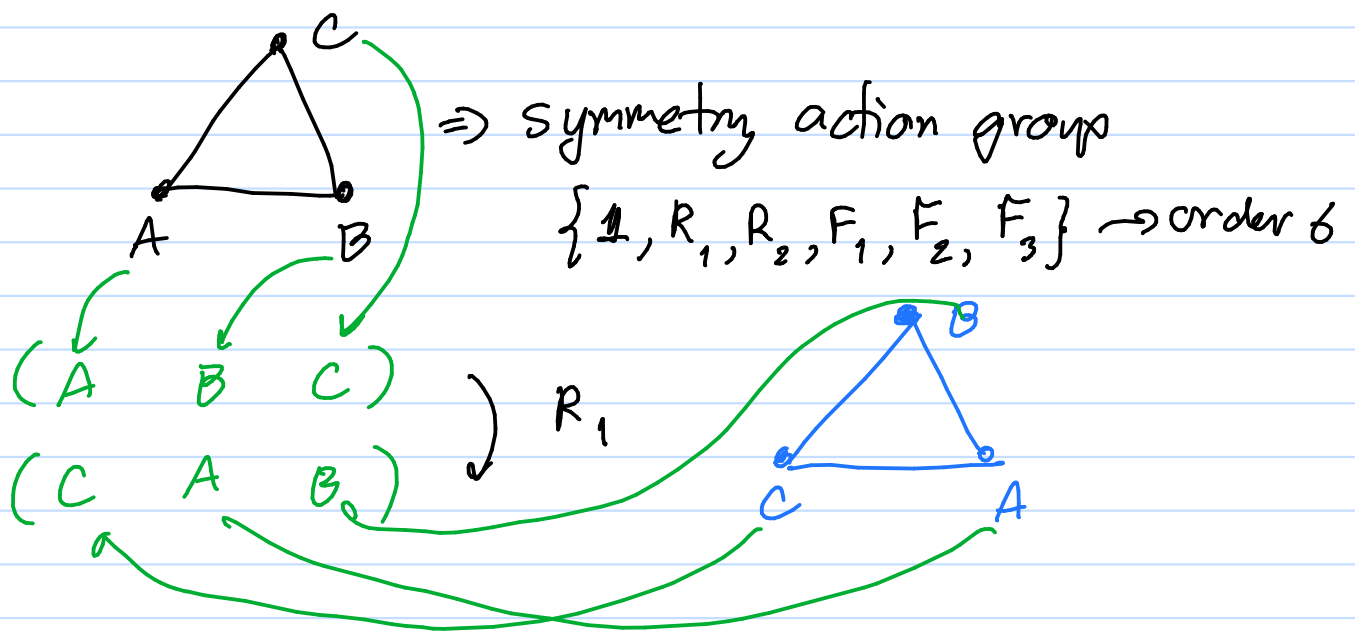
$F_3 F_1 = R_1$
 $F_1 F_3 = R_2$ ≠

↓
non Abelian group.

The permutation group

The permutation group

Let's get back to the first example



Changing the positions of vertices $\sim$ symmetry action

$$R_1: (A \ B \ C) \rightarrow (C \ A \ B)$$

$$R_2: (A \ B \ C) \rightarrow (B \ C \ A)$$

How many ways to arrange this? $\Rightarrow 3 \times 2 \times 1 = 6 = 3!$

$\Rightarrow$ Permutation group

3 objects

$\Rightarrow$ This group is called $S_3 = \{1, R_1, R_2, F_1, F_2, F_3\}$

$$S_2 \cong \mathbb{Z}_2$$

Sometimes, we use the notation of permutation

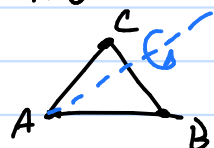
as an element

$$\begin{pmatrix} A & B & C \\ C & A & B \end{pmatrix}$$

$$R_1: \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

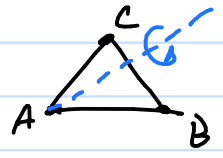
$$R_2: \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$F_1: \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$



and so on.

as an element



$$R_1: \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, R_2: \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, F_1: \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

and so on.

useful in multiplication

$$F_1 R_1: \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = F_3$$

Short-hand notation

$$R_1: \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \Rightarrow (1 \ 2 \ 3)$$

= cyclic permutation inside the parenthesis

$$F_1: \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (1)(2 \ 3)$$

$\Rightarrow$ cycle notation

Consider $S_3 = \{ \mathbb{1}, R_1, R_2, F_1, F_2, F_3 \}$

$\hookrightarrow$ permutation groups of 3 objects



Consider $\mathbb{Z}_3 = \{ \mathbb{1}, R_1, R_2 \}$

$\hookrightarrow$ rotation sym. act. of 3 objects (cyclic group)

★ $\mathbb{Z}_3$ is a subset of S_3

Since $\mathbb{Z}_3$ itself is a group $\rightarrow \mathbb{Z}_3$ is a subgroup of S_3

$\hookrightarrow$ subset that in itself is a group

$\{ \mathbb{1}, R_1 \}$ is not a group $\times$ subgroup

$\hookrightarrow 120^\circ$

just a subset

$R_1 \cdot R_1 = R_2$ no closure

$$S_3 = \{ \mathbb{1}, R_1, R_2, F_1, F_2, F_3 \}$$

$\{ \mathbb{1}, F_1 \}$ is a subset which is also a group

$\Rightarrow$ subgroup

$\{ \mathbb{1}, F_2 \}, \{ \mathbb{1}, F_3 \} = \text{subgroup}$

$\{ \mathbb{1}, R_1, F_1 \}$ is not a subgroup $R_1 F_1 = F_3$ no closure!

$\{ \mathbb{1}, F_1, F_2 \}$ is not a subgroup $F_1 F_2 = R_1 \dots$ no closure!

Consider $\mathbb{Z}_4 = \{ \mathbb{1}, R_1, R_2, R_3 \}$, $R_1 = \text{rotate } \frac{\pi}{2}, \dots$

$\mathbb{Z}_2 = \{ \mathbb{1}, R_2 \} = \text{subgroup of } \mathbb{Z}_4$

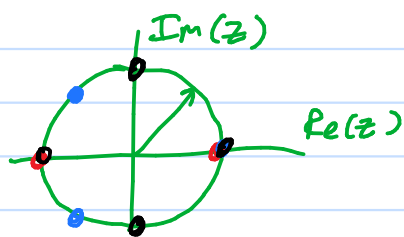
★ The trivial subgroup $\{ \mathbb{1} \}$

Consider the group of 2D rotation $\left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \right\}$

or $\{ e^{i\theta} \} = \{ z \mid |z| = 1, z \in \mathbb{C} \}$

$$e^{i\theta} \cdot e^{-i\theta} = 1$$

The set of all complex number with the unit magnitude



consider the subset $\{ 1, -1 \} = \mathbb{Z}_2 \subset R_{2D}$

consider the subset $\{ 1, e^{\frac{2\pi i}{3}}, e^{\frac{4\pi i}{3}} \} = \mathbb{Z}_3 \subset R_{2D}$

Subset $\{ 1, e^{\frac{2\pi i}{n}}, e^{\frac{2\pi i}{n} \cdot 2}, \dots, e^{\frac{2\pi i}{n} \cdot (n-1)} \} = \mathbb{Z}_n \subset R_{2D}$

There is a pattern in the subgroup structure!

Coset

Consider a subgroup H of a group G , $H \subset G$

We can define a coset as a set of multiplication of an element in G and subgroup H ↗ g

$$\text{coset} = H \cdot g = \{h \cdot g \mid \forall h \in H\}, g \in G$$

↖ multiplication

Example

$$\text{Consider } S_3 = \{1, R_1, R_2, F_1, F_2, F_3\}$$

$$\mathbb{Z}_3 \subset S_3, \mathbb{Z}_3 = \{1, R_1, R_2\}$$

$$\begin{aligned} \text{A coset } \mathbb{Z}_3 \cdot F_1 &= \{1 \cdot F_1, R_1 \cdot F_1, R_2 \cdot F_1\} \\ &= \{F_1, F_3, F_2\} \Rightarrow \text{is not a group} \end{aligned}$$

$$\begin{aligned} \text{A coset } \mathbb{Z}_3 \cdot R_1 &= \{1 \cdot R_1, R_1 \cdot R_1, R_2 \cdot R_1\} \\ &= \{R_1, R_2, 1\} = \mathbb{Z}_3 \Rightarrow \text{is a group} \end{aligned}$$

Notice that the multiplication can be done from the left or the right $\Rightarrow$ different sets!

$\Rightarrow$ We need to redefine a coset as a right coset & left coset

$\therefore$ A right coset $H \cdot g$

A left coset $g \cdot H$

$$\begin{aligned} \text{Ex } F_1 \cdot \mathbb{Z}_3 &= \{F_1 \cdot 1, F_1 \cdot R_1, F_1 \cdot R_2\} \\ \text{left coset} &= \{F_1, F_3, F_2\} \end{aligned}$$

Exercise Calculate all left-cosets & right cosets of

$$\mathbb{Z}_3 \subset S_3$$

$$\begin{array}{c|c|c} 1 \cdot \mathbb{Z}_3 = \mathbb{Z}_3 & R_1 \cdot \mathbb{Z}_3 = \dots & R_2 \cdot \mathbb{Z}_3 = \dots \\ \mathbb{Z}_3 \cdot 1 = \mathbb{Z}_3 & \mathbb{Z}_3 \cdot R_1 = \dots & \mathbb{Z}_3 \cdot R_2 = \dots \end{array}$$

... $\Rightarrow$ Homework

Notice

1). For $\mathbb{Z}_3$, all left coset = all right coset

$$R_1 \cdot \mathbb{Z}_3 = \mathbb{Z}_3 \cdot R_1, \quad R_2 \cdot \mathbb{Z}_3 = \mathbb{Z}_3 \cdot R_2 \dots$$

In this case, we call $\mathbb{Z}_3$ normal subgroup

(or invariant subgroup)

$$\{1, R_1, R_2\} \text{ \& \& } \{F_1, F_2, F_3\}$$

2). All cosets is either identical or disjoint

$$\{1, R_1, R_2\} \text{ \& } \{1, R_1, R_2\}$$

looks like
a $\mathbb{Z}_2$

all elements
are different

$$\left\{ \{1, R_1, R_2\}, \{F_1, F_2, F_3\} \right\}$$

Invariant (Normal) subgroup

= a subgroup that all left cosets = right cosets.

We can define a new group from invariant subgroup

If H is a normal subgroup of G , then the collection
of all cosets $= G/H$ is a group = quotient group

Ex The collection of all cosets of $\mathbb{Z}_3$ (factor group)

$$\left\{ \{1, R_1, R_2\}, \{F_1, F_2, F_3\} \right\} = S_3 / \mathbb{Z}_3$$

If H is a normal subgroup of G , then the collection of all cosets $= G/H$ is a group $=$ quotient group

Ex The collection of all cosets of $\mathbb{Z}_3$

$$\left\{ \{1, R_1, R_2\}, \{F_1, F_2, F_3\} \right\} = S_3 / \mathbb{Z}_3 \sim \mathbb{Z}_2$$

$\{1, R_1, R_2\} \star \{F_1, F_2, F_3\}$ quotient group

$$\begin{array}{l} \{1, R_1, R_2\} \cdot F_1 = \{F_1, F_2, F_3\} \\ \{1, R_1, R_2\} \cdot F_2 = \{F_1, F_2, F_3\} \\ \{1, R_1, R_2\} \cdot F_3 = \{F_1, F_2, F_3\} \end{array} \quad \left| \quad \begin{array}{l} \{1, R_1, R_2\} \cdot 1 = \{1, R_1, R_2\} \\ \quad \quad \quad \cdot R_1 = \{1, R_1, R_2\} \\ \quad \quad \quad \cdot R_2 = \{1, R_1, R_2\} \end{array} \right.$$

$$1 \times -1 = -1$$

$$1 \times 1 = 1$$

$$\begin{array}{l} \{F_1, F_2, F_3\} \cdot F_1 = \{1, R_1, R_2\} \\ \quad \quad \quad \cdot F_2 = \{1, R_1, R_2\} \\ \quad \quad \quad \cdot F_3 = \{1, R_1, R_2\} \end{array} \quad \left| \quad \begin{array}{l} -1 \times -1 = 1 \end{array} \right.$$

Example

$$\mathbb{Z}_4 = \{1, R_1, R_2, R_3\} \sim \left\{ 1, e^{\frac{i\pi}{2}}, e^{i\pi}, e^{\frac{3i\pi}{2}} \right\}$$

$$\bullet \mathbb{Z}_2 = \{1, R_2\} \sim \{1, e^{i\pi}\}$$

Consider the left coset

$$\begin{aligned} 1 \cdot \mathbb{Z}_2 &= \mathbb{Z}_2, & e^{\frac{i\pi}{2}} \cdot \mathbb{Z}_2 &= \left\{ e^{\frac{i\pi}{2}} \cdot 1, e^{\frac{i\pi}{2}} \cdot e^{i\pi} \right\} \\ & & &= \left\{ e^{\frac{i\pi}{2}}, e^{\frac{3i\pi}{2}} \right\} \end{aligned}$$

Consider the left coset



$$1 \cdot \mathbb{Z}_2 = \mathbb{Z}_2, \quad e^{\frac{i\pi}{2}} \cdot \mathbb{Z}_2 = \left\{ e^{\frac{i\pi}{2}} \cdot 1, e^{\frac{i\pi}{2}} \cdot e^{i\pi} \right\}$$

$$= \left\{ e^{\frac{i\pi}{2}}, e^{\frac{3i\pi}{2}} \right\}$$

$$\frac{3}{2} + 1 = \frac{5}{2}$$

$$\Rightarrow = 2 + \frac{1}{2}$$

$$e^{i\pi} \cdot \mathbb{Z}_2 = \left\{ e^{i\pi} \cdot 1, e^{i\pi} \cdot e^{i\pi} \right\}, \quad e^{\frac{3i\pi}{2}} \cdot \mathbb{Z}_2 = \left\{ e^{\frac{3i\pi}{2}} \cdot 1, e^{\frac{3i\pi}{2}} \cdot e^{i\pi} \right\}$$

$$= \left\{ e^{i\pi}, 1 \right\} = \mathbb{Z}_2$$

$$= \left\{ e^{\frac{3i\pi}{2}}, e^{\frac{i\pi}{2}} \right\}$$

Since this is an abelian group, i.e., $g_1 \cdot g_2 = g_2 \cdot g_1$

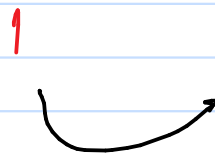
$$g \cdot \mathbb{Z}_2 = \mathbb{Z}_2 \cdot g \Rightarrow \text{left coset} = \text{right coset}$$

$\therefore \mathbb{Z}_2$ is a normal subgroup of $\mathbb{Z}_4$

$$\text{The collection of all cosets} = \left\{ \underbrace{\{1, e^{i\pi}\}}_1, \underbrace{\{e^{\frac{i\pi}{2}}, e^{\frac{3i\pi}{2}}\}}_{-1} \right\}$$

$$\sim \mathbb{Z}_2$$

$$\boxed{\mathbb{Z}_4 / \mathbb{Z}_2 \sim \mathbb{Z}_2}$$



Conjugacy

The conjugate element h' of h is an element

that is related with h by $\boxed{h' = ghg^{-1}}$, $g \in G$

Ex Consider $S_3 = \{1, R_1, R_2, F_1, F_2, F_3\}$

What are conjugate element of R_1

$$\text{Consider } R_2 = \underbrace{X}_{\text{try } R_1} R_1 \underbrace{X^{-1}}_{\text{try } R_2}$$

$$R_1^{-1} = R_2$$

try ~~R_1~~ ~~R_2~~

$$F_1^{-1} = F_1$$

$$R_2 \stackrel{?}{=} R_1 R_1 R_2 \times$$

	1	R_1	R_2	F_1	F_2	F_3
1	1	R_1	R_2	F_1	F_2	F_3
R_1	R_1	R_2	1	F_3	F_1	F_2
R_2	R_2	1	R_1	F_2	F_3	F_1
F_1	F_1	F_2	F_3	1	R_1	R_2
F_2	F_2	F_3	F_1	R_2	1	R_1
F_3	F_3	F_1	F_2	R_1	R_2	1

Ex Consider $S_3 = \{1, R_1, R_2, F_1, F_2, F_3\}$

What are conjugate element of $R_1 = R_1, R_2$

Consider $R_2 = \underbrace{X} \underbrace{R_1} \underbrace{X^{-1}}$

	1	R_1	R_2	F_1	F_2	F_3
1	1	R_1	R_2	F_1	F_2	F_3
R_1	R_1	R_2	1	F_3	F_1	F_2
R_2	R_2	1	R_1	F_2	F_3	F_1
F_1	F_1	F_2	F_3	1	R_1	R_2
F_2	F_2	F_3	F_1	R_2	1	R_1
F_3	F_3	F_1	F_2	R_1	R_2	1

$$R_1^{-1} = R_2$$

try $R_1 R_2$

$$F_1^{-1} = F_1$$

$$R_2 = R_1 R_1^{-1} R_2 X = R_1$$

$$R_1 R_1 R_1^{-1} = R_1 R_1 R_2 = R_1 \Rightarrow R_1 = \text{conjugate of } R_1$$

$$R_2 R_1 R_2^{-1} = R_2 R_1 R_1 = R_1 \Rightarrow \text{(obvious since } R_1 = 1 R_1 1^{-1} \text{)}$$

$$F_1 R_1 F_1^{-1} = F_1 (R_1 F_1) = F_1 F_3 = R_2 \Rightarrow R_1 \text{ \& } R_2 \text{ are conjugate}$$

$$F_2 R_1 F_2^{-1} = F_2 (R_1 F_2) = F_2 F_1 = R_2$$

$$F_3 R_1 F_3^{-1} = F_3 (R_1 F_3) = F_3 F_2 = R_2$$

How about 1?

Let's check ^{any} every elements

$$F_1 1 F_1^{-1} = 1, F_2 1 F_2^{-1} = 1$$

$$\Rightarrow X 1 X^{-1} = 1$$

Exercise

Find all conjugate elements of F_1, F_2, F_3

★ Notice that we can divide group from the conjugacy.

$$\Rightarrow \text{class } S_3 \Rightarrow \begin{cases} \{1\} \\ \{R_1, R_2\} \\ \{F_1, F_2, F_3\} \end{cases}$$

Elements belong to the same class are conjugate

$$R_1 F_1 R_2 = R_1 F_3 = F_2$$

Notice that the relation between conjugate elements ^(h, h')

$$h' = ghg^{-1} \Rightarrow \underline{h'g = gh}$$

Suppose that we collect all elements in the same class

$$\begin{aligned} \{h, h', h'', \dots\} * g &= g * \{h', h'', \dots\} \quad \leftarrow \text{left coset} \\ \left. \begin{aligned} \{hg, h'g, h''g, \dots\} &= \{gh', gh'', gh''', \dots\} \\ &= g \cdot \{h', h'', h''', \dots\} \end{aligned} \right\} \quad \text{right coset} \end{aligned}$$

$\therefore \{1\} + \text{classes that form subgroup}$
 $= \text{invariant (normal) subgroup}$

$$\begin{aligned} \{R_1, R_2\} * F_1 &= F_1 * \{R_2, R_1\} \\ F_1 R_1 F_1^{-1} &= R_2 \\ F_1 R_2 F_1^{-1} &= R_1 \end{aligned}$$

Quotient group

Consider group G with normal subgroup H

$\Rightarrow$ Collect all right-cosets of $H = G/H$ *

$\Rightarrow$ Forms group?

Example $S_3 = \{1, R_1, R_2, F_1, F_2, F_3\}$

$$Z_3 = \{1, R_1, R_2\}$$

$$\frac{S_3}{Z_3} = \left\{ \{1, R_1, R_2\}, \{F_1, F_2, F_3\} \right\}$$

Example $S_3 = \{1, R_1, R_2, F_1, F_2, F_3\}$

$$\mathbb{Z}_3 = \{1, R_1, R_2\}$$

$$\frac{S_3}{\mathbb{Z}_3} = \left\{ \{1, R_1, R_2\}, \{F_1, F_2, F_3\} \right\}$$

How do you define a group structure?

Let's go back to the collection of all right cosets

$$\{ \mathbb{Z}_3 \cdot 1, \mathbb{Z}_3 \cdot R_1, \mathbb{Z}_3 \cdot R_2, \mathbb{Z}_3 \cdot F_1, \mathbb{Z}_3 \cdot F_2, \mathbb{Z}_3 \cdot F_3 \}$$

We can define a multiplication between cosets

Suppose we have 2 cosets $H \cdot g_1, H \cdot g_2$

$$\text{The product of 2 cosets} = Hg_1 \star Hg_2 = H(g_1 \cdot g_2)$$

Example

$$\{ \mathbb{Z}_3 \cdot 1, \mathbb{Z}_3 \cdot R_1, \mathbb{Z}_3 \cdot R_2, \mathbb{Z}_3 \cdot F_1, \mathbb{Z}_3 \cdot F_2, \mathbb{Z}_3 \cdot F_3 \}$$

$$\text{Consider } \mathbb{Z}_3 \cdot 1 \star \mathbb{Z}_3 \cdot R_1 = \mathbb{Z}_3 \cdot (R_1 \cdot 1) = \mathbb{Z}_3 \cdot R_1$$

$$\mathbb{Z}_3 \cdot R_1 \star \mathbb{Z}_3 \cdot R_2 = \mathbb{Z}_3 \cdot (R_1 \cdot R_2) = \mathbb{Z}_3 \cdot 1$$

$\therefore$ This forms a group

Closure $\checkmark$ Asso $\checkmark$ Identity $\checkmark$ Inverse $\checkmark$

= Quotient group

collection of all cosets

$$\left\{ \{1, R_1, R_2\}, \{F_1, F_2, F_3\} \right\}$$

$$\mathbb{Z}_3 \cdot 1, \mathbb{Z}_3 \cdot R_1, \mathbb{Z}_3 \cdot R_2$$

$$\mathbb{Z}_3 \cdot F_1, \mathbb{Z}_3 \cdot F_2, \mathbb{Z}_3 \cdot F_3$$

$$\left\{ \left\{ \mathbb{1}, R_1, R_2 \right\}, \left\{ F_1, F_2, F_3 \right\} \right\}$$

$\nearrow \mathbb{Z}_3 \mathbb{1}, \mathbb{Z}_3 R_1, \mathbb{Z}_3 R_2$
 $\nwarrow \mathbb{Z}_3 F_1, \mathbb{Z}_3 F_2, \mathbb{Z}_3 F_3$

$$\text{Ex } \left\{ \mathbb{1}, R_1, R_2 \right\} \star \left\{ F_1, F_2, F_3 \right\}$$

$$\begin{aligned} \mathbb{Z}_3 \cdot \mathbb{1} \star \mathbb{Z}_3 F_1 &= \mathbb{Z}_3 \cdot (\mathbb{1} \cdot F_1) = \mathbb{Z}_3 \cdot F_1 \\ &= \left\{ F_1, F_2, F_3 \right\} \end{aligned}$$

Representation

= homomorphism between a group & $GL(n, \mathbb{R})$ matrix group

Review on Vector space

$$V = \text{vector space} \Rightarrow \text{basis} = \left\{ |v_1\rangle, |v_2\rangle, |v_3\rangle, \dots \right\}$$

$\nearrow$ $\nwarrow$ $\nearrow$
 vectors

$$\begin{aligned} \alpha_1 |v_1\rangle + \alpha_2 |v_2\rangle + \dots &= 0 \\ \alpha_1 = \alpha_2 = \dots &= 0 \end{aligned} \left. \vphantom{\begin{aligned} \alpha_1 |v_1\rangle + \alpha_2 |v_2\rangle + \dots \\ \alpha_1 = \alpha_2 = \dots \end{aligned}} \right\} \text{linearly independent}$$

Normally, we use orthonormal basis $\langle v | u \rangle = \vec{v} \cdot \vec{u}$

$$\left\{ |e_1\rangle, |e_2\rangle, \dots, |e_n\rangle \right\}$$

$$\Rightarrow \langle e_1 | e_1 \rangle = 1, \langle e_2 | e_2 \rangle = 1, \dots$$

$$\langle e_1 | e_2 \rangle = 0, \langle e_1 | e_3 \rangle = 0, \dots$$

$\nearrow$ length = 1

$\nearrow$ orthogonal with the rest

$$\Rightarrow \text{orthogonality condition} \Rightarrow \boxed{\langle e_i | e_j \rangle = \delta_{ij}}$$

$$\delta_{ij} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$$

A vector $|v\rangle = \sum_{i=1}^n v_i |e_i\rangle$ ← basis
 ↑ component

We can also write $\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = |v\rangle$ where the basis is assumed $\{|e_i\rangle\}$

Operator is a map $\hat{T}: V \rightarrow V$

$$|u\rangle \xrightarrow{\hat{T}} |v\rangle = \hat{T}|u\rangle$$

suppose $|u\rangle = \sum_{i=1}^n u_i |e_i\rangle$ $\left(|u\rangle = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} \right)$

We can look at the transformation (operator)

$$\hat{T}|u\rangle = \sum_{i=1}^n u_i \hat{T}|e_i\rangle$$

↑ If we know how each basis transform under $\hat{T}$

$$|v\rangle = \sum_{i=1}^n v_i |e_i\rangle \Rightarrow |v\rangle = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, |u\rangle = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$$

The new vector has components under $\{|e_i\rangle\}$

$$\tilde{M}_{n \times n} \rightarrow e^{\tilde{M}_{n \times n}}$$

$$\tilde{M} + \tilde{N} \rightarrow e^{\tilde{M} + \tilde{N}} \neq e^{\tilde{M}} + e^{\tilde{N}}$$

$$\begin{aligned} & \hat{T}(|v\rangle + |u\rangle) \\ &= \hat{T}|v\rangle + \hat{T}|u\rangle \end{aligned}$$

$$\hat{T}|u\rangle = \sum_{i=1}^n u_i \hat{T}|e_i\rangle$$

$$\hat{T}|u\rangle = |v\rangle$$

$$|v\rangle = \sum_{i=1}^n v_i |e_i\rangle$$

If we know how each basis transform under $\hat{T}$
 that is $\hat{T}|e_1\rangle = |e'_1\rangle = \underline{T_{11}}|e_1\rangle + \underline{T_{21}}|e_2\rangle + \underline{T_{31}}|e_3\rangle + \dots$
 new vector we can expand
 but on the same vector space in the old basis

$$\hat{T}|e_2\rangle = |e'_2\rangle = \underline{T_{12}}|e_1\rangle + \underline{T_{22}}|e_2\rangle + \underline{T_{32}}|e_3\rangle + \dots$$

$$\hat{T}|e_i\rangle = \underline{T_{1i}}|e_1\rangle + \underline{T_{2i}}|e_2\rangle + \underline{T_{3i}}|e_3\rangle + \dots$$

$$\hat{T}|e_i\rangle = \sum_{j=1}^n T_{ji} |e_j\rangle$$

sub into

$$|v\rangle = \sum_{i=1}^n v_i |e_i\rangle = \hat{T}|u\rangle = \sum_{i=1}^n u_i \hat{T}|e_i\rangle$$

$$= \sum_{i=1}^n u_i \sum_{j=1}^n T_{ji} |e_j\rangle = \sum_{j=1}^n u_i T_{ji} |e_j\rangle$$

$$3\hat{i} + 4\hat{j} = x\hat{i} + y\hat{j} \Rightarrow x=3, y=4$$

$$\hat{i} \cdot (3\hat{i} + 4\hat{j}) = \hat{i} \cdot (x\hat{i} + y\hat{j})$$

$$3 = x$$

$$i \leftrightarrow j$$

$$\therefore \sum_{i=1}^n v_i |e_i\rangle = \sum_{j,i=1}^n u_j T_{ij} |e_i\rangle$$

$$\sum_{i=1}^n \underbrace{v_i}_{\text{red wavy}} |e_i\rangle = \sum_{j,i=1}^n \underbrace{u_j T_{ij}}_{\text{green circle}} |e_i\rangle = \sum_{i=1}^n \underbrace{\left(\sum_{j=1}^n u_j T_{ij} \right)}_{\text{green circle}} |e_i\rangle$$

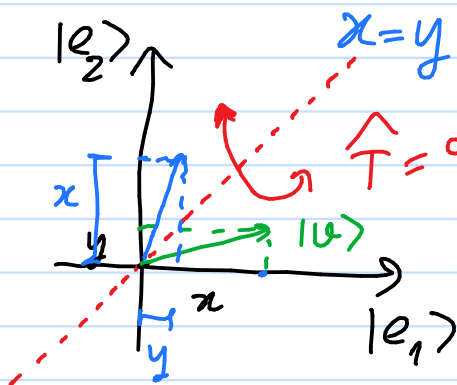
$$v_i = \sum_{j=1}^n T_{ij} u_j \Rightarrow v_1 = T_{11}u_1 + T_{12}u_2 + \dots$$

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} T_{11} & T_{12} & T_{13} & \dots \\ T_{21} & T_{22} & & \\ T_{31} & & \ddots & \\ \vdots & & & \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$$

~~~~~

This matrix is a matrix representation of transformation (operator)

Example Mirroring on 2D plane



$\hat{T}$  = operator of mirroring (flip plane)

$$|v\rangle = x|e_1\rangle + y|e_2\rangle$$

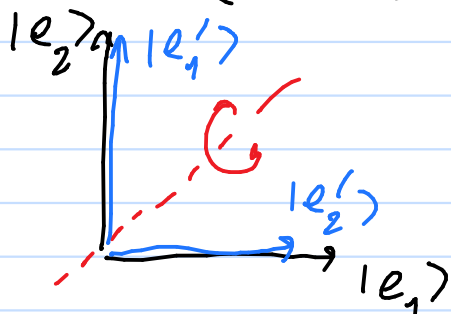
$$\hat{T}|v\rangle = x'|e_1\rangle + y'|e_2\rangle = y|e_1\rangle + x|e_2\rangle !$$

$\Rightarrow$  We can do this in matrix form!

$$|v\rangle = x|e_1\rangle + y|e_2\rangle = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\hat{T} = \begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \end{pmatrix} ?$$

$\Rightarrow$  consider how basis transform?



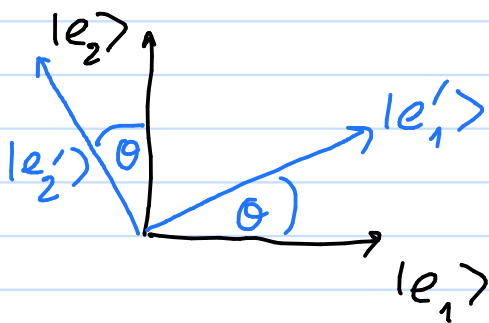
$$\hat{T}|e_1\rangle = |e'_1\rangle = \overset{\uparrow T_{11}}{0}|e_1\rangle + \overset{\uparrow T_{21}}{1}|e_2\rangle$$

$$\hat{T}|e_2\rangle = |e'_2\rangle = \underset{\hookleftarrow T_{12}}{1}|e_1\rangle + \underset{\hookleftarrow T_{22}}{0}|e_2\rangle$$

$$\therefore \hat{T} \text{ in matrix form is } = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\therefore \hat{T}|v\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$$

Ex rotation  $\hat{T}_\theta$



$\Rightarrow$  matrix representation of  $\hat{T}_\theta = ?$

$$\hat{T}_\theta |e_1\rangle = |e_1'\rangle = \cos\theta |e_1\rangle + \sin\theta |e_2\rangle$$

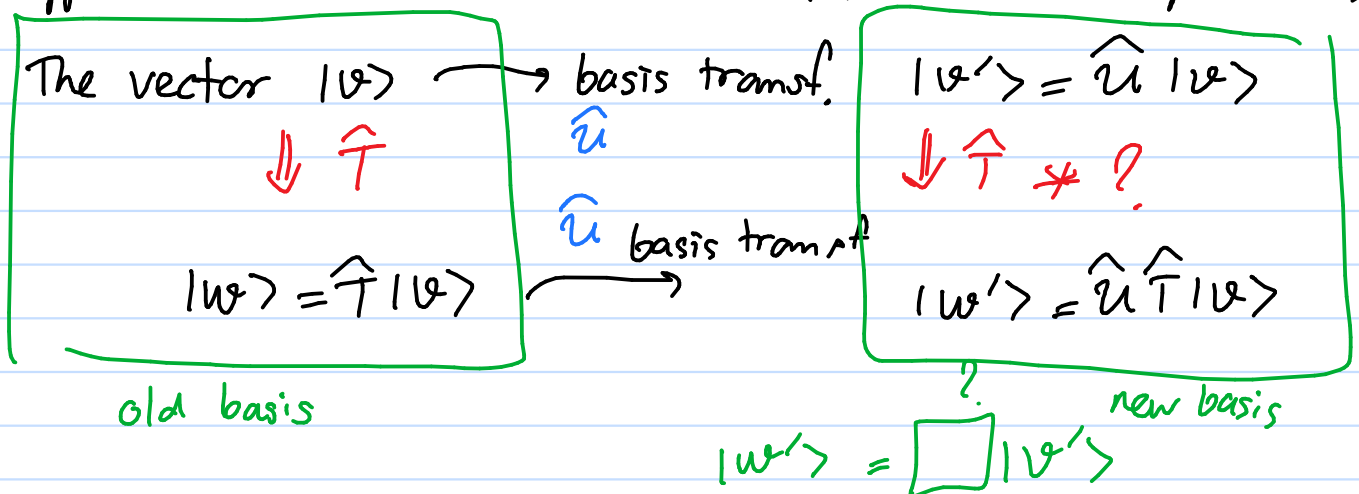
$$\hat{T}_\theta |e_2\rangle = |e_2'\rangle = -\sin\theta |e_1\rangle + \cos\theta |e_2\rangle$$

$$\therefore \hat{T}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \text{rotation matrix}$$

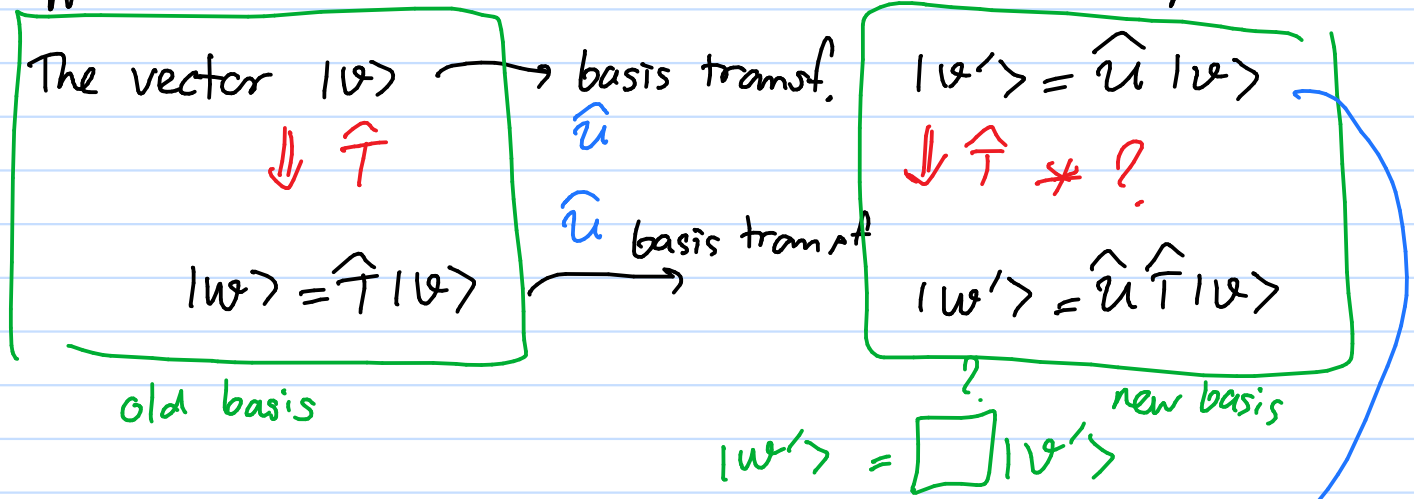
(operator)

- We have seen how basis transformation change vector components
- How about the change of an operator under the basis transformation?

Suppose that we have basis transformation  $\hat{U}$  (operator)



Suppose that we have basis transformation  $\hat{U}$  (operator)



Since  $|\psi\rangle = \hat{U}^{-1} |\psi'\rangle$

$$|\psi'\rangle = \hat{U} \hat{T} |\psi\rangle = \hat{U} \hat{T} \hat{U}^{-1} |\psi'\rangle$$

Operator  $\hat{T}$  in the new basis

$$\hat{T}' = \hat{U} \hat{T} \hat{U}^{-1}$$

Similarity transformation

Ex Consider the identity operator  $= \hat{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

What does it look like under rotation transf.?

(It should look the same)

Basis transf = rot.  $= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \hat{U}$

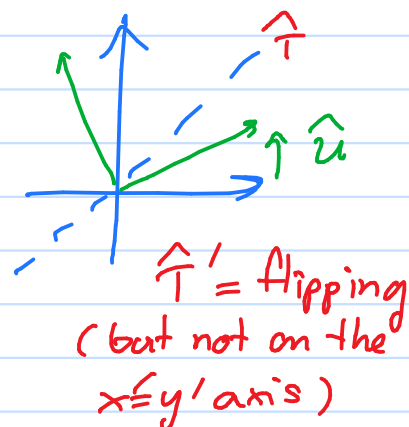
$$\Rightarrow \hat{U}^{-1} = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$\begin{aligned} \therefore \hat{I}' &= \hat{U} \hat{I} \hat{U}^{-1} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \\ &= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Exercise  $\hat{T} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  under rotation  $\hat{U} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

Find  $\hat{T}' = \hat{U}\hat{T}\hat{U}^{-1}$

Also find angle that  $\hat{T}' = \hat{T}$



\* Consider an orthonormal basis  $\{|e_i\rangle\}$

$(\langle e_i | e_j \rangle = \delta_{ij})$

$$\hat{T}|e_i\rangle = \sum_{j=1}^n T_{ji}|e_j\rangle \Rightarrow \langle e_k | \hat{T} | e_i \rangle = \sum_{j=1}^n T_{ji} \langle e_k | e_j \rangle$$

$k=j$   
 $\delta_{kj}$

$$= T_{ki}$$

$\therefore$   $T_{ki} = \langle e_k | \hat{T} | e_i \rangle$   $\Rightarrow$  matrix element of  $\hat{T}$

$T_{ij} = \langle e_i | \hat{T} | e_j \rangle$   $\Rightarrow$   $n \times n$  matrix  
 $\cap GL(n, \mathbb{R})$

Now let's get back to representation

"The numbers as a group" is a matrix group = general linear group

★ Homomorphism between a group and  $GL(n, \mathbb{R})$  is called "representation"

## Regular Representation

Let's consider a group  $G = \{g_1, g_2, \dots, g_n\}$

Now construct a vector space with basis labeled by

$$\{|g_1\rangle, |g_2\rangle, \dots, |g_n\rangle\} \Rightarrow \text{orthonormal } \langle g_i | g_j \rangle = \delta_{ij}$$

$\Rightarrow V$  has  $\dim = n = \text{order of group } G$

★ Consider a representation of  $G$ ,  $D: G \rightarrow GL(n, \mathbb{R})$

$$\Rightarrow D(g_i) = n \times n \text{ matrix} \quad \Rightarrow \quad D(g_i) D(g_j) = D(g_i g_j)$$

$\hookrightarrow$  operator on  $V$                       homomorphism.

Regular representation is defined by

$$D(g_i) |g_j\rangle = |g_i g_j\rangle$$

A basis vector that corresponds to  $g_i g_j$

or we can write  $D(g_i)$  in terms of matrix element.

$$[D(g_i)]_{kj} = \langle g_k | \underbrace{D(g_i)}_{\text{operator}} |g_j\rangle = \langle g_k | \underbrace{g_i g_j}_{\text{orthonormal}} \rangle = 0 \text{ or } 1$$
$$= \begin{cases} 1, & g_k = g_i g_j \\ 0, & \text{other cases} \end{cases}$$

Example  $\mathbb{Z}_3 = \{1, R_1, R_2\} = \{g_1, g_2, g_3\}$

|       | 1     | $R_1$ | $R_2$ |
|-------|-------|-------|-------|
| 1     | 1     | $R_1$ | $R_2$ |
| $R_1$ | $R_1$ | $R_2$ | 1     |
| $R_2$ | $R_2$ | 1     | $R_1$ |

$\Rightarrow$

|       | $g_1$ | $g_2$ | $g_3$ |
|-------|-------|-------|-------|
| $g_1$ | $g_1$ | $g_2$ | $g_3$ |
| $g_2$ | $g_2$ | $g_3$ | $g_1$ |
| $g_3$ | $g_3$ | $g_1$ | $g_2$ |

$$V = \{|g_1\rangle, |g_2\rangle, |g_3\rangle\}$$

Regular rep.  $D(g_1) = D(1) = ?$ ,  $D(g_2) = ?$ ,  $D(g_3) = ?$

Example  $\mathbb{Z}_3 = \{1, R_1, R_2\} = \{g_1, g_2, g_3\}$

|       | 1     | $R_1$ | $R_2$ |
|-------|-------|-------|-------|
| 1     | 1     | $R_1$ | $R_2$ |
| $R_1$ | $R_1$ | $R_2$ | 1     |
| $R_2$ | $R_2$ | 1     | $R_1$ |

$\Rightarrow$

|       | $g_1$ | $g_2$ | $g_3$ |
|-------|-------|-------|-------|
| $g_1$ | $g_1$ | $g_2$ | $g_3$ |
| $g_2$ | $g_2$ | $g_3$ | $g_1$ |
| $g_3$ | $g_3$ | $g_1$ | $g_2$ |

$$V = \{|g_1\rangle, |g_2\rangle, |g_3\rangle\}$$

Regular rep.  $D(g_1) = D(1) = ?$ ,  $D(g_2) = ?$ ,  $D(g_3) = ?$

$$\begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} \quad \begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} \quad \begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix}$$

$$[D(g_1)]_{ij} = \langle g_i | D(g_1) | g_j \rangle = \langle g_i | g_1 g_j \rangle \quad \text{reg. rep.}$$

$$[D(g_1)]_{11} = \langle g_1 | g_1 g_1 \rangle = \langle g_1 | g_1 \rangle = 1$$

$$[D(g_1)]_{12} = \langle g_1 | g_1 g_2 \rangle = \langle g_1 | g_2 \rangle = 0$$

$$[D(g_1)]_{13} = \langle g_1 | g_1 g_3 \rangle = \langle g_1 | g_3 \rangle = 0$$

$$[D(g_1)]_{21} = \langle g_2 | g_1 g_1 \rangle = \langle g_2 | g_1 \rangle = 0$$

$$[D(g_1)]_{22} = \langle g_2 | g_1 g_2 \rangle = \langle g_2 | g_2 \rangle = 1$$

$$[D(g_1)]_{23} = \langle g_2 | g_1 g_3 \rangle = \langle g_2 | g_3 \rangle = 0$$

$$[D(g_1)]_{13} = 0, [D(g_1)]_{23} = 0, [D(g_1)]_{33} = 1$$

$$D(g_1) = \begin{pmatrix} [D(g_1)]_{11} & [D(g_1)]_{12} & [D(g_1)]_{13} \\ [D(g_1)]_{21} & [D(g_1)]_{22} & [D(g_1)]_{23} \\ [D(g_1)]_{31} & [D(g_1)]_{32} & [D(g_1)]_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D(1) D(g) = D(1g) = D(g)$$

$1$  identity matrix

What about  $D(g_2)$

$$[D(g_2)]_{ij} = \langle g_i | D(g_2) | g_j \rangle = \langle g_i | g_2 g_j \rangle$$

|       | $g_1$                           | $g_2$                           | $g_3$                           |
|-------|---------------------------------|---------------------------------|---------------------------------|
| $g_1$ | $\langle g_1   g_2 g_1 \rangle$ | $\langle g_1   g_2 g_2 \rangle$ | $\langle g_1   g_2 g_3 \rangle$ |
| $g_2$ | $\langle g_2   g_2 g_1 \rangle$ | $\langle g_2   g_2 g_2 \rangle$ | $\langle g_2   g_2 g_3 \rangle$ |
| $g_3$ | $\langle g_3   g_2 g_1 \rangle$ | $\langle g_3   g_2 g_2 \rangle$ | $\langle g_3   g_2 g_3 \rangle$ |

$$[D(g_2)]_{11} = \langle g_1 | g_2 g_1 \rangle = \langle g_1 | g_2 \rangle = 0$$

$$[D(g_2)]_{12} = \langle g_1 | g_2 g_2 \rangle = \langle g_1 | g_3 \rangle = 0$$

$$[D(g_2)]_{13} = \langle g_1 | g_2 g_3 \rangle = \langle g_1 | g_1 \rangle = 1$$

$$[D(g_2)]_{21} = \langle g_2 | g_2 g_1 \rangle = \langle g_2 | g_2 \rangle = 1$$

$$[D(g_2)]_{22} = \langle g_2 | g_2 g_2 \rangle = \langle g_2 | g_3 \rangle = 0$$

$$[D(g_2)]_{23} = \langle g_2 | g_2 g_3 \rangle = \langle g_2 | g_1 \rangle = 0$$

$$[D(g_2)]_{31} = \langle g_3 | g_2 g_1 \rangle = \langle g_3 | g_2 \rangle = 0$$

$$[D(g_2)]_{32} = \langle g_3 | g_2 g_2 \rangle = \langle g_3 | g_3 \rangle = 1$$

$$[D(g_2)]_{33} = \langle g_3 | g_2 g_3 \rangle = \langle g_3 | g_1 \rangle = 0$$

$$\therefore D(g_2) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Exercise show that

$$D(g_3) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\therefore D(1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, D(R_1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, D(R_2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

We can also check homomorphism property,  $D(R_1)D(R_2) = D(1)$



## Homomorphism of Regular Representation

$$\Rightarrow [D(g_\alpha)D(g_\beta)]_{ij} = [D(g_\alpha g_\beta)]_{ij} \quad \checkmark$$

$$\begin{aligned} \text{RHS} &= \langle g_i | D(g_\alpha g_\beta) | g_j \rangle \\ &= \langle g_i | g_\alpha g_\beta g_j \rangle \quad \text{Regular rep.} \end{aligned}$$

$$\begin{aligned} \text{LHS} &= \langle g_i | D(g_\alpha) D(g_\beta) g_j \rangle \\ &= \langle g_i | D(g_\alpha) | g_\beta g_j \rangle \quad \text{Regular.} \\ &= \langle g_i | g_\alpha g_\beta g_j \rangle \quad \text{Regular} \end{aligned}$$

## Similarity transformation

Suppose that we have a representation  $D: G \rightarrow GL(n, \mathbb{R})$

$\Rightarrow D$  as a set of operators on  $V_n$

Under similarity transformation  $\Rightarrow \hat{U}$ , the matrices  $D$  change

$$D' = U^{-1} D U \Rightarrow D'(g_i) = U^{-1} D(g_i) U$$

Is  $D'$  also homomorphism, that is still a representation?

$$\text{Since } D(g_i) D(g_j) = D(g_i g_j)$$

$$U^{-1} D(g_i) D(g_j) U = U^{-1} D(g_i g_j) U$$

$$\underbrace{U^{-1} D(g_i) U}_{\text{sim. trans!}} \underbrace{U^{-1} D(g_j) U} = D'(g_i g_j)$$

$$D'(g_i) D'(g_j) = D'(g_i g_j) \Rightarrow \text{homomorphism}$$

$\therefore$  Similarity transformation relates between representations

## Reducible / Irreducible Representation

reducible  $\Rightarrow$  we can reduce representation down  
to the lower dimension

$$\mathbb{Z}_3 = \{1, R_1, R_2\} \quad D_1: \text{Regular rep} \Rightarrow 3 \times 3 \quad \text{reduce?}$$

$$D_1(1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D_1(R_1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad D_1(R_2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$D_2$ : a complex number  $\Rightarrow 1 \times 1$

$$D_2(1) = 1, \quad D_2(R_1) = e^{\frac{2\pi i}{3}}, \quad D_2(R_2) = e^{\frac{4\pi i}{3}}$$

$\Rightarrow$  We want to find a criteria to tell the difference  
between reducible/irreducible.

Recall that we have another way to construct a bigger  
representation.

$$D(1) = 1, \quad D(R_1) = e^{\frac{2\pi i}{3}}, \quad D(R_2) = e^{\frac{4\pi i}{3}}$$

Construct bigger matrices by stacking up

$$D'(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D'(R_1) = \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & e^{\frac{2\pi i}{3}} \end{pmatrix}, \quad D'(R_2) = \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & e^{\frac{4\pi i}{3}} \end{pmatrix}$$

$\Rightarrow D' = \text{representation}$

$\hookrightarrow$  stack up in diagonal  
direction.

$$D''(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D''(R_1) = \begin{pmatrix} e^{\frac{2\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix}, \quad D''(R_2) = \begin{pmatrix} e^{\frac{4\pi i}{3}} & 0 \\ 0 & 1 \end{pmatrix}$$

$D'' = \text{rep.}$

$\hookrightarrow$  trivial rep

In general, we can stack any representations together

$$\tilde{D}(1) = \begin{pmatrix} D_1(1) & 0 & 0 & \dots \\ 0 & \underline{D_2(1)} & 0 & \dots \\ 0 & 0 & D_3(1) & \\ \vdots & \vdots & & \ddots \end{pmatrix}$$

$D_1, D_2, \dots$   
could be  $n \times n$   
matrices!

$$\tilde{D}(R_1) = \begin{pmatrix} D_1(R_1) & 0 & 0 & \dots \\ 0 & \underline{D_2(R_1)} & 0 & \dots \\ 0 & 0 & D_3(R_1) & \\ \vdots & \vdots & & \ddots \end{pmatrix}$$

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & c & d \end{pmatrix} \begin{pmatrix} f & 0 & 0 \\ 0 & g & e \\ 0 & e & h \end{pmatrix}$$

$M_1 = \begin{pmatrix} a & f & 0 & 0 \\ 0 & & & \\ 0 & & M_1 & M_2 \\ 0 & & & \end{pmatrix}$

$M_2$

$$\tilde{D}(R_2) = \begin{pmatrix} D_1(R_2) & 0 & 0 & \dots \\ 0 & D_2(R_2) & 0 & \dots \\ 0 & 0 & D_3(R_2) & \\ \vdots & \vdots & & \ddots \end{pmatrix}$$

$\Rightarrow D_1, D_2, D_3, \dots$  could be  $n \times n$  matrices

$\Rightarrow$  "Block diagonal form"  $\Rightarrow$  reducible representation

$$\tilde{D}(g_i) = \begin{pmatrix} D_1(g_i)^{n_1 \times n_1} & 0^{n_1 \times n_2} & 0 \\ 0^{n_2 \times n_1} & D_2(g_i)^{n_2 \times n_2} & 0 \\ 0 & 0 & D_3(g_i)^{n_3 \times n_3} \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

\* Block diagonal form of rep  $\Rightarrow$  reducible!

? What about any rep that is not in this form?  
can we tell?

Ex Regular rep is not in block diagonal form

$$D_1(I) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D_1(R_1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad D_1(R_2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

We can use the fact that equivalent representations are related via similarity transformation

$\therefore$  If we can use similarity transformation relate

$D_1$  with a block diagonal form,  $D_1$  is reducible!

Example

$$D_1(I) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D_1(R_1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad D_1(R_2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

We can show that  $U = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$ ,  $\omega = e^{\frac{2\pi i}{3}}$

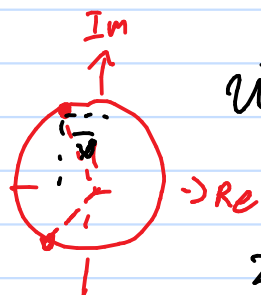
$U$  is the similarity transf that brings  $D_1 \Rightarrow$  block diagonal!

$$\Rightarrow U^{-1} D_1 U = \tilde{D} = \text{block diagonal!}$$

$$U^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$$

$$U U^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$$

$\omega = e^{\frac{2\pi i}{3}} = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = \frac{1}{2} + i \left(-\frac{\sqrt{3}}{2}\right) = \frac{1}{2} - i \frac{\sqrt{3}}{2}$



$$U^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}, \quad U U^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$$

$$\omega = e^{\frac{2\pi i}{3}} = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$\omega^2 = e^{\frac{4\pi i}{3}} = \cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) = -\frac{1}{2} - i \frac{\sqrt{3}}{2}$$

$$\omega^3 = \left(e^{\frac{2\pi i}{3}}\right)^3$$

$$= e^{2\pi i} = 1$$

$$1 + \omega + \omega^2 = 1 + \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2}\right) = 0$$

$$\omega^4 = \omega^3 \cdot \omega = \omega$$

$$U U^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 & 1+\omega+\omega^2 & 1+\omega^2+\omega \\ 1+\omega+\omega^2 & 1+\omega^3+\omega^3 & 1+\omega^2+\omega^2 \\ 1+\omega+\omega^2 & 1+\omega^2+\omega^2 & 1+\omega^3+\omega^3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$

$$U^{-1} D_1(I) U = \begin{pmatrix} U^{-1} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U \end{pmatrix} = U^{-1} U = I$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$U^{-1} D_1(R_1) U = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ \omega & \omega^2 & 1 \\ \omega^2 & \omega & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 & 1+\omega^2+\omega^0 & 1+\omega+\omega^2 \\ \omega+\omega^2+1 & \omega+\omega^3+\omega & \omega+\omega^3+\omega^2 \\ \omega^2+\omega+1 & \omega^2+\omega^3+\omega & \omega^2+\omega^3+\omega^2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}$$

$$U^{-1} D_1(R_2) U = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{pmatrix}$$

$$= \dots = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega^2 & 0 \\ 0 & 0 & \omega \end{pmatrix}$$

$$\therefore \tilde{D}(1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \tilde{D}(R_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \quad \tilde{D}(R_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega^2 & 0 \\ 0 & 0 & \omega \end{pmatrix}$$

⌋ block diagonal and the smaller reps are  $D_1$  &  $D_2$

$$D_1(1) = 1, \quad D_1(R_1) = \omega, \quad D_1(R_2) = \omega^2$$

$$\& \quad D_2(1) = 1, \quad D_2(R_1) = \omega^2, \quad D_2(R_2) = \omega$$


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## Schur's lemma

Let's go back to QM:  $\hat{H}$  Hamiltonian

Wave fn  $\hat{H}(\vec{x}, \vec{p})$  ( $\vec{x} \rightarrow \hat{x} = x, \vec{p} \rightarrow \hat{p} = -i\vec{\nabla}$ )

How can we realise symmetry of the system?

A group of symmetry of  $\hat{H}$  is a group  $G$  whose action leaves  $\hat{H}$  invariant.

$$g \in G, \quad g \cdot \vec{x} = \vec{x}' \quad \text{e.g.} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$$

$$\boxed{\hat{H}(g \cdot \vec{x}) = \hat{H}(\vec{x})}$$

Example  $\hat{H} = \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2} m \omega^2 x^2$  SHM

Example  $\hat{H} = \frac{-\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2} m \omega^2 x^2$  SHM in 1D

$\frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$

Suppose we consider 2 actions =  $\{1, F\}$

$1 \cdot x = x$  ,  $F \cdot x = -x$   $\mathbb{Z}_2$  group

$$\hat{H}(\vec{x}) = \frac{-\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2} m \omega^2 x^2$$

$$\hat{H}(1 \cdot \vec{x}) = \hat{H}(\vec{x}) , \quad \hat{H}(F \cdot \vec{x}) = \frac{-\hbar^2}{2m} \frac{\partial^2}{\partial (-x)^2} + \frac{1}{2} m \omega^2 (-x)^2 = \hat{H}(\vec{x})$$

$\therefore \mathbb{Z}_2$  is a group of symmetry of  $\hat{H}$  (SHM in 1D)

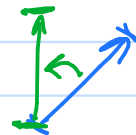
Example 2  $\hat{H} = \frac{-\hbar^2}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + \frac{1}{2} m \omega^2 (x^2 + y^2)$  (SHM in 2D)

A 2D rotation group  $\left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \right\} (= SO(2))$

$$\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix} , \quad \hat{R} \vec{r} = \vec{r}' = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\vec{p} = \begin{pmatrix} p_x \\ p_y \end{pmatrix} , \quad \vec{p}' = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} p_x' \\ p_y' \end{pmatrix}$$

We can check that a rotation preserve the length of all vectors, i.e.,  $\vec{r} \cdot \vec{r} = \vec{r}' \cdot \vec{r}'$



$$\vec{p} \cdot \vec{p} = \vec{p}' \cdot \vec{p}'$$

$$\therefore \hat{H} = \frac{-\hbar^2}{2m} \vec{p}^2 + \frac{1}{2} m \omega^2 \vec{r}^2$$

$$\therefore \hat{H} = \frac{\hbar^2}{2m} \vec{p}^2 + \frac{1}{2} m \omega^2 \vec{r}^2$$

$$\hat{H}(\hat{R}\vec{r}) = \frac{\hbar^2}{2m} \underbrace{\vec{p}'^2}_{\vec{p}^2} + \frac{1}{2} m \omega^2 \underbrace{\vec{r}'^2}_{\vec{r}^2} = \hat{H}(\vec{r})$$

$\therefore$  The  $SO(2)$  group is a group of symmetry of  $\hat{H}$   
(SHM in 2D)

$\Rightarrow$  The  $SO(3)$  group (rotation in 3D) is a group of symmetry of  $\hat{H}$  for  $V(\underbrace{|\vec{r}|^2})$  = central force problem

This is ok so far, but we normally use the wave  $\psi$  to describe the system. What is the group of symmetry of  $\hat{H}$  looks like in the wave  $\psi$  space?

★ We notice that the space of all wave  $\psi$  (normalisable)

= Vector space = Hilbert space

$\Rightarrow \psi(x)$  is a vector in the vector space

$$\Rightarrow \psi_1(x) + \psi_2(x), \quad a\psi_1(x), \quad a(\psi_1 + \psi_2) = a\psi_1 + a\psi_2$$

$$(a+b)\psi_1 = a\psi_1 + b\psi_1, \dots \quad \text{properties of a vector space}$$

$\Rightarrow$  We use a basis set in the form of Fourier transform

$$\underbrace{\psi(x)}_{\text{vector}} = \int \underbrace{\tilde{\psi}(k)}_{\text{component}} e^{ikx} dk \leftarrow \text{summation}$$

↖ basis

$$\Rightarrow \text{dot product} = \int \psi_1^*(x) \psi_2(x) dx$$



Since  $\psi(x)$  lives in a vector space  $= \mathcal{H}$

• A group of symmetry of  $\hat{H}$  should have a representation  $D(g) : \mathcal{H} \rightarrow \mathcal{H}$  (operator of  $\psi(x)$ )

In the spin  $\frac{1}{2}$  system,  $|\psi\rangle = \alpha|+\rangle + \beta|-\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$   
 $|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$   
 $= \alpha| \uparrow \rangle + \beta| \downarrow \rangle$   
 $\alpha, \beta \in \mathbb{C}$

$\therefore |\psi\rangle$  lives in  $\mathbb{C}^2$  vector space

We only need to find a representation of a  $g$  of symmetry of  $\hat{H}$  in  $GL(2, \mathbb{C})$   
 $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$   
 $a, b, c, d \in \mathbb{C}$

---

Example  $\mathbb{Z}_2$  is a group of symmetry of  $\hat{H}$  (SHM 1D)

$\Rightarrow$  We want to find representation of  $\mathbb{Z}_2$  for  $\psi(x)$

$\Rightarrow$  Homomorphism:  $\mathbb{Z} = \{1, F\}$

$$\hat{D}(1)\psi(x) = \psi(x), \quad \hat{D}(F)\psi(x) = -\psi(x)$$

$$\hat{D}(1) \cdot \hat{D}(F) = \hat{D}(1 \cdot F) = \hat{D}(F)$$

$$\hat{D}(F) \cdot \hat{D}(F) = \hat{D}(F \cdot F) = \hat{D}(1)$$

$$\begin{aligned} \hat{D}(1) \cdot \hat{D}(F) \psi(x) &= -\psi(x) = \hat{D}(F) \psi(x) \quad \checkmark \\ &\quad \begin{matrix} (+1) & (-1) \end{matrix} \end{aligned}$$

$$\hat{D}(F) \cdot \hat{D}(F) \psi(x) = +\psi(x) = \hat{D}(1) \psi(x) \quad \checkmark$$

Representation:  $\hat{D}(1) = +1, \hat{D}(F) = -1, GL(1, \mathbb{R}) = \mathbb{R}$

Since we know that under transformation  $g \in G$   
 $\psi(x) \rightarrow \hat{D}(g)\psi(x)$ . How does  $\hat{H}$  transform?

[This is how vectors transform]

$$\hat{H}' \rightarrow \hat{D}(g)^{-1} \hat{H} \hat{D}(g) \rightarrow \text{similarity transformation}$$

$\therefore$  If  $G$  is a group of symmetry of  $\hat{H}$ ,

$$\hat{D}(g)^{-1} \hat{H} \hat{D}(g) = \hat{H}$$

$$\hookrightarrow \hat{H}(g \cdot \vec{x}) = \hat{H}(\vec{x})$$

$$\hat{D}(g) \hat{D}(g)^{-1} \hat{H} \hat{D}(g) = \hat{D}(g) \hat{H}$$

$\Rightarrow$  consequences..

$$\hat{H} \hat{D}(g) = \hat{D}(g) \hat{H} \quad *$$

$$\Rightarrow [\hat{H}, \hat{D}(g)] = 0 \quad *$$

$$\leadsto \hat{L}_z, \hat{L}^2$$

$$|n, l, m\rangle$$

$\uparrow$  shared eigen states

QM is linear algebra on drugs

## Schur's lemma

$$GL(n, \mathbb{R}) \rightarrow n \times n$$

Theorem 1 If  $D_1(g)$  &  $D_2(g)$  are inequivalent irreducible representations &  $D_1(g)A = AD_2(g)$ ,  $\forall g \in G$ , then

$$A = 0_{n \times n}$$

$\hookrightarrow n \times n$  matrix

Notice  $A^{-1} D_1(g) A = D_2(g) \rightarrow$  similarity trans.f.

$\Rightarrow$  contradict  $\therefore A = 0$

$\Rightarrow$  equivalent

Theorem 2 If  $D_1(g)A = A D_2(g)$ , where  $D_1$  &  $D_2$  are equivalent irreducible representation of  $G$ , then

$$A \propto \mathbb{1}_{n \times n} \quad (A = \lambda \mathbb{1})$$

Notice  $A^{-1} D_1(g) A = D_2(g)$  sim transf.

Since  $D_1$  &  $D_2$  are equivalent  $\Rightarrow \underline{D_1(g) = D_2(g)}$

$\therefore A \propto \mathbb{1}$  is consistent

---

### Consequences of Schur's lemma

1). All irreducible representations of an abelian group are  $1 \times 1$  matrices (a number) (1D)

$g_1 g_2 = g_2 g_1, \quad g_1, g_2 \in G \Rightarrow \text{Abelian}$

Suppose that  $D(g)$  is irrep (irreducible representation)

\*  $\rightarrow \underbrace{D(g_1)}_A \underbrace{D(g_2)}_A = \underbrace{D(g_2)}_A \underbrace{D(g_1)}_A \Rightarrow D(g_2) = \lambda(g_2) \mathbb{1}$   
true for any  $g_2$

$\therefore D(g) = \lambda(g) \mathbb{1} \Rightarrow \begin{pmatrix} \lambda & & 0 \\ & \lambda & \\ 0 & & \lambda \end{pmatrix}$

$\Rightarrow D(g)$  is  $1 \times 1$  matrix

Example  $\mathbb{Z}_2$  irrep.  $= \{1, -1\}$  <sup>1Dim</sup> Abelian group

$\mathbb{Z}_3$  irrep  $= \{1, e^{2\pi i/3}, e^{4\pi i/3}\}$  <sup>1Dim</sup>

Consequence 2 Consider the symmetry group  $G$  of  $\hat{H}$

Suppose that  $D(g)$  is irrep of  $G$

Since  $G$  is sym. group of  $\hat{H} \Rightarrow [\hat{H}, D(g)] = 0$

$$\Rightarrow \underbrace{\hat{H}}_A D(g) = D(g) \underbrace{\hat{H}}_A \Rightarrow \hat{H} = \lambda \mathbb{1}$$

Schur's lemma

This looks familiar!

Consider  $\{|\psi\rangle\} = W \subset V$  (~~Ex~~ 3D  $\rightarrow$  1D)

&  $D(g)|\psi\rangle \in W$

$W$  is invariant subspace  
under  $D(g)$

if we consider  $W = 1D$   
 $D(g)|\psi\rangle = d(g)|\psi\rangle$   
eigenvector/value

★ If  $D(g)$  is irrep,  $\{|\psi\rangle\}$  is invariant subspace of  $\hat{H}$

$$\hat{H} D(g)|\psi\rangle = D(g) \hat{H} |\psi\rangle$$

$$\hat{H} (d(g)|\psi\rangle) = D(g) \lambda \mathbb{1} |\psi\rangle$$

Schur's lemma

$$\boxed{\hat{H} |\psi\rangle = \lambda |\psi\rangle}$$

$d(g)|\psi\rangle$

$\begin{pmatrix} & \\ & \\ & \end{pmatrix}_{3 \times 3}$

$\lambda = 0, 1, 1$

$|\psi_1\rangle, a|\psi_2\rangle + b|\psi_3\rangle$   
subspace

Another way to phrase this  $\Rightarrow$

$[\hat{A}, \hat{B}] = 0 \Rightarrow \hat{A} \& \hat{B}$  share eigenvectors!

Theorem Every representation of a finite group  $G$  is equivalent to a unitary representation

$$\hat{U}^\dagger \hat{U} = \hat{1}, \quad \hat{U}^\dagger = \hat{U}^{-1}$$

Unitary operator: an operator that preserves the length of vectors

$$|\psi\rangle \rightarrow |\psi'\rangle = \hat{U} |\psi\rangle$$

$$\langle \psi' | \psi' \rangle = \langle \psi | \psi \rangle \quad \nearrow \quad \hat{U}^\dagger = \hat{U}^{-1}$$

$$\langle \psi | \hat{U}^\dagger \hat{U} | \psi \rangle = \langle \psi | \psi \rangle \Rightarrow \hat{U}^\dagger \hat{U} = \hat{1}$$

Proof Consider  $D(g)$  is rep of  $G$  (finite group)

$$\text{Consider } T = D^\dagger(g_1) D(g_1) + D^\dagger(g_2) D(g_2) + \dots$$

$$n \times n \quad T = \sum_{g \in G} D^\dagger(g) D(g)$$

$$\underbrace{D^\dagger(h) T D(h)}_{\text{//}} = D^\dagger(h) \left( \sum_{g \in G} D^\dagger(g) D(g) \right) D(h)$$

$$= \sum_{g \in G} D^\dagger(h) D^\dagger(g) D(g) D(h)$$

$$= \sum_{g \in G} \underbrace{(D(g) D(h))^\dagger}_{D(g^\dagger h)} \underbrace{(D(g) D(h))}_{D(g h)} = \sum_{g \in G} D(g^\dagger h)^\dagger D(g h)$$

$$= \sum_{g \in G} D(g^\dagger h)^\dagger D(g h)$$

$\Downarrow$   
 $g^\dagger \in G$

$$g \rightarrow g^\dagger = \underline{g h}$$

$$= T$$

$$\therefore D^\dagger(h) T D(h) = T$$

$$\text{Now, let's define } T \equiv S^2 \quad (\hat{T} = \hat{S} \cdot \hat{S})$$

$$D^\dagger(h) S S D(h) = S S \quad \downarrow \times D^{-1}(h)$$

$$D^\dagger(h) S S = S S D^{-1}(h) \quad \downarrow \times S^{-1}$$

$$S^{-1} \times \left( D^\dagger(h) S = S S D^{-1}(h) S^{-1} \right)$$

$$\underline{S}^{-1} D^\dagger(h) \underline{S} = \underline{S} D^{-1}(h) \underline{S}^{-1}$$

$$\text{Since } T^\dagger = \left( \sum_{g \in G} D^\dagger(g) D(g) \right)^\dagger = \sum_{g \in G} D^\dagger(g) (D^\dagger(g))^\dagger = T \quad \text{Hermitian}$$

$$T^\dagger = (S S)^\dagger = S^\dagger S^\dagger = S S \Rightarrow S^\dagger = S$$

$$\left( \underline{S}^\dagger D(\underline{S}^{-1})^\dagger \right)^\dagger = \left( \underline{S} D(h) \underline{S}^{-1} \right)^{-1}$$

$$\underbrace{(S D(h) S^{-1})^\dagger}_{D'(h)} = \underbrace{(S D(h) S^{-1})^{-1}}_{D'(h)}$$

$\therefore D'$  is unitary representation!

$$D' = S D S^{-1}, \quad D' \& D \text{ are equivalent.}$$

$$\text{Ex } \mathbb{Z}_3 = \{1, e^{\frac{2\pi i}{3}}, e^{\frac{4\pi i}{3}}\} \text{ unitary.}$$

$$\left( e^{\frac{2\pi i}{3}} \right)^{-1} = \left( e^{\frac{2\pi i}{3}} \right)^*$$

# Orthogonality & Consequences of Schur's lemma

We can use Schur's lemma in order to tell properties of representations of group

Consider an operator

$$\hat{A} = \sum_{g \in G} D^{(\alpha)}(g) X D^{(\beta)}(g^{-1})$$

$n \times m$

sum over all elements in  $G$

$n \times n$

$n \times m$

$m \times m$

$\alpha, \beta = 1, \dots, ?$

$\alpha^{\text{th}}$  irreducible representation

$\beta^{\text{th}}$  irreducible rep.

if  $\alpha = \beta$  then equivalent

$$D^{(\alpha)}: V_n \rightarrow V_n$$

$\hookrightarrow n \times n$  matrix

$$D^{(\beta)}: V_m \rightarrow V_m$$

$\hookrightarrow m \times m$  matrix

$$\hat{A} = D^{(\alpha)}(g_1) X D^{(\beta)}(g_1^{-1}) + D^{(\alpha)}(g_2) X D^{(\beta)}(g_2^{-1}) + \dots$$

Consider  $h \in G$

$$\begin{aligned} D^{(\alpha)}(h) \hat{A} &= D^{(\alpha)}(h) \sum_{g \in G} D^{(\alpha)}(g) X D^{(\beta)}(g^{-1}) \\ &= \sum_{g \in G} \overbrace{D^{(\alpha)}(h) D^{(\alpha)}(g)}^{D^{(\alpha)}(h \cdot g)} X \overbrace{D^{(\beta)}(g^{-1}) D^{(\beta)}(h^{-1})}^{D^{(\beta)}(h^{-1} g^{-1})} D^{(\beta)}(h) \\ &= \sum_{g \in G} D^{(\alpha)}(h \cdot g) X D^{(\beta)}(g^{-1} h^{-1}) D^{(\beta)}(h) \end{aligned}$$

is equivalent to sum over  $g'$  in  $G$

$$\begin{aligned} &= \sum_{g' \in G} \underbrace{D^{(\alpha)}(g') X D^{(\beta)}(g'^{-1})}_{\hat{A}} D^{(\beta)}(h) = \hat{A} D^{(\beta)}(h) \end{aligned}$$

$$\therefore D^{(\alpha)}(h) \hat{A} = \hat{A} D^{(\beta)}(h)$$

if  $\alpha \neq \beta \Rightarrow D^{(\alpha)} \& D^{(\beta)}$  are inequivalent  $\Rightarrow \hat{A} = 0$

if  $\alpha = \beta \Rightarrow D^{(\alpha)} \& D^{(\beta)}$  are equivalent  $\Rightarrow \hat{A} = \lambda \mathbb{1}_{n \times n}$   
(m x m)

$$\therefore \hat{A} = \lambda(x) \cdot \mathbb{1} \cdot \delta_{\alpha\beta}$$

(X is an arbitrary matrix in  $\hat{A} = \sum_{g \in G} D^{(\alpha)}(g) X D^{(\beta)}(g^{-1})$ )

$$\sum_{g \in G} D^{(\alpha)}(g) X D^{(\beta)}(g^{-1}) = \lambda(x) \mathbb{1} \delta_{\alpha\beta}$$

Let's specify the X,  $X = \begin{pmatrix} \ddots & & \\ & 1 & \\ & & \ddots \end{pmatrix}_{n \times m}$  the rest is zero

$$X = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots \\ 0 & \ddots & & & \\ \vdots & & 1 & & \\ \vdots & & & \ddots & \end{pmatrix}$$

such that  $X_{lp} = 1$

and the rest of  $X_{ij} = 0$

$l \& p$  position

$$\left( \sum_{g \in G} D^{(\alpha)}(g) X D^{(\beta)}(g^{-1}) \right)_{ij} = \lambda(x) \delta_{\alpha\beta} \delta_{ij}$$

$$\sum_{g \in G} \left( \sum_r \sum_s [D^{(\alpha)}(g)]_{ir} [X]_{rs} [D^{(\beta)}(g^{-1})]_{sj} \right) = \lambda(x) \delta_{\alpha\beta} \delta_{ij}$$

only  $X_{lp} = 1$  & the rest is zero!

$$\sum_{g \in G} D^{(\alpha)}(g)_{il} \cdot X_{lp} \cdot D^{(\beta)}(g^{-1})_{pj} = \lambda_{lp} \delta_{\alpha\beta} \delta_{ij}$$

$$\sum_{g \in G} D^{(\alpha)}(g)_{il} D^{(\beta)}(g^{-1})_{pj} = \lambda_{lp} \delta_{\alpha\beta} \delta_{ij}$$



$$\sum_{g \in G} D^{(\alpha)}(g)_{il} D^{(\beta)}(g^{-1})_{pj} = \lambda_{lp} \delta_{\alpha\beta} \delta_{ij}$$

Now we can find  $\lambda_{lp}$

Let's set  $i=j$  & sum over  $i$

$$n = 1 + 1 + 1 + \dots \\ \delta_{11} + \delta_{22} + \dots$$

$$\sum_{g \in G} \sum_i D^{(\alpha)}(g)_{il} D^{(\beta)}(g^{-1})_{pi} = \lambda_{lp} \delta_{\alpha\beta} \sum_i \delta_{ii}$$

dim of  $D^{(\alpha)}$  or  $D^{(\beta)}$

$$\sum_{g \in G} \sum_i D^{(\beta)}(g^{-1})_{pi} D^{(\alpha)}(g)_{il} = \lambda_{lp} \delta_{\alpha\beta} \cdot n$$

Now let's set  $\alpha = \beta$

same irrep!

$$\sum_{g \in G} \sum_i D^{(\alpha)}(g^{-1})_{pi} D^{(\alpha)}(g)_{il} = \lambda_{lp} \cdot n$$

$$\sum_{g \in G} \left[ D^{(\alpha)}(g^{-1}g) \right]_{pl} = \lambda_{lp} \cdot n$$

homomorphism

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & & \\ 0 & & \ddots & \\ 0 & & & \ddots \end{pmatrix}$$

$$\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & \ddots \end{pmatrix}$$

$$D^{(\alpha)}(\mathbb{1}) = \mathbb{1}$$

$$D(g)D(\mathbb{1}) = D(g \cdot \mathbb{1}) \\ D(g)D(\mathbb{1}) = D(g) \\ \hookrightarrow \mathbb{1}_{n \times n}$$

$$\sum_{g \in G} \delta_{pl} = \lambda_{lp} \cdot n$$

$|G| = \text{order}$

$= \# \text{ of elements in } G$

$$\delta_{pl} |G| = \lambda_{lp} \cdot n$$

$$\therefore \boxed{\lambda_{lp} = \frac{|G|}{n} \delta_{pl}}$$

$$\therefore \sum_{g \in G} D^{(\alpha)}(g)_{il} D^{(\beta)}(g^{-1})_{pj} = \lambda_{lp} \delta_{\alpha\beta} \delta_{ij} = \frac{|G|}{n} \delta_{pl} \delta_{\alpha\beta} \delta_{ij}$$

$$\sum_{g \in G} D^{(\alpha)}(g)_{il} D^{(\beta)}(g^{-1})_{pj} = \frac{|G|}{n} \delta_{pl} \delta_{\alpha\beta} \delta_{ij}$$

Since we know that for a finite group,  $D \rightarrow$  unitary

$$\Rightarrow D^{(\beta)}(g^{-1})_{pj} = [D^{(\beta)}(g)]^{-1} = [D^{(\beta)}(g)]^+ \rightsquigarrow \text{Transpose and conjugate}$$

$$\underbrace{D(g)}_{[D(g)]^{-1}} \underbrace{D(g^{-1})}_{[D(g)]^{-1}} = D(gg^{-1}) = D(\mathbb{1}) = \mathbb{1}$$

$p \leftrightarrow j$   $\checkmark$   $\star$

$$\sum_{g \in G} D^{(\alpha)}(g)_{il} D^{(\beta)*}(g)_{jp} = \frac{|G|}{n} \delta_{pl} \delta_{\alpha\beta} \delta_{ij}$$

Character (of representation)

$\rightarrow$  orthogonality.

$\hookrightarrow$  character

$$\chi^{(\alpha)}(g) \equiv \text{Tr}(D^{(\alpha)}(g)) = \sum_i D^{(\alpha)}(g)_{ii}$$

\* The character is invariant under similarity transf.

$$\chi(g) = \text{Tr}(D(g)) \quad \left. \begin{array}{l} \\ \end{array} \right\} D'(g) = U^{-1} D(g) U$$

$$\begin{aligned} \chi'(g) &= \text{Tr}(D'(g)) = \text{Tr}(U^{-1} D(g) U) \\ &= \text{Tr}(U U^{-1} D(g)) \\ &= \text{Tr}(D(g)) = \chi(g) \end{aligned}$$

$\therefore$  Character is the same for equivalent reps.

$$\star \chi(\mathbb{1}) = \text{Tr}(D(\mathbb{1})) = \text{Tr}(\mathbb{1}) = n$$

Let's go back to

$$\sum_{g \in G} D^{(\alpha)}(g)_{il} D^{(\beta)}(g^{-1})_{pj} = \frac{|G|}{n} \delta_{pl} \delta_{\alpha\beta} \delta_{ij}$$

Let's set  $i=l$  & sum over i and set  $p=j$   
& sum over j

$$\sum_{g \in G} \left( \sum_i D^{(\alpha)}(g)_{ii} \right) \left( \sum_j D^{(\beta)}(g^{-1})_{jj} \right) = \frac{|G|}{n} \delta_{\alpha\beta} \sum_i \sum_j \delta_{ji} \delta_{ij}$$

$$\begin{aligned} \sum_{g \in G} \chi^{(\alpha)}(g) \chi^{(\beta)}(g^{-1}) &= \frac{|G|}{n} \delta_{\alpha\beta} \sum_i \sum_j \delta_{ij} \delta_{ji} \\ &= \frac{|G|}{n} \delta_{\alpha\beta} \left[ \sum_i \delta_{ii} \right] \rightarrow n \end{aligned}$$

$j=i$

$$\sum_{g \in G} \chi^{(\alpha)}(g) \chi^{(\beta)}(g^{-1}) = |G| \delta_{\alpha\beta}$$

or we can use the unitary version

$$\sum_{g \in G} \chi^{(\alpha)}(g) \chi^{(\beta)*}(g) = |G| \delta_{\alpha\beta}$$

Vector space  
with dim = |G|

$$|\chi^{(\alpha)}\rangle = (\chi^{(\alpha)}(g_1), \chi^{(\alpha)}(g_2), \dots, \chi^{(\alpha)}(g_{|G|}))$$

$$|\chi^{(\beta)}\rangle = (\chi^{(\beta)}(g_1), \chi^{(\beta)}(g_2), \dots, \chi^{(\beta)}(g_{|G|}))$$

$$\langle \chi^{(\beta)} | \chi^{(\alpha)} \rangle = |G| \delta_{\alpha\beta}$$

$|\chi^{(\alpha)}\rangle$  are a set  
of orthogonal  
vectors

$$\sum_{g \in G} \chi^{(\alpha)}(g) \chi^{(\beta)*}(g) = |G| \delta_{\alpha\beta}$$

Vector space  
with  $\dim = |G|$

$$|\chi^{(\alpha)}\rangle = (\chi^{(\alpha)}(g_1), \chi^{(\alpha)}(g_2), \dots, \chi^{(\alpha)}(g_{|G|}))$$

$$|\chi^{(\beta)}\rangle = (\chi^{(\beta)}(g_1), \chi^{(\beta)}(g_2), \dots, \chi^{(\beta)}(g_{|G|}))$$

$$\langle \chi^{(\beta)} | \chi^{(\alpha)} \rangle = |G| \delta_{\alpha\beta}$$

$|\chi^{(\alpha)}\rangle$  are a set  
of orthogonal  
vectors

Since the maximum number of orthogonal vectors  
is the dimension of  $V$  ( $|G|$ )

$\therefore$  the number of irreducible representations  $\leq |G|$

inequivalent

Since some of them could be equivalent (via  
similarity transformation)  $\Rightarrow$  # of inequivalent irreps  
 $\leq |G|$

This bound is much stronger.

Consider a conjugacy class  
 $= \{h \mid h = g^{-1}h^*g, g \in G\}$

if  $h$  &  $h^*$  belong to  
the same class

$$\chi^{(\alpha)}(h) = \sum_i D^{(\alpha)}(h)_{ii}$$

then there is a  $g$  such  
that  $gh = h^*g$

This bound is much stronger.

Consider a conjugacy class  $\left\{ h \mid h = g^{-1} h^* g, g \in G \right\}$  if  $h$  &  $h^*$  belong to the same class

then there is a  $g$  such that  $gh = h^*g$

*Def of character*

$$\chi^{(\alpha)}(h) = \sum_i D^{(\alpha)}(h)_{ii}$$

$$= \sum_i D^{(\alpha)}(g^{-1} h^* g)_{ii} \quad \text{homomorphism}$$

$$= \sum_i \left( D^{(\alpha)}(g^{-1}) D^{(\alpha)}(h^*) D^{(\alpha)}(g) \right)_{ii} \quad \text{Trace } \text{Tr}(ABC) = \text{Tr}(CAB)$$

$$= \sum_i \left( D^{(\alpha)}(g) D^{(\alpha)}(g^{-1}) D^{(\alpha)}(h^*) \right)_{ii}$$

$D^{(\alpha)}(gg^{-1}) \Rightarrow 1$

$$= \sum_i D^{(\alpha)}(h^*)_{ii} = \chi^{(\alpha)}(h^*)$$

★ Characters of 2 elements in the same conjugacy class are equal.

$$\therefore \sum_{g \in G} \chi^{(\alpha)}(g) \chi^{(\beta)}(g)^* = |G| \delta_{\alpha\beta}$$

$|G|$   $\rightarrow$  # of classes

$$\sum_r C_r \chi_r^{(\alpha)}(g_r) \chi_r^{(\beta)}(g_r)^* = |G| \delta_{\alpha\beta}$$

$C_r$   $\rightarrow$  how many elements in class  $r^{th}$

$r$  = index for class  $r$

$\alpha = \beta$  case

$$\chi^{(\alpha)}(g_1) \chi^{(\alpha)}(g_1)^* + \dots + \chi^{(\alpha)}(g_p) \chi^{(\alpha)}(g_p)^* + \dots$$

Class  $r$  # of elements =  $C_r \Rightarrow C_r \chi(g_r) \chi^{(\alpha)}(g_r)$

$$\sum_r^c C_r \chi_r^{(\alpha)}(g_r) \chi_r^{(\beta)}(g_r)^* = |G| \delta_{\alpha\beta}$$

$\xrightarrow{\text{\# of classes}}$   
 $\xrightarrow{\text{how many elements in class } r^{\text{th}}}$

$$\therefore |\chi^{(\alpha)}\rangle = (\chi_1^{(\alpha)}(g_1), \chi_2^{(\alpha)}(g_2), \dots, \chi_c^{(\alpha)}(g_c))$$

We can look at this as an inner product (dot)

$$\langle \chi^{(\alpha)} | \chi^{(\beta)} \rangle = |G| \delta_{\alpha\beta}$$

$\therefore$  The number of irreducible representations must be less than the number of conjugacy classes.

or equal to

★ We can also check which representation is irreducible by calculating the character.

Consider a reducible representation  $D(g)$

$$D(g) \xrightarrow{\text{sim trans}^t} \begin{pmatrix} D^{(1)}(g) & & 0 \\ & D^{(2)}(g) & \\ 0 & & D^{(n)}(g) \end{pmatrix} \rightarrow \text{irrep.}$$

$$\therefore \chi(g) = \text{Tr}(D(g)) = \text{Tr}(D^{(1)}(g)) + \text{Tr}(D^{(2)}(g)) + \dots$$

$$= \chi^{(1)}(g) + \chi^{(2)}(g) + \dots$$

$$= \sum_m C_m \chi^{(m)}(g)$$

$\hookrightarrow$  # of times that  $D^{(m)}$  appears

$$\therefore \chi(g) = \sum_m C_m \chi^{(m)}(g)$$

Let's consider

$$\begin{aligned} \sum_{g \in G} \chi(g) \chi(g)^* &= \sum_{g \in G} \left( \sum_m C_m \chi^{(m)}(g) \right) \left( \sum_{m'} C_{m'} \chi^{(m')}(g) \right)^* \\ &= \sum_{m, m'} C_m C_{m'} \left[ \sum_{g \in G} \chi^{(m)}(g) \chi^{(m')}(g)^* \right] \rightarrow |G| \delta_{mm'} \end{aligned}$$

$$= \sum_{m, m'} C_m C_{m'} |G| \delta_{mm'} = \sum_m C_m^2 |G| \geq |G|$$

$C_m \in \mathbb{N}$

$$\therefore \sum_{g \in G} |\chi(g)|^2 = \sum_m C_m^2 |G| \geq |G|$$

Reducible:  $\sum_{g \in G} |\chi(g)|^2 > |G| \quad (m > 1)$

Irreducible:  $\sum_{g \in G} |\chi(g)|^2 = |G| \quad (m=1, C_m=1)$

Example

$$\mathbb{Z}_3 = \{1, R_1, R_2\} \quad D_1: \text{Regular rep} \Rightarrow 3 \times 3$$

$$D_1(1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D_1(R_1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad D_1(R_2) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Let's calculate the characters

$$\chi(1) = \text{Tr}(D_1(1)) = 3, \quad \chi(R_1) = \text{Tr}(D_1(R_1)) = 0$$

$$\chi(R_2) = \text{Tr}(D_1(R_2)) = 0$$

Let's calculate the characters

$$\chi(1) = \text{Tr}(D(1)) = 3, \quad \chi(R_1) = \text{Tr}(D(R_1)) = 0$$

$$\chi(R_2) = \text{Tr}(D(R_2)) = 0$$

$$\begin{aligned} \sum_{g \in G} |\chi(g)|^2 &= \chi(1)^2 + \chi(R_1)^2 + \chi(R_2)^2 \\ &= 9 \end{aligned}$$

but the order of  $\mathbb{Z}_3 = 3$

$\therefore$  The regular rep has  $\sum_{g \in G} |\chi(g)|^2 = 9 > |G| = 3$   
 $\hookrightarrow$  reducible

How about irreducible representation of  $\mathbb{Z}_3$ ?

$$\text{Trivial: } D^0(1) = 1, \quad D^0(R_1) = 1, \quad D^0(R_2) = 1$$

$$D^1: \quad D^1(1) = 1, \quad D^1(R_1) = e^{2\pi i/3}, \quad D^1(R_2) = e^{4\pi i/3}$$

$$\chi^0(1) = 1, \quad \chi^0(R_1) = 1, \quad \chi^0(R_2) = 1$$

$$\begin{aligned} \therefore \sum_{g \in G} |\chi(g)|^2 &= \chi^0(1)^2 + \chi^0(R_1)^2 + \chi^0(R_2)^2 \\ &= 1 + 1 + 1 = 3 = \text{order of } \mathbb{Z}_3 \end{aligned}$$

$\therefore$  Trivial = irrep.

$$\chi^1(1) = 1, \quad \chi^1(R_1) = e^{2\pi i/3}, \quad \chi^1(R_2) = e^{4\pi i/3}$$

$$\begin{aligned} \therefore \sum_{g \in G} |\chi(g)|^2 &= |\chi^1(1)|^2 + |\chi^1(R_1)|^2 + |\chi^1(R_2)|^2 \\ &= 1 + e^{2\pi i/3} \cdot e^{-2\pi i/3} + e^{4\pi i/3} \cdot e^{-4\pi i/3} \end{aligned}$$

$\therefore D^1 = \text{irrep.}$

$$= 1 + 1 + 1 = 3 = \text{order of } \mathbb{Z}_3$$



How many classes in  $\mathbb{Z}_3$ ?

$$\{1\} \quad \text{since} \quad g1g^{-1} = 1$$

$$\{R_1\} \quad \text{since} \quad R_2 R_1 R_2^{-1} = R_2 R_1 R_1 = R_2 R_2 = R_1$$

$$R_1 R_1 R_1^{-1} = R_1$$

$$\{R_2\} \quad \text{since} \quad R_1 R_2 R_1^{-1} = R_1 R_2 R_2 = R_1 R_1 = R_2$$

$$R_2 R_2 R_2^{-1} = R_2$$

$\therefore$  # of classes = 3

$\therefore$  # of irreps of  $\mathbb{Z}_3 \leq 3$

consistent with our findings!

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Midterm \* Exam coming up!

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## Lie group & Lie algebra

Consider the rotation group of 2D plane.

$$\Rightarrow \text{continuous} = \{ e^{i\theta} \mid \theta \in [0, 2\pi) \}$$

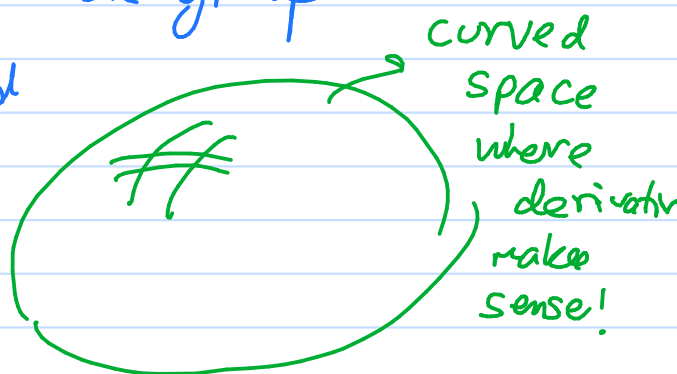
$$\text{or} = \left\{ \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \mid \theta \in [0, 2\pi) \right\}$$

The group elements depend on the continuous parameters. The order of group is infinite  
"infinite group"

The group elements depend on the continuous parameters. The order of group is infinite  
"infinite group"

In our example, we can do calculus on the set of parameters.  
↳ smooth or differentiable

⇒ The space of parameters of Lie group is a differentiable manifold



Differentiable (can perform derivative)

$$g(\theta) = e^{i\theta}$$

$$\frac{\partial g(\theta)}{\partial \theta} = ie^{i\theta} \quad \text{define over } \theta \in [0, 2\pi)$$

Suppose the group  $G$  with elements parametrized by parameters  $\vec{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_n)$

$$g(\vec{\alpha}) = g(\alpha_1, \alpha_2, \dots, \alpha_n) \in G$$

\* We assume that the way we parametrize is that

$$g(\vec{\alpha}=0) = g(0, 0, \dots, 0) = \mathbb{1}$$

We consider a representation  $D(g(\vec{\alpha})) \equiv D(\vec{\alpha})$

$$\Rightarrow D(\vec{\alpha}=0) = D(\mathbb{1}) = \hat{\mathbb{1}} \quad (*)$$

↑ identity matrix

Ex Rotation group in 2D

$$D(\theta) = e^{i\theta} \rightarrow D(\theta=0) = e^{(0)} = 1 \quad \checkmark$$

$$D'(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \rightarrow D'(\theta=0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \checkmark$$

Let's do the calculus on  $\vec{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_n)$

Consider a neighborhood of the identity element

$g(0) \rightarrow g(0 + d\vec{\alpha})$  must be close to  $\mathbb{1}$

how close? how to measure?  
representation  
numbers  
Taylor's expansion  
it doesn't make sense to measure the distance between group elements.

$$D(0) = \mathbb{1} \rightarrow D(0 + d\vec{\alpha}) = \mathbb{1} + \sum_a i d\alpha_a \left( -i \frac{\partial D(\vec{\alpha})}{\partial \alpha_a} \right) \Big|_{\vec{\alpha}=0} + \dots$$

$$f(dx) = f(0) + dx \cdot \left( \frac{df}{dx} \Big|_{x=0} \right) + \frac{1}{2!} (dx)^2 \left( \frac{d^2 f}{dx^2} \Big|_{x=0} \right) + \dots$$

$$f(d\vec{r}) = f(0) + dx \cdot \frac{\partial f}{\partial x} \Big|_{\vec{r}=0} + dy \cdot \frac{\partial f}{\partial y} \Big|_{\vec{r}=0} + \dots$$

$$D(d\vec{\alpha}) = \mathbb{1} + \sum_a i d\alpha_a \left( -i \frac{\partial D(\vec{\alpha})}{\partial \alpha_a} \Big|_{\vec{\alpha}=0} \right)$$

parameters  
generators  $\hat{X}_a$   
Matrix (operator) independent from  $\vec{\alpha}$

$$D(d\vec{\alpha}) = \hat{1} + \sum_a i d\alpha_a \left( \underbrace{-i \frac{\partial D(\vec{\alpha})}{\partial \alpha_a} \Big|_{\vec{\alpha}=0}}_{\text{Matrix (operator) independent from } \vec{\alpha}} \right)$$

parameters

generators  $\hat{X}_a$

Matrix (operator)  
independent from  $\vec{\alpha}$

$$D(d\vec{\alpha}) = \hat{1} + \sum_a i d\alpha_a \hat{X}_a$$

$$\hat{X}_a \equiv -i \frac{\partial D(\vec{\alpha})}{\partial \alpha_a} \Big|_{\vec{\alpha}=0}$$

generators

# of generators = # parameters

EX Rotation on 2D

$$D(\theta) = e^{i\theta}$$

generator:  $\hat{X} = -i \frac{\partial D(\theta)}{\partial \theta} \Big|_{\theta=0} = -i \cdot i = 1$

Taylor expansion  $D(d\theta) = e^{id\theta} \cong 1 + id\theta \text{ (1 generator)}$

$$D'(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

generator:  $\hat{X} = -i \frac{\partial D'(\theta)}{\partial \theta} \Big|_{\theta=0} = -i \begin{pmatrix} \frac{\partial \cos\theta}{\partial \theta} & \frac{\partial \sin\theta}{\partial \theta} \\ \frac{\partial (-\sin\theta)}{\partial \theta} & \frac{\partial \cos\theta}{\partial \theta} \end{pmatrix} \Big|_{\theta=0}$

$$= -i \begin{pmatrix} -\sin\theta & \cos\theta \\ -\cos\theta & -\sin\theta \end{pmatrix} \Big|_{\theta=0} = -i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

generator:  $\hat{X} = -i \frac{\partial}{\partial \theta} D'(\theta) \Big|_{\theta=0} = -i \begin{pmatrix} \frac{\partial \cos \theta}{\partial \theta} & \frac{\partial \sin \theta}{\partial \theta} \\ \frac{\partial (-\sin \theta)}{\partial \theta} & \frac{\partial \cos \theta}{\partial \theta} \end{pmatrix} \Big|_{\theta=0}$

$$= -i \begin{pmatrix} -\sin \theta & \cos \theta \\ -\cos \theta & -\sin \theta \end{pmatrix} \Big|_{\theta=0} = -i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\hat{X} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$\therefore D'(d\theta) = \begin{pmatrix} \cos(d\theta) & \sin(d\theta) \\ -\sin(d\theta) & \cos(d\theta) \end{pmatrix} \approx \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + i d\theta \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

Taylor expansion of the above relation

$$\begin{pmatrix} \cos(d\theta) & \sin(d\theta) \\ -\sin(d\theta) & \cos(d\theta) \end{pmatrix} = \begin{pmatrix} 1 - \frac{d\theta^2}{2!} + \dots & d\theta - \frac{d\theta^3}{3!} + \dots \\ -(d\theta - \frac{d\theta^3}{3!} + \dots) & 1 - \frac{d\theta^2}{2!} + \dots \end{pmatrix} \quad \text{LHS.}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & d\theta \\ -d\theta & 0 \end{pmatrix} = \begin{pmatrix} 1 & d\theta \\ -d\theta & 1 \end{pmatrix} \quad \text{RHS.}$$

This is true for the first power of  $d\theta$ .

Can we do better than the first order?

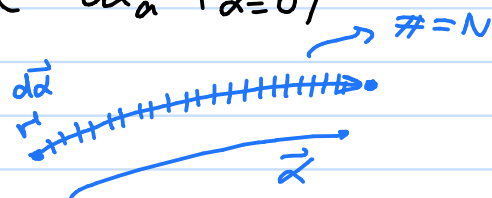
$\Rightarrow$  Can we do a general set of parameters?

Yes! we can keep all orders of expansion!

Consider the expansion in first order

$$D(d\vec{\alpha}) = \hat{1} + \sum_a i d\alpha_a \left( -i \frac{\partial}{\partial \alpha_a} D(\vec{\alpha}) \Big|_{\vec{\alpha}=0} \right)$$

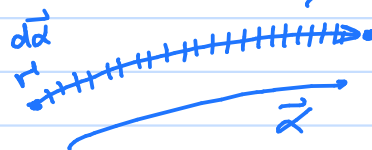
then consider  $\vec{\alpha} = N d\vec{\alpha}$



Consider the expansion in first order

$$D(d\vec{\alpha}) = \hat{1} + \sum_a i d\alpha_a \left( -i \frac{\partial D(\vec{\alpha})}{\partial \alpha_a} \bigg|_{\vec{\alpha}=0} \right)$$

then consider  $\vec{\alpha} = N d\vec{\alpha}$



$$\therefore D(\vec{\alpha}) = \underbrace{D(d\vec{\alpha}) \cdot D(d\vec{\alpha}) \cdot D(d\vec{\alpha}) \cdot \dots \cdot D(d\vec{\alpha})}_N$$

$$g(\vec{\alpha}) = g(d\vec{\alpha}) \cdot g(d\vec{\alpha}) \cdot \dots \cdot g(d\vec{\alpha})$$

homomorphism

successive operation group "multiplication"

$$D(\vec{\alpha}) = D\left(\frac{\vec{\alpha}}{N}\right)^N$$

$$= \left( \hat{1} + \sum_a i \left( \frac{\alpha_a}{N} \right) \hat{X}_a \right)^N$$

Definition of e

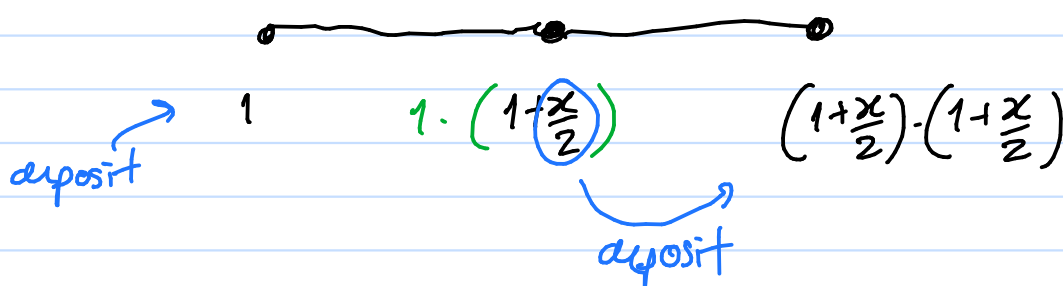
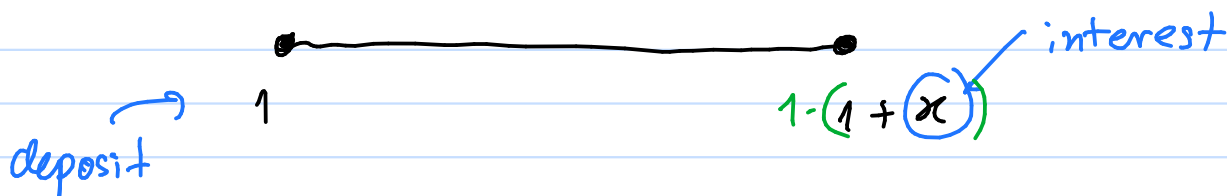
$$= \lim_{N \rightarrow \infty} \left( \hat{1} + \left( \frac{\sum_a i \alpha_a \hat{X}_a}{N} \right) \right)^N$$

$$\left( \lim_{N \rightarrow \infty} \left( 1 + \frac{x}{N} \right)^N = e^x \right)$$

Side note

$$\lim_{N \rightarrow \infty} \left( 1 + \frac{x}{N} \right)^N = e^x$$

Infinite interest rate



deposit  $\rightarrow$  1  $\xrightarrow{\quad}$   $1 \cdot (1 + x)$  interest

deposit  $\rightarrow$  1  $\xrightarrow{\quad}$   $1 \cdot (1 + \frac{x}{2})$   $\xrightarrow{\quad}$   $(1 + \frac{x}{2}) \cdot (1 + \frac{x}{2})$  deposit

1  $\xrightarrow{\quad}$   $1 + \frac{x}{3}$   $\xrightarrow{\quad}$   $(1 + \frac{x}{3})(1 + \frac{x}{3})$   $\xrightarrow{\quad}$   $(1 + \frac{x}{3})^3$

$\dots \xrightarrow{\quad}$   $(1 + \frac{x}{N}) \dots \xrightarrow{\quad}$   $(1 + \frac{x}{N})^N = e^x$   
 $\uparrow$   
 $\lim_{N \rightarrow \infty}$

Back to the main story:

$$D(\vec{\alpha}) = D\left(\frac{\vec{\alpha}}{N}\right)^N$$

$$= \left( \hat{1} + \sum_a i \frac{\alpha_a}{N} \hat{X}_a \right)^N$$

$$= \lim_{N \rightarrow \infty} \left( \hat{1} + \frac{\sum_a i \alpha_a \hat{X}_a}{N} \right)^N$$

$$D(\vec{\alpha}) = e^{\sum_a i \alpha_a \hat{X}_a}$$

Exponential  
parametrization.

Ex Rotation on 2D

$$D(\theta) = e^{i\theta \hat{1}} \rightarrow \text{generator} \rightarrow \text{exponential parametrization}$$

Ex Rotation on 2D

$$D(\theta) = e^{i\theta \hat{X}} \rightarrow \text{generator} \rightarrow \text{exponential parametrisation}$$

$$D'(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \quad \hat{X} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$D'(\theta) = e^{i\theta \hat{X}} = e^{i\theta \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}}$$

\*  $e^{\hat{M}} = \hat{1} + \hat{M} + \frac{\hat{M} \cdot \hat{M}}{2!} + \frac{\hat{M} \cdot \hat{M} \cdot \hat{M}}{3!} + \dots$  Definition of exponential of a matrix

$$\therefore e^{i\theta \hat{X}} = \hat{1} + i\theta \hat{X} + \frac{(i\theta)^2}{2!} \hat{X} \cdot \hat{X} + \frac{(i\theta)^3}{3!} \hat{X} \cdot \hat{X} \cdot \hat{X} + \dots$$

$$\hat{X} \cdot \hat{X} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hat{1}$$

$$\underbrace{\hat{X} \cdot \hat{X}}_{\hat{1}} \cdot \hat{X} = \hat{1} \cdot \hat{X} = \hat{X}, \quad \hat{X}^4 = \hat{1}, \quad \hat{X}^5 = \hat{X}$$

$$e^{i\theta \hat{X}} = \hat{1} + i\theta \hat{X} + \frac{(i\theta)^2}{2!} \hat{1} + \frac{(i\theta)^3}{3!} \hat{X}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + i\theta \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \frac{(-\theta^2)}{2!} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{-i\theta^3}{3!} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\cong \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \quad \checkmark \text{ convince yourself on this!}$$

$$= \begin{pmatrix} 1 - \frac{\theta^2}{2!} + \dots & \theta - \frac{\theta^3}{3!} + \dots \\ -\theta + \frac{\theta^3}{3!} + \dots & 1 - \frac{\theta^2}{2!} + \dots \end{pmatrix}$$



## More on infinite (continuous) groups

### 1) Matrix group / General linear group, $GL(n, \mathbb{R})$

(invertible)

$$A \in GL(n, \mathbb{R}), \quad A = \text{matrix } n \times n \quad (\det A \neq 0)$$

The subgroups of  $GL(n, \mathbb{R})$  are important in physics

### 2). Special linear group, $SL(n, \mathbb{R}) \subset GL(n, \mathbb{R})$

(unimodular group)

$$A \in SL(n, \mathbb{R}), \quad A \text{ is } n \times n \text{ matrix with } \underline{\det A = 1}$$

$$\text{Closure: } A, B \in SL(n, \mathbb{R}) \rightarrow AB \in SL(n, \mathbb{R})$$

$$\det(AB) = \det(A) \cdot \det(B) = 1 \cdot 1 = 1$$

$$\text{Asso: } \checkmark \quad (AB)C = A(BC)$$

$$\text{Identity: } \det(\mathbb{1}) = \det \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} = 1, \quad \mathbb{1} \in SL(n, \mathbb{R})$$

$$\text{Inverse: } A \in SL(n, \mathbb{R}) \rightarrow A^{-1} \in SL(n, \mathbb{R})$$

$$\det(AA^{-1}) = \det(\mathbb{1})$$

$$\underbrace{\det(A)}_1 \underbrace{\det(A^{-1})}_1 = \underbrace{\det(\mathbb{1})}_1 \Rightarrow A^{-1} \in SL(n, \mathbb{R})$$

$\therefore SL(n, \mathbb{R})$  is also a group  $\Rightarrow SL(n, \mathbb{R})$  is a subgroup of  $GL(n, \mathbb{R})$

### 3). Orthogonal group, $O(n)$

$$R \in O(n), \quad RR^T = R^T R = \mathbb{1}$$

3). Orthogonal group,  $O(n)$   <sup>$\subset GL(n, \mathbb{R})$</sup>   $R$  is  $n \times n$  matrix  
 $R \in O(n)$ ,  $\boxed{RR^T = R^T R = \mathbb{1}}$   $\rightarrow$  orthogonal property

Closure:  $R, S \in O(n) \rightarrow RS \in O(n)$

$$(RS)(RS)^T = \mathbb{1}$$

$$\underbrace{RS S^T}_{=\mathbb{1}} R^T = \underbrace{RR^T}_{=\mathbb{1}} = \mathbb{1}$$

since  $S \in O(n)$        $R \in O(n)$

Asso:  $\checkmark$

Identity:  $\mathbb{1} \mathbb{1}^T = \mathbb{1} \quad \checkmark \quad \mathbb{1} \in O(n)$

Inverse:  $R \in O(n) \rightarrow R^{-1} \in O(n)$

$$RR^T = \mathbb{1} \rightarrow (RR^T)^{-1} = \mathbb{1}$$

$$\therefore \mathbb{1} = (RR^T)^{-1} = (R^T)^{-1} R^{-1} = (R^{-1})^T R^{-1}$$

$\therefore O(n)$  is a subgroup of  $GL(n, \mathbb{R})$

\*  $R \in O(n) \rightarrow \det R = \pm 1$

$$RR^T = \mathbb{1} \rightarrow \det(RR^T) = \det \mathbb{1} = 1$$

$$\det(R) \det(R^T) = 1$$

$$(\det(R))^2 = 1 \Rightarrow \boxed{\det R = \pm 1}$$

The meaning of orthogonal group on the vector space

Consider  $R \in O(n)$ ,  $R: V^n \rightarrow V^n$

"rotation"  $\rightarrow$  operator on vector space  <sup>$\dim = n$</sup>

Under the rotation, the length of vector is invariant.

The meaning of orthogonal group on the vector space

Consider  $R \in O(n)$ ,  $R: V^n \rightarrow V^n$

"rotation"

operator on vector space  
dim = n

Under the rotation, the length of vector is invariant.

Consider  $\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ ,  $\vec{v} \cdot \vec{v} = v_1^2 + v_2^2 + \dots + v_n^2$

$$(v_1, \dots, v_n) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = |\vec{v}|^2 = \vec{v}^T \vec{v} \quad \leftarrow \text{column vector form}$$

$$\therefore R\vec{v} = \begin{pmatrix} R_{11} & \dots \\ \vdots & \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \vec{v}' \quad \checkmark \text{ a new vector after rotation. } (R)$$

$$|\vec{v}'|^2 = \vec{v}'^T \vec{v}' = (R\vec{v})^T (R\vec{v})$$

$$= \vec{v}^T R^T R \vec{v}$$

$$= \vec{v}^T \vec{v} = |\vec{v}|^2$$

Since R does not change the length of vector.

$$\Rightarrow \boxed{R^T R = \mathbb{I}} \Rightarrow \text{orthogonality condition.}$$

$\therefore$  The set of all rotation that preserve the length of vector forms a group called  $O(n)$

Orthogonal group.

Similar to  $GL(n, \mathbb{R}) \rightarrow SL(n, \mathbb{R})$ ,  $\det = 1$

We can construct a subgroup of  $O(n)$  with  $\det = 1$

Since  $R \in O(n)$ ,  $\det R = \pm 1$

Similar to  $GL(n, \mathbb{R}) \rightarrow SL(n, \mathbb{R})$ ,  $\det = 1$

We can construct a subgroup of  $O(n)$  with  $\det = 1$

Since  $R \in O(n)$ ,  $\det R = \pm 1$

The subgroup of  $O(n)$  is called  $SO(n)$

Special Orthogonal group,  $R \in SO(n)$ ,  $RR^T = 1$   
&  $\det R = 1$

identity  $1 \in SO(n)$

Sometimes we call part of  $O(n)$  with  $\det = 1$ , proper

&  $\det = -1$   $\Rightarrow$  improper

4). Pseudo Orthogonal group of order  $p, q$ ,  $O(p, q)$

Let's define the metric tensor,  $G =$  diagonal matrix  $n \times n$

$$G = \begin{pmatrix} G_{11} & & 0 \\ & \ddots & \\ 0 & & \end{pmatrix}_{n \times n}$$

$$p + q = n$$

$$\& G_{ii} = \begin{cases} -1 & , i = 1, \dots, p \\ 1 & , i = p+1, \dots, n \end{cases}$$

e.g.

$$G = \begin{pmatrix} -1 & & & & & & 0 \\ & -1 & & & & & \\ & & \ddots & & & & \\ & & & -1 & & & \\ 0 & & & & 1 & & \\ & & & & & 1 & \\ & & & & & & \ddots \end{pmatrix} \begin{matrix} p \\ q \end{matrix} \Bigg\} n$$

Pseudo orthogonal group  $O(p, q)$   $\nearrow \Lambda^T \Lambda = \mathbb{1}$  orthogonal

$$\Lambda \in O(p, q), \quad \boxed{\Lambda^T G \Lambda = G} \quad \text{Pseudo orthogonal condition}$$

Closure:  $\Lambda_1, \Lambda_2 \in O(p, q) \rightarrow \Lambda_1 \Lambda_2 \in O(p, q)$

$$\begin{aligned} (\Lambda_1 \Lambda_2)^T G (\Lambda_1 \Lambda_2) &= \Lambda_2^T (\underbrace{\Lambda_1^T G \Lambda_1}_G \text{ since } \Lambda_1 \in O(p, q)) \Lambda_2 \\ &= \underbrace{\Lambda_2^T G \Lambda_2}_{\Lambda_2 \in O(p, q)} = G \end{aligned}$$

Asso:  $\checkmark$

Identity:  $\mathbb{1}^T G \mathbb{1} = G \quad \checkmark$

Inverse:  $\Lambda_1 \in O(p, q) \rightarrow \Lambda_1^{-1} \in O(p, q)$

$$\begin{aligned} (\Lambda_1^{-1})^T G (\Lambda_1^{-1}) &= (\Lambda_1^T)^{-1} G (\Lambda_1)^{-1} \\ &= (\Lambda_1^T)^{-1} G^{-1} \Lambda_1^{-1} \\ &= (\Lambda_1 G \Lambda_1^T)^{-1} \\ &= \left( (\Lambda_1^T G^T \Lambda_1)^T \right)^{-1} \\ &= \left( (\underbrace{\Lambda_1^T G \Lambda_1}_G)^T \right)^{-1} \text{ , } \Lambda_1 \in O(p, q) \\ &= G \end{aligned}$$

$$\therefore \Lambda_1^{-1} \in O(p, q).$$

$$G = G^{-1}$$

$$G G = \mathbb{1}$$

$$G^T = G$$

What does the pseudo orthogonality means in the vector space?

$\Rightarrow \Lambda \in O(p, q) \Rightarrow$  operator that preserve the "length" of vectors

$$\Lambda: V^n \rightarrow V^n$$

↳ not the usual length.

Consider  $\vec{v} \in V^n$ ,  $\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$

Let's define a new definition of length

$$\begin{aligned} |\vec{v}|^2 &= \vec{v}^T G \vec{v} = (v_1 \dots v_n) \begin{pmatrix} -1 & & \\ & -1 & \\ & & \ddots & \\ & & & -1 & \\ & & & & 1 & \dots & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \\ &= -v_1^2 - v_2^2 - \dots - v_p^2 + v_{p+1}^2 + \dots + v_n^2 \end{aligned}$$

(This is familiar to us,  $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$ )  
(In Special Relativity, this is called "distance")

The transformation (operator)  $\Lambda \in O(p, q)$

leaves the "length" invariant

$$\vec{v}' = \Lambda \vec{v} \quad \rightarrow \text{transformation on vector}$$

$$|\vec{v}'|^2 = \vec{v}'^T G \vec{v}' = (\Lambda \vec{v})^T G (\Lambda \vec{v})$$

$$= \vec{v}^T \underbrace{\Lambda^T G \Lambda}_G \vec{v} \quad \text{since } \Lambda \in O(p, q)$$

$$= \vec{v}^T G \vec{v} = |\vec{v}|^2$$

The Lorentz transformations in Special Relativity form "Lorentz group" =  $O(1,3)$

- Note that we can also define the metric tensor the other way around:

$$G = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 0 \\ 0 & & & -1 & \\ & & & & -1 & \\ & & & & & -1 & \\ & & & & & & \ddots & \\ & & & & & & & -1 \end{pmatrix}$$

Since  $\Lambda^T G \Lambda = G \Rightarrow \underbrace{\Lambda^T}_{\equiv G'} (-G) \underbrace{\Lambda}_{\equiv G'} = (-G)$

We still have the same property.

## 5. Symplectic group $Sp(2n)$

Consider the matrix  $2n \times 2n$

$$J = \begin{pmatrix} O_{n \times n} & \mathbb{I}_{n \times n} \\ -\mathbb{I}_{n \times n} & O_{n \times n} \end{pmatrix}$$

$* \sum_{i=1}^{2n} 2 \underbrace{(-1)(-1)}_{(-1)(-1)} \underbrace{1 \dots 1}_{2n} = 1$

Element of  $Sp(2n)$  is a matrix  $A$  such that

$$A^T J A = J$$

$\therefore Sp(2n)$  is a group of

symmetry that preserve the "length"  $|\vec{v}|^2 = \vec{v}^T J \vec{v}$

Note that  $\det A = \pm 1$

$$\det(A^T) \det(J) \det(A) = \det(J)$$

$\stackrel{=1}{=}$

Similar to

$$\Lambda^T G \Lambda = G$$

$$-(-1) \cdot 1 = 1$$

6). General linear complex group  $GL(n, \mathbb{C})$

An invertible matrix of dim  $n \times n$  where its components are complex numbers.  $\Rightarrow$  form a group

$$\left\{ \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix}_{n \times n} \right\}$$

7). Special Linear group  $SL(n, \mathbb{C}) \subset GL(n, \mathbb{C})$

$$\det A = 1, \quad A \in SL(n, \mathbb{C})$$

8). Unitary group  $U(n)$

$$U(n) = \left\{ A \mid (A^T)^* A = A (A^T)^* = \mathbb{1} \right\}$$

where  $(A^T)^* \equiv A^\dagger$  adjoint operator (dagger)

$$\det(A^\dagger) \det(A) = \det(\mathbb{1})$$

$$(\det A)^* \det A = 1 \Rightarrow |\det A|^2 = 1$$

$$\therefore \det A = e^{i\phi} \rightarrow \textcircled{1}, -1, \dots$$

9). Special Unitary group  $SU(n)$

$$SU(n) \subset U(n)$$

$$SU(n) = \left\{ A \mid A^\dagger A = A A^\dagger = \mathbb{1} \text{ \& } \det A = 1 \right\}$$

Interpretation in physics

We can look  $U \in U(n)$  as an operator on  $V = \mathbb{C}^n$

$$|\psi\rangle \in \mathbb{C}^n, \quad |\psi\rangle = (\psi_1, \dots, \psi_n), \quad \psi_1, \dots, \psi_n \in \mathbb{C}$$

$$|\psi\rangle \rightarrow |\psi'\rangle = \hat{U} |\psi\rangle$$



## Interpretation in physics

We can look  $U \in U(n)$  as an operator on  $V = \mathbb{C}^n$   
 $|\psi\rangle \in \mathbb{C}^n$ ,  $|\psi\rangle = (\psi_1, \dots, \psi_n)$ ,  $\psi_1, \dots, \psi_n \in \mathbb{C}$

$$|\psi\rangle \rightarrow |\psi'\rangle = \hat{U}|\psi\rangle$$

$\therefore$  The length in complex vector space  $\neq \psi_1^2 + \psi_2^2 + \dots + \psi_n^2$

$$= \psi_1^* \psi_1 + \psi_2^* \psi_2 + \dots + \psi_n^* \psi_n$$

$$\| |\psi\rangle \|^2 = \langle \psi | \psi \rangle = (\psi_1^* \ \psi_2^* \ \dots \ \psi_n^*) \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_n \end{pmatrix}$$

$$\begin{array}{lcl} |\psi\rangle \rightarrow \langle \psi| & \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_n \end{pmatrix} \rightarrow & (\psi_1^* \ \dots \ \psi_n^*) \\ \text{ket} \rightarrow \text{bra} & & \end{array}$$

adjoint operator.  
transpose & complex conjugate

$$|\psi'\rangle = \hat{U}|\psi\rangle \rightarrow \langle \psi'| = \langle \psi| \hat{U}^\dagger$$

$$\therefore \langle \psi' | \psi' \rangle = \langle \psi | \hat{U}^\dagger \hat{U} | \psi \rangle = \langle \psi | \psi \rangle$$

Requirement that the length of the complex vector is invariant  $\Rightarrow$  transformations = unitary group

$$\therefore \boxed{\hat{U}^\dagger \hat{U} = \mathbb{I}} \quad \text{Unitary condition.}$$

---

Let's go back to Lie group

Consider a group of 2D rotation  $\left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \right\}$

This group is just  $SO(2)$

Let's go back to Lie group

Consider a group of 2D rotation  $\left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \right\}$

This group is just  $SO(2)$

$$R^T R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{pmatrix} = \mathbb{1} \quad \checkmark \quad O(2)$$

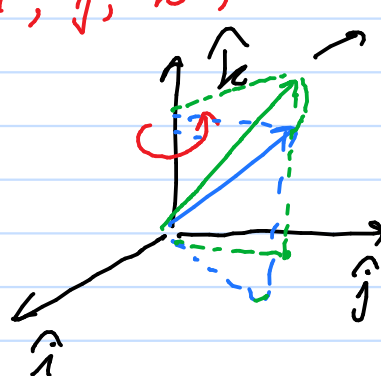
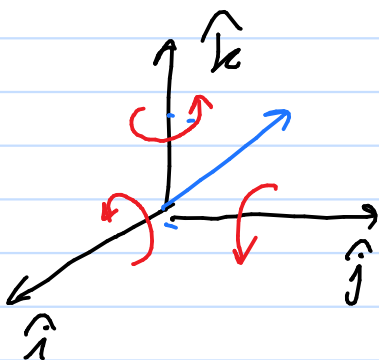
$$\det(R) = \cos^2\theta - (-\sin\theta)\sin\theta = 1 \quad \checkmark \quad SO(2)$$

Let's consider  $SO(3)$   $\Rightarrow$  rotation group in 3D space.

(The length of 3D vector is invariant)

Normally we quantify the 3D rotation by rotations in

3 different axes ( $\hat{i}, \hat{j}, \hat{k}$ )



The  $\hat{k}$  component is not changed under rotation around  $\hat{k}$  axis

$$\vec{v} = (x \ y \ z)$$

$$x' = x \cos\theta + y \sin\theta$$

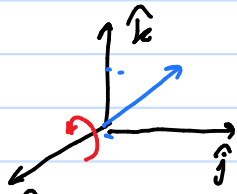
$$\hat{R}_k \vec{v} = (x' \ y' \ z)$$

$$y' = -x \sin\theta + y \cos\theta$$

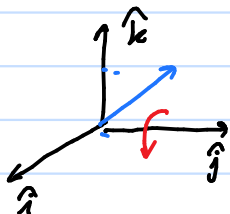
$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\hat{R}_k} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$z' = z$$

Rotation around  $\hat{i} \Rightarrow \hat{R}_i = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}$



Rotation around  $\hat{j} \Rightarrow \hat{R}_j = \begin{pmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix}$



\* rotation on  $\hat{k} \hat{i}$  plane  $\rightarrow \theta \rightarrow -\theta$   
not  $\hat{i} \hat{k}$  plane.

A general rotation = 3 successive rotations

$$\begin{pmatrix} \end{pmatrix} = \begin{pmatrix} \cos\theta_z & \sin\theta_z & 0 \\ -\sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & \sin\theta_x \\ 0 & -\sin\theta_x & \cos\theta_x \end{pmatrix} \begin{pmatrix} \cos\theta_y & 0 & -\sin\theta_y \\ 0 & 1 & 0 \\ \sin\theta_y & 0 & \cos\theta_y \end{pmatrix}$$

\* The order is important here  $\Rightarrow$  This group is non abelian  
representation (3D)

$$g(\vec{\theta}) \text{ or } D(\vec{\theta}) = D(\theta_x, \theta_y, \theta_z)$$

The group element is parametrized by 3 parameters

Generators of  $SO(3)$  (3D representation)

From definition

$$D(d\vec{\alpha}) = \mathbb{1} + \sum_a i d\alpha_a \hat{X}_a, \quad \hat{X}_a \equiv -i \frac{\partial D(\vec{\alpha})}{\partial \alpha_a} \Big|_{\vec{\alpha}=0}$$

$$D(d\vec{\theta} = d\theta_x, d\theta_y, d\theta_z) = \mathbb{1} + i d\theta_x \hat{X}_x + i d\theta_y \hat{X}_y + i d\theta_z \hat{X}_z$$

$$\hat{X}_x \equiv -i \frac{\partial D(\vec{\theta})}{\partial \theta_x} \Big|_{\vec{\theta}=0} = -i \frac{\partial}{\partial \theta_x} \begin{pmatrix} \theta_z \\ \theta_x \\ \theta_y \end{pmatrix} \begin{pmatrix} \theta_z \\ \theta_z \\ \theta_y \end{pmatrix} \begin{pmatrix} \theta_y \end{pmatrix}$$

$$D(d\vec{\theta} = d\theta_x, d\theta_y, d\theta_z) = \mathbb{1} + i d\theta_x \hat{X}_x + i d\theta_y \hat{X}_y + i d\theta_z \hat{X}_z$$

$$\hat{X}_x = -i \frac{\partial D(\vec{\theta})}{\partial \theta_x} \Big|_{\vec{\theta}=0} = -i \frac{\partial}{\partial \theta_x} \left( \underbrace{\theta_z}_{\mathbb{1}} \right) \left( \underbrace{\theta_x}_{\mathbb{1}} \right) \left( \underbrace{\theta_y}_{\mathbb{1}} \right)$$

$$= \left( \underbrace{\theta_z}_{\mathbb{1}} \right) \left( -i \frac{\partial}{\partial \theta_x} \left( \underbrace{\theta_x}_{\mathbb{1}} \right) \right) \left( \underbrace{\theta_y}_{\mathbb{1}} \right) \Big|_{\vec{\theta}=0}$$

$$= -i \frac{\partial}{\partial \theta_x} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & \sin \theta_x \\ 0 & -\sin \theta_x & \cos \theta_x \end{pmatrix} \Big|_{\theta_x=0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$\hat{X}_y = -i \frac{\partial D(\vec{\theta})}{\partial \theta_y} \Big|_{\vec{\theta}=0} = \left( \underbrace{\theta_z}_{\mathbb{1}} \right) \left( \underbrace{\theta_x}_{\mathbb{1}} \right) -i \frac{\partial}{\partial \theta_y} \begin{pmatrix} \cos \theta_y & 0 & -\sin \theta_y \\ 0 & 1 & 0 \\ \sin \theta_y & 0 & \cos \theta_y \end{pmatrix} \Big|_{\vec{\theta}=0}$$

$$= \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}$$

Similar

$$\hat{X}_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\therefore$   $SO(3)$  in 3D rep has 3 generators

$$\hat{X}_x, \hat{X}_y, \hat{X}_z = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

In the case of many generators case, we can do the exponential parametrization:

$$1 \text{ generator} : D(d\alpha) = \hat{1} + i d\alpha \hat{X} \Rightarrow D(\alpha) = e^{i\alpha \hat{X}}$$

$$\text{many generator} : D(d\vec{\alpha}) = \hat{1} + \sum_a i d\alpha_a \hat{X}_a \Rightarrow D(\alpha) = e^{i \sum_a \alpha_a \hat{X}_a}$$

---

## Lie algebra

Consider the exponential parametrization of group with many generators :

$$D(\vec{\alpha}) = e^{i \sum_a \alpha_a \hat{X}_a}$$

$$a=1, \dots, n, \quad n = \# \text{ of generators}$$

Suppose that we want to multiply 2 elements in this form.

$$D(\vec{\alpha}) D(\vec{\beta}) = e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b} : a, b=1, \dots, n$$

Since we know the product is also a group element

$$\begin{aligned} D(\vec{\alpha}) D(\vec{\beta}) &= D(\vec{\gamma}) \\ &= e^{i \sum_c \gamma_c \hat{X}_c} \end{aligned}$$

Is this the case? The exponential of matrix

$$e^M = \hat{1} + M + \frac{M^2}{2!} + \dots \Rightarrow$$

$$e^M e^N \neq e^{M+N}$$

$$e^M e^N = \left( \hat{1} + M + \frac{M^2}{2!} + \dots \right) \left( \hat{1} + N + \frac{N^2}{2!} + \dots \right) \neq \hat{1} + (M+N) + \frac{(M+N)^2}{2!} + \dots$$

$$e^M e^N = \left(1 + M + \frac{M^2}{2!} + \dots\right) \left(1 + N + \frac{N^2}{2!} + \dots\right) \neq 1 + (M+N) + \frac{(M+N)^2}{2!} + \dots$$

no N-M (M+N)(M+N)  
N.M

$e^M e^N \neq e^{M+N}$  except that  $M$  &  $N$  are numbers

How do we make sure that  $D(\vec{\alpha}) D(\vec{\beta}) = D(\vec{\gamma})$  ?

Since we don't have

$$e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b} \neq e^{i \sum_a (\alpha_a + \beta_a) \hat{X}_a}$$

★ Let's figure the condition that this is true.

Suppose  $D(\vec{\alpha}) D(\vec{\beta}) = D(\vec{\gamma})$

$$e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b} = e^{i \sum_c \gamma_c \hat{X}_c}$$

$$\ln \left( e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b} \right) = i \sum_c \gamma_c \hat{X}_c$$

$$i \sum_c \gamma_c \hat{X}_c = \ln \left( \underbrace{1 + e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b} - 1}_{= y} \right)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$\ln(1+y) = y - \frac{y^2}{2} + \mathcal{O}(y^3)$$

$$y \ll 1, \vec{\alpha}, \vec{\beta} \rightarrow 0$$

Define  $y = e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b} - 1$

$$y = \left( \hat{1} + i \sum_a \alpha_a \hat{X}_a + \frac{(i)^2}{2!} \left( \sum_a \alpha_a \hat{X}_a \right)^2 + \dots \right) \left( \hat{1} + i \sum_b \beta_b \hat{X}_b + \frac{(i)^2}{2!} \left( \sum_b \beta_b \hat{X}_b \right)^2 + \dots \right) - \hat{1}$$

Define  $y = e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b} - 1$

$$y = (\hat{1} + i \sum_a \alpha_a \hat{X}_a + \frac{(i)^2}{2!} (\sum_a \alpha_a \hat{X}_a)^2 + \dots) (\hat{1} + i \sum_b \beta_b \hat{X}_b + \frac{(i)^2}{2!} (\sum_b \beta_b \hat{X}_b)^2 + \dots) - \hat{1}$$

$$= \cancel{\hat{1}} + i \sum_a \alpha_a \hat{X}_a + i \sum_b \beta_b \hat{X}_b + (i)^2 \sum_{a,b} \alpha_a \beta_b \hat{X}_a \hat{X}_b$$

$$+ \frac{(i)^2}{2!} (\sum_a \alpha_a \hat{X}_a)^2 + \frac{(i)^2}{2!} (\sum_b \beta_b \hat{X}_b)^2 + \dots - \cancel{\hat{1}}$$

$$y = i \sum_a \alpha_a \hat{X}_a + i \sum_b \beta_b \hat{X}_b - \sum_{a,b} \alpha_a \beta_b \hat{X}_a \hat{X}_b$$

$$- \frac{1}{2} (\sum_a \alpha_a \hat{X}_a)^2 - \frac{1}{2} (\sum_b \beta_b \hat{X}_b)^2$$

plug back into

$$i \sum_c \gamma_c \hat{X}_c = \ln \left( 1 + \overbrace{e^{i \sum_a \alpha_a \hat{X}_a} e^{i \sum_b \beta_b \hat{X}_b}}^y - 1 \right)$$

$$= y - \frac{y^2}{2} + \mathcal{O}(y^3)$$

$$i \sum_c \gamma_c \hat{X}_c = i \sum_a \alpha_a \hat{X}_a + i \sum_b \beta_b \hat{X}_b - \sum_{a,b} \alpha_a \beta_b \hat{X}_a \hat{X}_b$$

$$- \frac{1}{2} (\sum_a \alpha_a \hat{X}_a)^2 - \frac{1}{2} (\sum_b \beta_b \hat{X}_b)^2 + \dots$$

$$- \frac{1}{2} \left( i \sum_a \alpha_a \hat{X}_a + i \sum_b \beta_b \hat{X}_b - \sum_{a,b} \alpha_a \beta_b \hat{X}_a \hat{X}_b \right)$$

$$- \frac{1}{2} (\sum_a \alpha_a \hat{X}_a)^2 - \frac{1}{2} (\sum_b \beta_b \hat{X}_b)^2$$

$$\times \left( i \sum_a \alpha_a \hat{X}_a + i \sum_b \beta_b \hat{X}_b - \sum_{a,b} \alpha_a \beta_b \hat{X}_a \hat{X}_b \right)$$

$$- \frac{1}{2} (\sum_a \alpha_a \hat{X}_a)^2 - \frac{1}{2} (\sum_b \beta_b \hat{X}_b)^2$$

$$i \sum_c \hat{x}_c = i \sum_a \alpha_a \hat{x}_a + i \sum_b \beta_b \hat{x}_b - \sum_{a,b} \alpha_a \beta_b \hat{x}_a \hat{x}_b - \frac{1}{2} \left( \sum_a \alpha_a \hat{x}_a \right)^2 - \frac{1}{2} \left( \sum_b \beta_b \hat{x}_b \right)^2 + \dots \quad \left. \vphantom{\sum_c} \right\} y$$

$$- \frac{1}{2} \left( i \sum_a \alpha_a \hat{x}_a + i \sum_b \beta_b \hat{x}_b - \sum_{a,b} \alpha_a \beta_b \hat{x}_a \hat{x}_b - \frac{1}{2} \left( \sum_a \alpha_a \hat{x}_a \right)^2 - \frac{1}{2} \left( \sum_b \beta_b \hat{x}_b \right)^2 \right) \quad \left. \vphantom{\sum_c} \right\} y$$

$$\times \left( i \sum_a \alpha_a \hat{x}_a + i \sum_b \beta_b \hat{x}_b - \sum_{a,b} \alpha_a \beta_b \hat{x}_a \hat{x}_b - \frac{1}{2} \left( \sum_a \alpha_a \hat{x}_a \right)^2 - \frac{1}{2} \left( \sum_b \beta_b \hat{x}_b \right)^2 \right) \quad \left. \vphantom{\sum_c} \right\} y$$

$$= i \sum_a \alpha_a \hat{x}_a + i \sum_b \beta_b \hat{x}_b - \sum_{a,b} \alpha_a \beta_b \hat{x}_a \hat{x}_b - \frac{1}{2} \left( \sum_a \alpha_a \hat{x}_a \right)^2 - \frac{1}{2} \left( \sum_b \beta_b \hat{x}_b \right)^2 + \dots - \frac{1}{2} i^2 \left( \sum_a \alpha_a \hat{x}_a \right)^2 - \frac{1}{2} i^2 \left( \sum_b \beta_b \hat{x}_b \right)^2 - \frac{1}{2} i^2 \sum_{a,b} \alpha_a \beta_b \hat{x}_a \hat{x}_b + \mathcal{O}(\hat{x}^3)$$

$$= i \sum_a \alpha_a \hat{x}_a + i \sum_b \beta_b \hat{x}_b - \frac{1}{2} \sum_{a,b} \alpha_a \beta_b \hat{x}_a \hat{x}_b + \frac{1}{2} \sum_{a,b} \beta_b \alpha_a \hat{x}_b \hat{x}_a$$

$$= i \sum_a \alpha_a \hat{x}_a + i \sum_b \beta_b \hat{x}_b - \frac{1}{2} \left[ \sum_a \alpha_a \hat{x}_a, \sum_b \beta_b \hat{x}_b \right]$$

Suppose that we drop the higher order terms

Requirement that l.h.s gives  $i \sum_c \hat{x}_c$

leads to

commutator  
of 2 matrices



$$= i \sum_a \alpha_a \hat{X}_a + i \sum_b \beta_b \hat{X}_b - \frac{1}{2} \left[ \sum_a \alpha_a \hat{X}_a, \sum_b \beta_b \hat{X}_b \right]$$

Suppose that we drop the higher order terms

Requirement that l.h.s gives  $i \sum_c \gamma_c \hat{X}_c$

commutator of 2 matrices

leads to

$$i \sum_c \gamma_c \hat{X}_c = i \sum_a \alpha_a \hat{X}_a + i \sum_b \beta_b \hat{X}_b - \frac{1}{2} \left[ \sum_a \alpha_a \hat{X}_a, \sum_b \beta_b \hat{X}_b \right]$$

$$\frac{1}{2} \sum_{a,b} \alpha_a \beta_b [\hat{X}_a, \hat{X}_b] = i \sum_c (\alpha_c + \beta_c - \gamma_c) \hat{X}_c$$

This requires  $[\hat{X}_a, \hat{X}_b] \propto \hat{X}_c \dots$

$\therefore$  Suppose

$$* \left[ \hat{X}_a, \hat{X}_b \right] = i \sum_c f_{abc} \hat{X}_c \quad *$$

numbers depending on a, b, c

L.H.S.

$$\frac{1}{2} \sum_{a,b} \alpha_a \beta_b \sum_c f_{abc} \hat{X}_c = \sum_c \left( i \frac{1}{2} \sum_{a,b} \alpha_a \beta_b f_{abc} \right) \hat{X}_c$$

R.H.S.  $\sum_c \left( i (\alpha_c + \beta_c - \gamma_c) \right) \hat{X}_c$

$\therefore$  The assumption can be made true by setting

$$i \frac{1}{2} \sum_{a,b} \alpha_a \beta_b f_{abc} = i (\alpha_c + \beta_c - \gamma_c)$$

consequence of  $\gamma$  free

$\therefore$  The conclusion is we need to require that

$$[\hat{X}_a, \hat{X}_b] = i \sum_c f_{abc} \hat{X}_c$$

Consistency requirement.

The conclusion is we need to require that

$$[\hat{X}_a, \hat{X}_b] = i \sum_c f_{abc} \hat{X}_c$$

This is a structure called algebra (Lie algebra)

★  $f_{abc}$  is called structure constants

$f_{abc}$  depend on the group & representation

Example  $SO(3)$  has 3 generators

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

$$\begin{aligned} [\hat{X}_1, \hat{X}_2] &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} &= \cancel{0\hat{X}_1} + \cancel{0\hat{X}_2} + (-i)\hat{X}_3 = i(0\hat{X}_1 + 0\hat{X}_2 + (-1)\hat{X}_3) \\ &\quad \begin{matrix} \uparrow & \uparrow & \uparrow \\ f_{121}=0 & f_{122}=0 & f_{123}=-i \end{matrix} \quad \begin{matrix} \uparrow & \uparrow & \uparrow \\ f_{121}=0 & f_{122}=0 & f_{123}=-1 \end{matrix} \end{aligned}$$

$$[\hat{X}_2, \hat{X}_3] = \dots \text{ try it yourself}$$

One can see that we don't need to do  $[\hat{X}_2, \hat{X}_1]$

since we know that  $[\hat{X}_1, \hat{X}_2] = -[\hat{X}_2, \hat{X}_1]$

$$\therefore f_{abc} = -f_{bac}$$

$$[\hat{X}_a, \hat{X}_b] = i \sum_c f_{abc} \hat{X}_c$$

= vector

What is algebra?

we can look at this as  
an operator between  $\hat{X}_a$  &  $\hat{X}_b$   
vector vector

It looks like a cross product!

$$\vec{A} \times \vec{B} = \vec{C} \Rightarrow \hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \dots$$

Algebra: cross product / tensor product / commutator /  $\wedge$

Properties of cross product

1)  $\vec{A}, \vec{B} \in V, \vec{A} \times \vec{B} \in V$  ( $\vec{A} \times \vec{B} = \vec{C}$ )

2)  $\vec{A} \times (b\vec{B} + c\vec{C}) = b(\vec{A} \times \vec{B}) + c(\vec{A} \times \vec{C})$

3). non-commutative

$$\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$$

In fact we have  $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$

4) non-associative

$$\vec{A} \times (\vec{B} \times \vec{C}) \neq (\vec{A} \times \vec{B}) \times \vec{C}$$

Structure constants of cross product

$$\hat{e}_1 = \hat{i}, \hat{e}_2 = \hat{j}, \hat{e}_3 = \hat{k}$$

$$\hat{e}_a \times \hat{e}_b = \sum_c f_{abc} \hat{e}_c$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 = f_{121} \hat{e}_1 + f_{122} \hat{e}_2 + f_{123} \hat{e}_3$$

$$\hat{e}_2 \times \hat{e}_3 = \hat{e}_1 = \dots$$

Algebra.

$\Rightarrow$  similarity between cross product &  $[, ]$

Another important property is the Jacobi identity

$$[A, [B, C]] + [C, [A, B]] + [B, [C, A]] = 0$$

We can check this by expanding

$$[A, BC - CB] + [C, AB - BA] + [B, CA - AC] = 0$$

$$0 = A(\cancel{BC} - \cancel{CB}) - (\cancel{BC} - \cancel{CB})A + \underline{C(AB - BA)} - \underline{(AB - BA)C} \\ + B(\underline{CA - AC}) + \underline{(CA - AC)B}$$

Another algebra in physics = Poisson bracket.

CM: Poisson Bracket

$$\{ \underbrace{f(q, p)}_{f^u}, \underbrace{g(q, p)}_{f^v} \}_{p.b.} = \frac{\partial f}{\partial q} \frac{\partial g}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial g}{\partial q} \Rightarrow \underline{f^u f^v}$$

We can check that this satisfies all properties of algebra

Exercise

## Representations of Lie algebra (group)

There are many representations of a Lie group.

Representation of a group  $\Rightarrow$  homomorphism between  
the group &  $GL(n, \mathbb{R})$   
(matrix group)

Sometimes we just identify generators in terms of  
the matrix form.

Example  $SO(2) \Rightarrow$  2D representation

$$\left\{ \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \right\}$$

$$D(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

$$\text{generator: } \hat{X} = -i \left. \frac{\partial D(\theta)}{\partial \theta} \right|_{\theta=0} = -i \left. \begin{pmatrix} \frac{\partial \cos \theta}{\partial \theta} & \frac{\partial \sin \theta}{\partial \theta} \\ \frac{\partial (-\sin \theta)}{\partial \theta} & \frac{\partial \cos \theta}{\partial \theta} \end{pmatrix} \right|_{\theta=0}$$

$$= -i \left. \begin{pmatrix} -\sin \theta & \cos \theta \\ -\cos \theta & -\sin \theta \end{pmatrix} \right|_{\theta=0} = -i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\hat{X} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \rightarrow \text{2D representation}$$

---

Similar to the construction of regular rep.

There is a standard way to construct rep of Lie group.

### Adjoint Representation

Consider an algebra  $\{\hat{X}_a\}$ ,  $a, b, c = 1, \dots, n$

$$[\hat{X}_a, \hat{X}_b] = i \sum_c f_{abc} \hat{X}_c$$

# generators

Consider  $\hat{X}_a$  as a matrix  $n \times n$

$[\hat{X}_a]_{bc} \equiv -i f_{abc}$

 Adjoint representation

We can prove that this satisfies the algebra above

## Adjoint Representation

Consider an algebra  $\{\hat{X}_a\}$ ,  $a, b, c = 1, \dots, n$

$$[\hat{X}_a, \hat{X}_b] = i \sum_c f_{abc} \hat{X}_c$$

# generators

Consider  $\hat{X}_a$  as a matrix  $n \times n$

$$[\hat{X}_a]_{bc} \equiv -i f_{abc}$$

Adjoint representation

We can prove that this satisfies the algebra above

Consider Jacobi identity + algebra (structure constant)

$$\begin{aligned} [\hat{X}_a, [\hat{X}_b, \hat{X}_c]] + [\hat{X}_c, [\hat{X}_a, \hat{X}_b]] + [\hat{X}_b, [\hat{X}_c, \hat{X}_a]] &= 0 \\ &= \sum_d f_{bcd} \hat{X}_d \quad = \sum_d f_{abd} \hat{X}_d \quad = \sum_d f_{cad} \hat{X}_d \end{aligned}$$

$$\begin{aligned} \sum_d \left( f_{bcd} [\hat{X}_a, \hat{X}_d] + f_{abd} [\hat{X}_c, \hat{X}_d] + f_{cad} [\hat{X}_b, \hat{X}_d] \right) &= 0 \\ &= \sum_e f_{ade} \hat{X}_e \quad \sum_e f_{cde} \hat{X}_e \quad \sum_e f_{bde} \hat{X}_e \end{aligned}$$

$$\sum_{d,e} \left( f_{bcd} f_{ade} + f_{abd} f_{cde} + f_{cad} f_{bde} \right) \hat{X}_e = 0$$

$$\sum_e \left( \sum_d (f_{bcd} f_{ade} + f_{abd} f_{cde} + f_{cad} f_{bde}) \right) \hat{X}_e = 0$$

= 0

$$\sum_d f_{bcd} f_{ade} + f_{abd} f_{cde} + f_{cad} f_{bde} = 0$$

Jacobi identity.

Let's get back to the proof

We choose  $[\hat{X}_a]_{bc} = -if_{abc}$

$$\left( [\hat{X}_a, \hat{X}_b] = \sum_c f_{abc} \hat{X}_c \right)_{gh} \quad \text{consider component } gh \text{ of the matrix}$$

$$[\hat{X}_a \hat{X}_b - \hat{X}_b \hat{X}_a]_{gh} = i \sum_c f_{abc} [\hat{X}_c]_{gh} \quad \text{Adjoint rep}$$

$$[\hat{X}_a \hat{X}_b]_{gh} - [\hat{X}_b \hat{X}_a]_{gh} = i \sum_c f_{abc} (-i) f_{cgh}$$

$$\sum_e [\hat{X}_a]_{ge} [\hat{X}_b]_{eh} - \sum_e [\hat{X}_b]_{ge} [\hat{X}_a]_{eh} = (-i) \sum_c f_{abc} f_{cgh}$$

$$(AB)_{ij} \Rightarrow \begin{pmatrix} \dots \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ i \end{pmatrix} = \sum_k A_{ik} B_{kj}$$

$$\begin{bmatrix} A & B \end{bmatrix}_{ij}$$

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

$$= \begin{pmatrix} (AB)_{11} & (AB)_{12} \\ (AB)_{21} & (AB)_{22} \end{pmatrix}$$

Matrix multiplication

in the component form.

$$(AB)_{11} = A_{11} B_{11} + A_{12} B_{21}$$

$$= \sum_k A_{1k} B_{k1}$$

$$\sum_e (-i) f_{age} (-i) f_{beh} - \sum_e (-i) f_{bge} (-i) f_{aeh} = \sum_c f_{abc} f_{cgh}$$

$$- \sum_e f_{age} f_{beh} + \sum_e f_{bge} f_{aeh} = \sum_e f_{abe} f_{egh} \quad c \rightarrow e$$

$$\sum_e (f_{gae} f_{beh} + f_{bge} f_{aeh} + f_{bae} f_{egh}) = 0 \quad f_{abc} = -f_{bac}$$

$$\sum_e (f_{gae} f_{beh} + f_{bge} f_{aeh} + f_{bae} f_{egh}) = 0 \quad \text{--- (1)}$$

Compare this with Jacobi identity:

$$\sum_a (f_{bcd} f_{ade} + f_{abd} f_{cde} + f_{cad} f_{bde}) = 0$$

$d \rightarrow e$   
 $e \rightarrow g$

$$\sum_e (f_{bce} f_{aeg} + f_{abe} f_{ceg} + f_{cae} f_{beg}) = 0$$

$f_{cae} f_{beg} = f_{ace} f_{ebg}$

$$\sum_e (f_{bce} f_{aeg} + f_{abe} f_{ceg} + f_{ace} f_{ebg}) = 0 \quad \text{--- (2)}$$

$$\sum_e (f_{gae} f_{beh} + f_{bge} f_{aeh} + f_{bae} f_{egh}) = 0 \quad \text{--- (1)}$$

$h \rightarrow g, a \rightarrow c, b \rightarrow a, g \rightarrow b$

$$\sum_e (f_{bce} f_{aeg} + f_{abe} f_{ceg} + f_{ace} f_{ebg}) = 0 \quad \Rightarrow \underline{\underline{(2)}}$$

$$\therefore \textcircled{1} = \textcircled{2}$$

The conclusion is that  $[\hat{X}_a]_{bc} = -if_{abc}$  is a generator of Lie group (algebra)  $\Rightarrow$  adjoint representation

Example  $SO(3)$  algebra (group)

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ -1 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, (-1) \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

$$[\hat{X}_1, \hat{X}_2] = \hat{X}_1 \hat{X}_2 - \hat{X}_2 \hat{X}_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$



Example  $SO(3)$  algebra (group)

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

$$[\hat{X}_1, \hat{X}_2] = \hat{X}_1 \hat{X}_2 - \hat{X}_2 \hat{X}_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} *$$

$$= i \sum_c f_{12c} \hat{X}_c = i \left( \underset{=}{f_{121}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix} + \underset{=}{f_{122}} \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} + \underset{1}{f_{123}} \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right)$$

$$= i (0 \hat{X}_1 + 0 \hat{X}_2 + 1 \hat{X}_3)$$

$$\therefore f_{121} = 0, f_{122} = 0, f_{123} = 1$$

$$[\hat{X}_2, \hat{X}_3] = \hat{X}_2 \hat{X}_3 - \hat{X}_3 \hat{X}_2$$

$$= \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$= i \sum_c f_{23c} \hat{X}_c = i \left( \underset{=}{f_{231}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix} + \cancel{f_{232}} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} + \cancel{f_{233}} \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right)$$

$$\therefore f_{231} = 1, f_{232} = 0, f_{233} = 0$$

$$[\hat{X}_1, \hat{X}_3] = ..$$

All commutators give

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

$$f_{123} = f_{312} = f_{231} = 1, \quad f_{213} = f_{321} = f_{132} = -1$$

$$\text{the rest is zero } f_{111}, f_{112}, f_{113}, \dots = 0$$

All  $f$ 's are combined into Levi-Civita symbol

$$\varepsilon_{abc} = f_{abc}, \quad 1). \quad \varepsilon_{123} = 1$$

$$2). \quad \varepsilon_{ijk} = -\varepsilon_{ikj} \quad \begin{array}{l} \text{1 permutation} \\ = (-1) \end{array}$$

$$3). \quad \varepsilon_{iik} = 0, \quad \text{repeating index} \Rightarrow \text{vanish}$$

$\hat{X}_1, \hat{X}_2, \hat{X}_3$  are already in the adjoint representation.

$$[\hat{X}_1]_{ij} = -i f_{1ij} = -i \begin{pmatrix} f_{111} & f_{112} & f_{113} \\ f_{121} & f_{122} & f_{123} \\ f_{131} & f_{132} & f_{133} \end{pmatrix}$$

$$= -i \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$[\hat{X}_2]_{ij} = -i f_{2ij} = -i \begin{pmatrix} f_{211} & f_{212} & f_{213} \\ f_{221} & f_{222} & f_{223} \\ f_{231} & f_{232} & f_{233} \end{pmatrix} = -i \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}$$

$$[\hat{X}_3]_{ij} = -if_{3ij} = -i \begin{pmatrix} f_{311} & f_{312} & f_{313} \\ f_{321} & f_{322} & f_{323} \\ f_{331} & f_{332} & f_{333} \end{pmatrix} = -i \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$


---

Adjoint representation property:

Let's consider a trace of product of adjoint rep.

$$\text{Tr}(\hat{X}_a \hat{X}_b)$$

This is similar to an inner product (dot product) of 2 vectors, matrix · matrix  $\Rightarrow$  scalar

Suppose we are free to do the transformation on these matrices  $\hat{X}'_1 = a\hat{X}_1 + b\hat{X}_2 + \dots$

In general, we choose

$$\hat{X}'_a = \sum_b L_{ab} \hat{X}_b$$

The structure constant is also changed by this transf.

$$[\hat{X}_a, \hat{X}_b] = i \sum_c f_{abc} \hat{X}_c$$

$$[\hat{X}'_a, \hat{X}'_b] = i \sum_c f'_{abc} \hat{X}'_c$$

$L_{ab}$

$$[\hat{X}'_a, \hat{X}'_b] = \left[ \sum_c L_{ac} \hat{X}_c, \sum_d L_{bd} \hat{X}_d \right]$$

$$= \sum_c \sum_d L_{ac} L_{bd} [\hat{X}_c, \hat{X}_d]$$

$$[\hat{X}'_a, \hat{X}'_b] = \left[ \sum_c L_{ac} \hat{X}_c, \sum_d L_{bd} \hat{X}_d \right]$$

$$= \sum_c \sum_d L_{ac} L_{bd} [\hat{X}_c, \hat{X}_d] \quad i \sum_e f_{cde} \hat{X}_e$$

$$= \sum_c \sum_d \sum_e L_{ac} L_{bd} i f_{cde} \hat{X}_e$$

We can use inverse  $\hat{X}'_a = \sum_b L_{ab} \hat{X}_b$ ,  $\hat{X}_a = \sum_b L_{ab}^{-1} \hat{X}'_b$

$$[\hat{X}'_a, \hat{X}'_b] = \sum_c \sum_d \sum_e L_{ac} L_{bd} i f_{cde} \sum_g L_{eg}^{-1} \hat{X}'_g$$

$$= i \sum_g \left( \sum_c \sum_d \sum_e L_{ac} L_{bd} L_{eg}^{-1} f_{cde} \right) \hat{X}'_g$$

$\equiv f'_{abg}$

$$f'_{abg} = \sum_c \sum_d \sum_e L_{ac} L_{bd} f_{cde} L_{eg}^{-1}$$

(This transf. is similar to the tensor

$f_{ab}^g \xrightarrow{\text{invert}}$   
 $f_{ab}^g \xrightarrow{\text{normal}}$

Since  $f_{abc}$  is basically the element of  $\hat{X}_a$  (in the adjoint rep)

$$\therefore -i f'_{abg} = \sum_c \sum_d \sum_e L_{ac} L_{bd} (-i f_{cde}) L_{eg}^{-1}$$

$$[\hat{X}'_a]_{bg} = \sum_c \sum_d \sum_e L_{ac} L_{bd} [\hat{X}_c]_{de} L_{eg}^{-1}$$

$$= \sum_c L_{ac} [L \hat{X}_c L^{-1}]_{bg}$$

$$\begin{aligned}
 [\hat{X}'_a]_{bg} &= \sum_c \sum_d \sum_e L_{ac} L_{bd} [\hat{X}_c]_{de} L_{eg}^{-1} \\
 &= \sum_c L_{ac} [L \hat{X}_c L^{-1}]_{bg}
 \end{aligned}$$

$$\hat{X}'_a = \sum_c L_{ac} (L \hat{X}_c L^{-1})$$

$\therefore$  The transformation can change the Trace by

$$\begin{aligned}
 \text{Tr}(\hat{X}'_a \hat{X}'_b) &= \text{Tr}\left(\sum_c L_{ac} (L \hat{X}_c L^{-1}) \sum_d L_{bd} (L \hat{X}_d L^{-1})\right) \\
 &= \sum_c \sum_d L_{ac} L_{bd} \text{Tr}(L \hat{X}_c L^{-1} L \hat{X}_d L^{-1})
 \end{aligned}$$

$$\text{Tr}(\hat{X}'_a \hat{X}'_b) = \sum_c \sum_d L_{ac} L_{bd} \text{Tr}(\hat{X}_c \hat{X}_d)$$

[matrix]<sub>ab</sub>
[matrix]<sub>cd</sub>

$$\begin{aligned}
 \text{Tr}(ABC) \\
 = \text{Tr}(CAB)
 \end{aligned}$$

$$M'_{ab} = \sum_c \sum_d L_{ac} M_{cd} [L^T]_{db}$$

$$M' = L M L^T \Rightarrow \begin{matrix} \text{Rotation} \\ \text{Diagonalization} \end{matrix}$$

$\Rightarrow$  We can always choose

$$\text{Tr}(\hat{X}_a \hat{X}_b) = \lambda_a \delta_{ab} = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \end{pmatrix}$$

★ If all value of  $\lambda_a$  are positive

⇒ We can always choose

$$\text{Tr}(\hat{X}_a \hat{X}_b) = \lambda_a \delta_{ab} = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \end{pmatrix}$$

★ If all value of  $\lambda_a$  are positive (true for groups in physics)

We can use  $L$  to rescale  $\lambda$

$$L = \begin{pmatrix} \frac{1}{\sqrt{\lambda_1}} & & 0 \\ & \frac{1}{\sqrt{\lambda_2}} & \\ 0 & & \ddots \end{pmatrix}, L^T = L$$

$$M' = \begin{pmatrix} \frac{1}{\sqrt{\lambda_1}} & & \\ & \ddots & \\ & & \frac{1}{\sqrt{\lambda_n}} \end{pmatrix} \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{\lambda_1}} & & \\ & \frac{1}{\sqrt{\lambda_2}} & \\ & & \ddots \end{pmatrix} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}$$

The most general form of  $\text{Tr}(\hat{X}_a \hat{X}_b)$  for adjoint rep:

$$\text{Tr}(\hat{X}_a \hat{X}_b) = \lambda \delta_{ab}$$

Example

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

$$\text{Tr}(\hat{X}_1 \hat{X}_1) = \text{Tr} \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \right) = \text{Tr} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= 2$$

$$\text{Tr}(\hat{X}_2 \hat{X}_2) = \text{Tr}(\hat{X}_3 \hat{X}_3) = 2 \quad \left\{ \begin{array}{l} \text{Tr}(\hat{X}_a \hat{X}_b) = 2 \delta_{ab} \\ \text{Tr}(\hat{X}_1 \hat{X}_2) = 0 \end{array} \right.$$

$$\text{Tr}(WW) = \dots$$

Let's Play with the exponential of matrix

The goal is to derive the Baker-Campbell-Hausdorff formula:

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{[\hat{A}, [\hat{A}, \hat{B}]]}{2!} + \frac{[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]]}{3!} + \dots$$

Proof

$$\text{Consider } F(t) = e^{t\hat{A}} \hat{B} e^{-t\hat{A}}$$

Let's consider the Taylor expansion of  $F(t)$

$$F(0) = e^0 \hat{B} e^0 = \hat{B}$$

$$F(t) = F(0) + t \cdot \left. \frac{dF(t)}{dt} \right|_{t=0} + \frac{t^2}{2!} \left. \frac{d^2 F(t)}{dt^2} \right|_{t=0} + \dots$$

$$\frac{dF(t)}{dt} = \frac{d}{dt} (e^{t\hat{A}} \hat{B} e^{-t\hat{A}})$$

$$= \left( \frac{d e^{t\hat{A}}}{dt} \right) \hat{B} e^{-t\hat{A}} + e^{t\hat{A}} \hat{B} \left( \frac{d e^{-t\hat{A}}}{dt} \right)$$

$$\frac{d e^{t\hat{A}}}{dt} = \frac{d}{dt} \left( \hat{1} + t\hat{A} + \frac{t^2}{2!} \hat{A}^2 + \dots \right) = \hat{A} + \frac{2t}{2!} \hat{A}^2 + \frac{3t^2}{3!} \hat{A}^3 + \dots$$

$$= \hat{A} + t\hat{A}^2 + \frac{t^2}{2!} \hat{A}^3 + \dots = \hat{A} \left( 1 + t\hat{A} + \frac{t^2}{2!} \hat{A}^2 + \dots \right)$$

$$= \hat{A} e^{t\hat{A}} = e^{t\hat{A}} \hat{A}$$

$$\frac{dF}{dt} = e^{t\hat{A}} \hat{A} \hat{B} e^{-t\hat{A}} + e^{t\hat{A}} \hat{B} (-\hat{A}) e^{-t\hat{A}} = e^{t\hat{A}} (\hat{A} \hat{B} - \hat{B} \hat{A}) e^{-t\hat{A}}$$

$$\frac{dF}{dt} = e^{t\hat{A}} \hat{A} \hat{B} e^{-t\hat{A}} + e^{t\hat{A}} \hat{B} (-\hat{A}) e^{-t\hat{A}} = e^{t\hat{A}} (\hat{A}\hat{B} - \hat{B}\hat{A}) e^{-t\hat{A}}$$

$$\left. \frac{dF}{dt} \right|_{t=0} = e^0 [\hat{A}, \hat{B}] e^0 = [\hat{A}, \hat{B}]$$

$$\begin{aligned} \frac{d^2 F}{dt^2} &= \left( \frac{d}{dt} e^{t\hat{A}} \right) [\hat{A}, \hat{B}] e^{-t\hat{A}} + e^{t\hat{A}} [\hat{A}, \hat{B}] \left( \frac{d}{dt} e^{-t\hat{A}} \right) \\ &= e^{t\hat{A}} \hat{A} [\hat{A}, \hat{B}] e^{-t\hat{A}} + e^{t\hat{A}} [\hat{A}, \hat{B}] (-\hat{A}) e^{-t\hat{A}} \\ &= e^{t\hat{A}} ([\hat{A}, [\hat{A}, \hat{B}]]) e^{-t\hat{A}} \end{aligned}$$

$$\left. \frac{d^2 F}{dt^2} \right|_{t=0} = [\hat{A}, [\hat{A}, \hat{B}]]$$

$$\therefore F(t) = F(0) + t \left. \frac{dF}{dt} \right|_{t=0} + \frac{t^2}{2!} \left. \frac{d^2 F}{dt^2} \right|_{t=0} + \dots$$

$$e^{t\hat{A}} \hat{B} e^{-t\hat{A}} = \hat{B} + t [\hat{A}, \hat{B}] + \frac{t^2}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \dots$$

$$t=1$$

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{[\hat{A}, [\hat{A}, \hat{B}]]}{2!} + \dots$$

#

## Simple Algebras & Simple Groups

Consider an algebra =  $\{\hat{X}_i, \hat{Y}_j\}$

contains 2 types of generators =  $\hat{X}$ 's,  $\hat{Y}$ 's

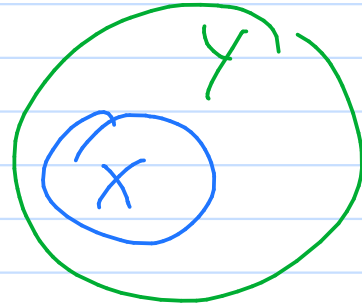


# Simple Algebras & Simple Groups

Consider an algebra  $= \{\hat{Y}_i\}$ ,  $\{\hat{X}_i\} \subset \{\hat{Y}_i\}$

contains a type of generators  $= \hat{X}$ 's

such that  $[\hat{X}_i, \hat{Y}_j] = \sum_k C_{jk} \hat{X}_k$



$\{\hat{X}_i\} =$  invariant subalgebra

Suppose  $h = e^{i\hat{X}}$ ,  $g = e^{i\hat{Y}}$   
 generated by invariant subgroup      any element

Consider

$$g^{-1} h g = e^{-i\hat{Y}} e^{i\hat{X}} e^{i\hat{Y}}$$

$$= e^{-i\hat{Y}} \left( \hat{1} + i\hat{X} + \frac{(i)^2 \hat{X}^2}{2!} + \dots \right) e^{i\hat{Y}}$$

$$= \hat{1} + i(e^{-i\hat{Y}} \hat{X} e^{i\hat{Y}}) + \frac{(i)^2}{2!} (e^{-i\hat{Y}} \hat{X} \hat{X} e^{i\hat{Y}}) + \dots$$

$e^{i\hat{Y}} e^{-i\hat{Y}}$

Define  $\hat{X}' = e^{-i\hat{Y}} \hat{X} e^{i\hat{Y}}$

$$= \hat{1} + i\hat{X}' + \frac{(i)^2}{2!} \hat{X}'^2 + \dots = e^{i\hat{X}'}$$

Consider  $\hat{X}' = e^{-i\hat{Y}} \hat{X} e^{i\hat{Y}}$        $\hat{A} = -i\hat{Y}, \hat{B} = \hat{X}$

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{[\hat{A}, [\hat{A}, \hat{B}]]}{2!} + \dots$$

Consider

$$\hat{X}' = e^{-i\hat{Y}} \hat{X} e^{i\hat{Y}}$$

$$\hat{A} = -i\hat{Y}, \hat{B} = \hat{X}$$

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{[\hat{A}, [\hat{A}, \hat{B}]]}{2!} + \dots$$

$$\hat{X}' = \sum c \hat{X}$$

$$\hat{X}' = \hat{X} - i [\hat{Y}, \hat{X}] + \frac{(-i)^2}{2!} [\hat{Y}, [\hat{Y}, \hat{X}]] + \dots$$

If  $\hat{X}$  is a generator in the invariant subalgebra

$$[\hat{Y}, \hat{X}] \in \{\hat{X}\} \Rightarrow \hat{X}' \in \{\hat{X}\}$$

$\therefore \hat{X}'$  is in the invariant subalgebra

$$\Rightarrow g^{-1} h g = e^{-i\hat{Y}} e^{i\hat{X}} e^{i\hat{Y}} = e^{i\hat{X}'} = h'$$

$h$  &  $h'$  are conjugate elements

$\therefore \{h = e^{i\hat{X}}\}$  is invariant subgroup

still  $h$   
is generated by  
invariant subalgebra

Ex 1). trivial subalgebra  $\Rightarrow$  the whole algebra  $\{\hat{Y}\}$

$\Rightarrow$  This is a subalgebra &  $[\hat{Y}_i, \hat{Y}_j] = \sum_k f_{ijk} \hat{Y}_k$

$\therefore$  The whole algebra is a trivial invariant  
 $\Rightarrow$  generate the whole group subalgebra.

2).  $\{\hat{O}\}$  is also trivial subalgebra  $\hat{O}|0\rangle = |0\rangle$

$[\hat{Y}, \hat{O}] = \hat{O} \checkmark \Rightarrow$  generate the trivial subgroup

$$\{e^{i\hat{O}} = \mathbb{1}\}$$

$\hookrightarrow$  always there

We are normally interested in an algebra that has no nontrivial

invariant subalgebra  $\Rightarrow$  Simple algebra

(Simple group)

not  $\{\hat{Y}\}$  &  $\{\hat{O}\}$

Notice that  $\Rightarrow$  The Abelian group  $\Rightarrow$  not simple group

$$gh = hg \Rightarrow \hat{X} = \begin{pmatrix} \cdot & & \\ & \cdot & \\ & & \cdot \end{pmatrix} = \text{diagonal}$$

$$e^{i\hat{X}_1} e^{i\hat{X}_2} = e^{i\hat{X}_2} e^{i\hat{X}_1} \Rightarrow [\hat{X}, \hat{X}] = 0$$

$$\Rightarrow U(1) \text{ group} = \{e^{i\theta}\}$$

## SU(2) group & algebra

$SU(2) \subset \underline{SL(2, \mathbb{C})}$  and  $SU(2)$  has a unitarity property

\*  $g \in SU(2)$ , where the general form

$$g = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \alpha, \beta \in \mathbb{C} \quad \& \quad \boxed{|\alpha|^2 + |\beta|^2 = 1}$$

$$\det g = \alpha \cdot \alpha^* - (-\beta^*)\beta = |\alpha|^2 + |\beta|^2 = 1 \Rightarrow \text{Special Linear}$$

$$\begin{aligned} g^\dagger g &= \begin{pmatrix} \alpha^* & -\beta \\ \beta^* & \alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & \alpha^* \beta - \beta \alpha^* \\ \alpha \beta^* - \alpha^* \beta & |\alpha|^2 + |\beta|^2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

1 constraint

$|\alpha|^2 + |\beta|^2 = \underbrace{\text{Re}(\alpha)^2 + \text{Im}(\alpha)^2 + \dots + \dots}_{=1}$

$\Rightarrow$  How many parameters?  $\alpha = \underline{\text{Re}(\alpha)} + i \underline{\text{Im}(\alpha)}$

$$\beta = \underline{\text{Re}(\beta)} + i \underline{\text{Im}(\beta)}$$

4 parameters + 1 constraint

$\Rightarrow$  3 parameters

Alternatively, we can start with  $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$

$\alpha, \beta, \gamma, \delta \in \mathbb{C} \Rightarrow 8$  parameters

Unitary condition  $g^\dagger g = \mathbb{1}_{2 \times 2} \Rightarrow 4$  constraints

$$\begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow$$

+  $\det g = 1 \Rightarrow 1$  constraint

$8 - 5 = 3$  parameters

Consider a generator  $\hat{X} \Rightarrow$  generates  $SU(2)$  element.

$g = e^{i\hat{X}} \Rightarrow$  Unitary condition  $\Rightarrow g^\dagger g = \mathbb{1}$

$$(e^{i\hat{X}})^\dagger (e^{i\hat{X}}) = e^{-i\hat{X}^\dagger} e^{i\hat{X}} = \mathbb{1} = e^{i\hat{0}}$$

$$e^{i(\hat{X} - \hat{X}^\dagger)} = e^{i\hat{0}} \Rightarrow \hat{X} - \hat{X}^\dagger = \hat{0}$$

$$\boxed{\hat{X} = \hat{X}^\dagger} \Rightarrow \text{Hermitian operators.}$$

Hermitian generators  $\Rightarrow$  generates unitary element.

How about  $\det = 1$ ?

$$\det(g) = \det(e^{i\hat{X}}) \Rightarrow \text{we use } \det(e^{\hat{M}}) = e^{\text{Tr}(\hat{M})}$$

Digress

$$\det(e^{\hat{M}}) = e^{\text{Tr}(\hat{M})}$$

$\det(e^{\hat{M}}) \Rightarrow$  diagonalization =

Digress

$$\det(e^{\hat{M}}) = e^{\text{Tr}(\hat{M})}$$

$\det(e^{\hat{M}}) \Rightarrow$  diagonalization = product of eigenvalues

$$\det(A) = \det(\overbrace{U^{-1}UA}^{\det(U^{-1})\det(U)\det A}) = \det(UAU^{-1}) = \prod_{i=1}^n \lambda_i$$

choose  $U$  such that  $UAU^{-1} = A_D$  (diagonalized.)

$$A_D = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix} \text{ where } \lambda_1, \dots, \lambda_n \text{ are eigenvalues of } A$$

$$\text{Tr}(A) = \text{Tr}(U^{-1}UA) = \text{Tr}(\underbrace{UAU^{-1}}_{A_D}) = \sum_{i=1}^n \lambda_i$$

$$A_D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \quad \Downarrow \quad \sum_i (ABC)_{ii} = \sum_i \sum_j \sum_k \overbrace{A_{ij} B_{jk} C_{ki}}^{\text{cyclic}} \\ = \sum_k \sum_i \sum_j \underbrace{C_{ki} A_{ij} B_{jk}}_{\text{cyclic}} \\ = \sum_k (CAB)_{kk}$$

We can justify the formula

$$\boxed{\det(e^{\hat{M}}) = e^{\text{Tr}(\hat{M})}} \quad , \quad \text{eigenvalue of } \hat{M} = m_i$$

$$e^{\text{Tr}(\hat{M})} = e^{\sum_{i=1}^n m_i} = e^{m_1 + m_2 + \dots} = (e^{m_1})(e^{m_2}) \dots = \prod_{i=1}^n e^{m_i}$$

Since the eigenvalue of  $e^{\hat{M}} = e^{m_i}$   $\Rightarrow \det(e^{\hat{M}})$

$$e^{\hat{M}_D} = e^{\begin{pmatrix} m_1 & & 0 \\ & \ddots & \\ 0 & & m_n \end{pmatrix}} = 1 + \begin{pmatrix} m_1 & & 0 \\ & \ddots & \\ 0 & & m_n \end{pmatrix} + \frac{1}{2!} \begin{pmatrix} m_1 & & 0 \\ & \ddots & \\ 0 & & m_n \end{pmatrix} \begin{pmatrix} m_1 & & 0 \\ & \ddots & \\ 0 & & m_n \end{pmatrix} + \dots \\ = \begin{pmatrix} 1 + m_1 + \frac{m_1^2}{2!} + \dots & & 0 \\ & \ddots & \\ 0 & & 1 + m_n + \frac{m_n^2}{2!} + \dots \end{pmatrix} = \begin{pmatrix} e^{m_1} & & 0 \\ & \ddots & \\ 0 & & e^{m_n} \end{pmatrix}$$

Let's get back to  $SU(2)$

How about  $\det = 1$ ?

$$\det(g) = \det(e^{i\hat{X}}) = e^{i\text{Tr}(\hat{X})} = 1$$

$$\therefore \boxed{\text{Tr}(\hat{X}) = 0}$$

→ The requirement of  $\hat{X}$  of  $SU(2)$

- 1). There are 3 of them (3 parameters)
- 2).  $\hat{X}^\dagger = \hat{X}$  (Hermitian)
- 3).  $\text{Tr}(\hat{X}) = 0$  (Traceless)

The most general form of Hermitian  $2 \times 2$  matrix

$$\hat{M} = \begin{pmatrix} a & b-ic \\ b+ic & d \end{pmatrix}$$

real number  $a^* = a, d^* = d$

$(b-ic)^* = b+ic$   
 $(b+ic)^* = b-ic$

Traceless requirement  $\Rightarrow d = -a$

$$\therefore \hat{M} = \begin{pmatrix} a & b-ic \\ b+ic & -a \end{pmatrix}$$

$$= a \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$a, b, c$  can be absorbed into parameters

$\therefore$  3 generators = 3 matrices

$$\therefore \hat{X}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \hat{X}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \hat{X}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\Rightarrow$  These are known as Pauli matrices!

$$\hat{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \hat{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \hat{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

They are the well-known generators of  $SU(2)$

(Although this is just 1 representation (2 dim))

Let's calculate the structure constant of  $SU(2)$  algebra

$\rightarrow$  we normally use

$$\left\{ \hat{X}_1 = \frac{\hat{\sigma}_1}{2}, \hat{X}_2 = \frac{\hat{\sigma}_2}{2}, \hat{X}_3 = \frac{\hat{\sigma}_3}{2} \right\}$$

$$\left[ e^{i\vec{\sigma} \cdot \vec{\theta}} = \cos\left(\frac{\theta}{2}\right) \mathbb{1} + i\vec{\sigma} \cdot \vec{\theta} \sin\left(\frac{\theta}{2}\right) \right]?$$

$\theta \rightarrow 4\pi$

$$1). [\hat{X}_1, \hat{X}_2] = \left[ \frac{\hat{\sigma}_1}{2}, \frac{\hat{\sigma}_2}{2} \right] = \frac{1}{4} \left\{ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$$

$$[\hat{X}_i, \hat{X}_j] = i \sum_k f_{ijk} \hat{X}_k = \frac{1}{4} \left( \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \right) = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

$$= i \frac{\hat{\sigma}_3}{2} = i \hat{X}_3 = \underbrace{i f_{123}}_{=1} \hat{X}_3 + \underbrace{i f_{121}}_{=0} \hat{X}_1 + \underbrace{i f_{122}}_{=0} \hat{X}_2$$

$$2). [\hat{X}_2, \hat{X}_3] = \dots = i \hat{X}_1 = \underbrace{i f_{231}}_{=1} \hat{X}_1 + \underbrace{i f_{232}}_{=0} \hat{X}_2 + \underbrace{i f_{233}}_{=0} \hat{X}_3$$

$$3). [\hat{X}_3, \hat{X}_1] = \dots = i \hat{X}_2 = \underbrace{i f_{312}}_{=1} \hat{X}_2 + \underbrace{i f_{311}}_{=0} \hat{X}_1 + \underbrace{i f_{313}}_{=0} \hat{X}_3$$

$\therefore$  We end up with  $f_{ijk} = \epsilon_{ijk} = \text{Levi-Civita symbol}$

SU(2) Algebra: 
$$[\hat{X}_i, \hat{X}_j] = i \sum_k \epsilon_{ijk} \hat{X}_k$$

This is very familiar to physicists.

1). Angular momentum (QM)

$$[\hat{L}_i, \hat{L}_j] = i \sum_k \epsilon_{ijk} \hat{L}_k \quad (\hbar)$$

2). Spin (QM)

$$[\hat{S}_i, \hat{S}_j] = i \sum_k \epsilon_{ijk} \hat{S}_k \quad (\hbar)$$

3). Simple Harmonic Oscillator.\*

---

The question is how do we generate a representation of this SU(2) algebra.?

$\Rightarrow$  We can start by using the lesson from QM

$\Rightarrow$  This problem turns out to be the construction of states from  $\hat{L}_i$ 's

•  $\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} \Rightarrow \{\hat{L}_1, \hat{L}_2, \hat{L}_3\}$

• In QM, we start from construct another operator  
= total angular momentum

$$\hat{L}^2 \equiv \hat{L}_1^2 + \hat{L}_2^2 + \hat{L}_3^2$$

$$\hat{C}_2 \equiv \hat{X}_1^2 + \hat{X}_2^2 + \hat{X}_3^2$$

$\rightarrow$  Casimir operator  
(2 = order of polynomial of  $\hat{X}$ )



- We then check  $[\hat{L}^2, \hat{L}_i] = 0, \forall i$

$$\Rightarrow [\hat{L}_2, \hat{X}_i] = 0, \forall i$$

The property of Casimir operator (invariant)

- In QM, we select the set of mutually commuting operators  $\Rightarrow$  then they share eigenstates

- $[\hat{A}, \hat{B}] = 0, \hat{A}|a, b\rangle = a|a, b\rangle, \hat{B}|a, b\rangle = b|a, b\rangle$

We choose  $\Rightarrow \{\hat{L}^2, \hat{L}_3\}$

(Can we choose  $\{\hat{L}^2, \hat{L}_1\}$  ? but  $[\hat{L}_3, \hat{L}_1] = -i\hat{L}_2 \neq 0$   
 $[\hat{L}^2, \hat{L}_3] = 0$   $[\hat{L}^2, \hat{L}_1] = 0$

Then the eigenstates are  $|j, m\rangle$

such that  $\hat{L}^2|j, m\rangle = j(j+1)|j, m\rangle$  ( $\hbar = 1$ )

$$\hat{L}_3|j, m\rangle = m|j, m\rangle$$

- Then, we use the set  $\{\hat{L}_2, \hat{X}_3\}$

$$\hat{L}_2|j, m\rangle = j(j+1)|j, m\rangle$$

$$\hat{X}_3|j, m\rangle = m|j, m\rangle$$

- We then construct the raising/lowering operator from

$$\hat{L}_1 \& \hat{L}_2, \quad \hat{L}_{\pm} = \frac{1}{\sqrt{2}}(\hat{L}_1 \pm i\hat{L}_2)$$

We found that

$$[\hat{L}_3, \hat{L}_{\pm}] = \pm \hat{L}_{\pm}$$

- We then construct the raising/lowering operator from

$$\hat{L}_1 \& \hat{L}_2, \quad \hat{L}_{\pm} = \frac{1}{\sqrt{2}}(\hat{L}_1 \pm i\hat{L}_2)$$

We found that

$$[\hat{L}_3, \hat{L}_{\pm}] = \pm \hat{L}_{\pm}$$

$$\begin{aligned} [\hat{L}_3, \frac{1}{\sqrt{2}}(\hat{L}_1 \pm i\hat{L}_2)] &= \frac{1}{\sqrt{2}}[\hat{L}_3, \hat{L}_1] \pm \frac{i}{\sqrt{2}}[\hat{L}_3, \hat{L}_2] \\ &= \frac{1}{\sqrt{2}}i\hat{L}_2 \pm \frac{i}{\sqrt{2}}(-i\hat{L}_1) \quad (\pm)^2 = 1 \\ &= \frac{1}{\sqrt{2}}i\hat{L}_2 \pm \frac{1}{\sqrt{2}}\hat{L}_1 = \pm \frac{1}{\sqrt{2}}(\hat{L}_1 \pm i\hat{L}_2) \\ &= \pm \hat{L}_{\pm} \end{aligned}$$

Also  $[\hat{L}_+, \hat{L}_-] = \hat{L}_3$

$$\begin{aligned} [\hat{L}_+, \hat{L}_-] &= \frac{1}{2}[\hat{L}_1 + i\hat{L}_2, \hat{L}_1 - i\hat{L}_2] \\ &= \frac{1}{2}(-i[\hat{L}_1, \hat{L}_2] + i[\hat{L}_2, \hat{L}_1]) \\ &= \hat{L}_3 \end{aligned}$$

- We have then the raising/lowering operator

$$\hat{X}_{\pm} = \frac{1}{\sqrt{2}}(\hat{X}_1 \pm i\hat{X}_2)$$

$$[\hat{X}_3, \hat{X}_{\pm}] = \pm \hat{X}_{\pm} \& [\hat{X}_+, \hat{X}_-] = \hat{X}_3$$

(Sidenote for SHM,  $\{\hat{a}_+, \hat{a}_-, \hat{N}\}$  which form the same algebra as above)

- In OM, we argue that there is a highest possible value of  $m$  for a value of  $j$

$$\hat{L}^2 = \hat{L}_1^2 + \hat{L}_2^2 + \hat{L}_3^2 \Rightarrow \text{eigenvalue of } m \text{ is bounded from } j$$

$\hat{L}^2 \xrightarrow{j} j(j+1)$ 
 $\hat{L}_3^2 \xrightarrow{m} m$

There exists the highest value of  $m = m_h$

- The interpretation of  $\hat{L}_{\pm}$  is the raising/lowering of value of  $m$

$$\begin{aligned}
 \hat{L}_3(\hat{L}_+ |j, m\rangle) &= (\hat{L}_3 \hat{L}_+) |j, m\rangle \quad \xrightarrow{[\hat{L}_3, \hat{L}_+] = +\hat{L}_+} \hat{L}_3 \hat{L}_+ - \hat{L}_+ \hat{L}_3 \\
 &= ([\hat{L}_3, \hat{L}_+] + \hat{L}_+ \hat{L}_3) |j, m\rangle \\
 &= (+\hat{L}_+ + \hat{L}_+ \hat{L}_3) |j, m\rangle \\
 &= (\hat{L}_+ + \hat{L}_+ m) |j, m\rangle \quad \text{--- } |j, m_h\rangle \\
 &= (m+1) \hat{L}_+ |j, m\rangle \quad \text{--- } |j, m+1\rangle \\
 \hat{L}_+ |j, m\rangle &\propto |j, m+1\rangle \quad \xrightarrow{\hat{L}_+} \hat{L}_+ |j, m+1\rangle \quad \dots \quad \hat{L}_+ |j, m_h\rangle = 0
 \end{aligned}$$

$\hat{L}_+ |j, m\rangle \propto |j, m+1\rangle$ 
 $\hat{L}_- |j, m\rangle \propto |j, m-1\rangle$

$\therefore$  The definition of highest possible state  $|j, m_h\rangle$

$$\boxed{\hat{L}_+ |j, m_h\rangle = 0}$$

- We have the highest possible state of  $|j, m\rangle \equiv |j, m_h\rangle$  such that  $\hat{L}_+ |j, m_h\rangle = 0$

• In order to check the value  $m_h$ , we write

$$\hat{L}^2 = \hat{L}_3^2 + \hat{L}_1^2 + \hat{L}_2^2$$

$$\hat{L}_- \hat{L}_+ = \cancel{\frac{1}{2}} (\hat{L}_1 - i\hat{L}_2)(\hat{L}_1 + i\hat{L}_2) \quad \leftarrow \hat{L}_1^2 + \hat{L}_2^2 = \cancel{2} \hat{L}_- \hat{L}_+ + \hat{L}_3^2$$

$$= \cancel{\frac{1}{2}} \left( \hat{L}_1^2 - \underbrace{i\hat{L}_2\hat{L}_1 + i\hat{L}_1\hat{L}_2}_{+i[\hat{L}_1, \hat{L}_2]} + \hat{L}_2^2 \right) = \cancel{\frac{1}{2}} (\hat{L}_1^2 + \hat{L}_2^2 - \hat{L}_3^2)$$

$$\boxed{\hat{L}^2 = \hat{L}_3^2 + \cancel{2} \hat{L}_- \hat{L}_+ + \hat{L}_3^2}$$

$$\hat{L}_+ |j, m_h\rangle = 0$$

$$\hat{L}^2 |j, m_h\rangle = (\hat{L}_3^2 + \cancel{2} \hat{L}_- \hat{L}_+ + \hat{L}_3^2) |j, m_h\rangle$$

$$= (m_h^2 + 0 + m_h) |j, m_h\rangle$$

$$= j(j+1) |j, m_h\rangle$$

$$\therefore m_h(m_h+1) = j(j+1) \Rightarrow \boxed{m_h = j} \quad (m_h = \cancel{j-1})$$

$\therefore$  The highest possible state  $\underbrace{|j, j\rangle}_{|j, m_h\rangle}$  is determined by the  $\hat{L}^2$  operator.

• In the algebra case, we have the highest possible state  $= |j, j\rangle$  where the value of  $j$  is determined by the Casimir operator  $\hat{C}_2$

- Similarly, the same argument can be applied to the lowest value of  $m$ ,  $\hat{L}^2 = \hat{L}_1^2 + \hat{L}_2^2 + \hat{L}_3^2$

\* the negative value of  $m$   $\hookrightarrow (-\dots)^2 = \dots^2$

must be bound in the same fashion.

$$\hat{L}_- |j, m_\ell\rangle = 0$$

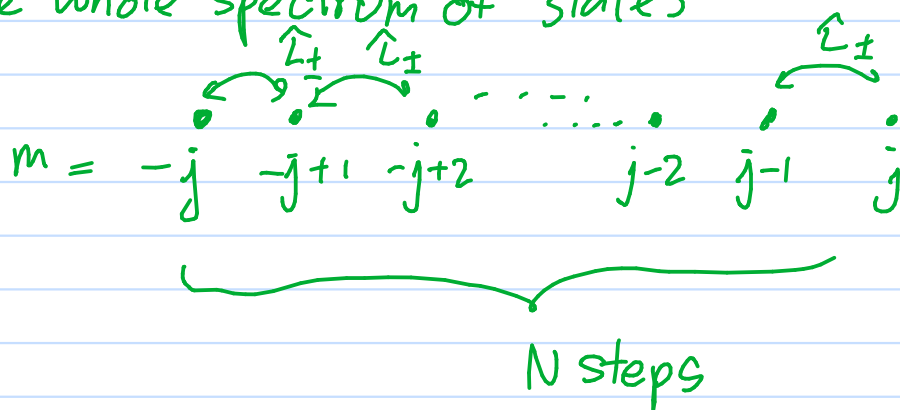
no state lower than this

$\downarrow$   $|j, m_\ell\rangle$

$$\hat{L}^2 = \hat{L}_z^2 + \cancel{\hat{L}_+ \hat{L}_-} - \hat{L}_z$$

$$j(j+1) = m_\ell(m_\ell - 1) \Rightarrow m_\ell = -j$$

$\Rightarrow$  The whole spectrum of states



$$-j + N = j \Rightarrow N = 2j \Rightarrow j = \frac{N}{2}$$

$\therefore$  The whole set of states are characterized by  $j = \frac{N}{2}$

$N=0$  :  $|0,0\rangle$  Singlet  $j=0$

$N=1$  :  $|\frac{1}{2}, -\frac{1}{2}\rangle \quad |\frac{1}{2}, \frac{1}{2}\rangle$  Doublet  $j=\frac{1}{2}$

∴ The whole set of states are characterized by  $j = \frac{N}{2}$

$N=0$  :  $|0,0\rangle$  Singlet  $j=0$

$N=1$  :  $|\frac{1}{2}, -\frac{1}{2}\rangle$   $|\frac{1}{2}, \frac{1}{2}\rangle$  Doublet  $j = \frac{1}{2}$

$N=2$  :  $|1, -1\rangle$   $|1, 0\rangle$   $|1, 1\rangle$  Triplet  $j=1$

• We have the whole spectrum characterized by

$N=0$  :  $|0,0\rangle$   
 $N=1$  :  $|\frac{1}{2}, -\frac{1}{2}\rangle$   $|\frac{1}{2}, \frac{1}{2}\rangle$   $\Rightarrow \hat{S}_i = \frac{\hat{\sigma}_i}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \dots$   
 $N=2$  :  $|1, -1\rangle$   $|1, 0\rangle$   $|1, 1\rangle$  2D representation  
 representation

∴ The representation of  $SU(2)$  algebra can be found by finding the eigenstates (spectrum) of  $\{\hat{X}_3, \hat{C}_2\}$

In  $SU(2)$ , we normally write  $|>$  as a column vector

$$\Rightarrow j = \frac{1}{2} \Rightarrow \text{states} = \left\{ \left| \frac{1}{2}, \frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \right\}$$

$$= \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

Since we know that  $\hat{L}_3 \left| \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{1}{2} \left| \frac{1}{2}, \frac{1}{2} \right\rangle$

$$\hat{L}_3 \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{2} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\therefore \hat{L}_3 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Matrix element  $\langle \frac{1}{2}, \frac{1}{2} | \hat{L}_3 | \frac{1}{2}, \frac{1}{2} \rangle = \frac{1}{2} \langle \frac{1}{2}, \frac{1}{2} | \frac{1}{2}, \frac{1}{2} \rangle = \frac{1}{2}$

$$\langle \frac{1}{2}, -\frac{1}{2} | \hat{L}_3 | \frac{1}{2}, -\frac{1}{2} \rangle = -\frac{1}{2} \langle \frac{1}{2}, -\frac{1}{2} | \frac{1}{2}, -\frac{1}{2} \rangle = -\frac{1}{2}$$

$$\langle \frac{1}{2}, -\frac{1}{2} | \hat{L}_3 | \frac{1}{2}, \frac{1}{2} \rangle = \langle \frac{1}{2}, \frac{1}{2} | \hat{L}_3 | \frac{1}{2}, -\frac{1}{2} \rangle = 0$$

$$\hat{L}_3 = \begin{pmatrix} \langle \frac{1}{2}, \frac{1}{2} | \hat{L}_3 | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, \frac{1}{2} | \hat{L}_3 | \frac{1}{2}, -\frac{1}{2} \rangle \\ \langle \frac{1}{2}, -\frac{1}{2} | \hat{L}_3 | \frac{1}{2}, \frac{1}{2} \rangle & \langle \frac{1}{2}, -\frac{1}{2} | \hat{L}_3 | \frac{1}{2}, -\frac{1}{2} \rangle \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$
$$= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{2} \hat{\sigma}_3$$

$$\hat{L}_1 \& \hat{L}_2 = ? \quad \hat{L}_+ \left| \frac{1}{2}, \frac{1}{2} \right\rangle = 0, \quad \hat{L}_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

We forgot to mention the normalization of  $\hat{L}_+$ ,  $\hat{L}_-$  states

Digress

$$\hat{L}_1 \& \hat{L}_2 = ? \quad \hat{L}_+ | \frac{1}{2}, \frac{1}{2} \rangle = 0, \quad \hat{L}_+ | \frac{1}{2}, -\frac{1}{2} \rangle \propto | \frac{1}{2}, \frac{1}{2} \rangle$$

We forgot to mention the normalization of  $\hat{L}_+, \hat{L}_-$  states

Digress We know that  $|j, m\rangle \rightarrow \hat{L}_+ |j, m\rangle \propto |j, m+1\rangle$

We can find the normalisation factor here

Recall that  $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$

$$\hat{L}_- \hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hat{L}_z$$

We assume that  $\hat{L}_+ |j, m\rangle = C_{j,m}^+ |j, m+1\rangle$  → normalised.

$$\therefore \langle j, m | \hat{L}_+^\dagger \hat{L}_+ | j, m \rangle = \langle j, m | \hat{L}_- \hat{L}_+ | j, m \rangle$$

$$\begin{aligned} \hat{L}_+^\dagger &= (\hat{L}_1 + i\hat{L}_2)^\dagger \\ &= \hat{L}_1 - i\hat{L}_2 \\ &= \hat{L}_- \end{aligned}$$

$$= \langle j, m+1 | (C_{j,m}^+)^\dagger C_{j,m}^+ | j, m+1 \rangle$$

$$= |C_{j,m}^+|^2$$

$$\begin{aligned} \hat{L}^2 |j, m\rangle &= j(j+1) |j, m\rangle \\ \hat{L}_z |j, m\rangle &= m |j, m\rangle \end{aligned}$$

$$|C_{j,m}^+|^2 = \langle j, m | (\hat{L}^2 - \hat{L}_z^2 - \hat{L}_z) | j, m \rangle$$

$$= \frac{1}{2} (j(j+1) - m^2 - m) \langle j, m | j, m \rangle$$

$$\therefore |C_{j,m}^+| = \frac{1}{\sqrt{2}} \sqrt{j(j+1) - m(m+1)}$$

Similarly we can calculate  $\hat{L}_- |j, m\rangle = C_{j,m}^- |j, m-1\rangle$

$$|C_{j,m}^-| = \sqrt{j(j+1) + m(m+1)}$$



Let's get back to our calculation

$$\hat{L}_1 \& \hat{L}_2 = ? \quad \hat{L}_+ | \frac{1}{2}, \frac{1}{2} \rangle = 0, \quad \hat{L}_+ | \frac{1}{2}, -\frac{1}{2} \rangle = C_{\frac{1}{2}, -\frac{1}{2}}^+ | \frac{1}{2}, \frac{1}{2} \rangle$$

$$C_{\frac{1}{2}, -\frac{1}{2}}^+ = \sqrt{\frac{1}{2}(\frac{1}{2}+1) - (-\frac{1}{2})(-\frac{1}{2}+1)} = \sqrt{\frac{1}{2} \cdot \frac{3}{2} + \frac{1}{2} \cdot \frac{1}{2}} = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$$

The matrix element for  $\hat{L}_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \& \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Similarly  $\hat{L}_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

$$\therefore \hat{L}_1 = \frac{\hat{L}_+ + \hat{L}_-}{2} = \frac{(\hat{L}_1 + i\hat{L}_2) + (\hat{L}_1 - i\hat{L}_2)}{2} = \frac{2\hat{L}_1}{2} = \hat{L}_1$$

$$\hat{L}_1 = \frac{1}{2} \left\{ \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} = \frac{1}{2} \hat{\sigma}_1$$

$$\& \hat{L}_2 = \frac{\hat{L}_+ - \hat{L}_-}{2i} = \frac{(\hat{L}_1 + i\hat{L}_2) - (\hat{L}_1 - i\hat{L}_2)}{2i} = \frac{2i\hat{L}_2}{2i} = \hat{L}_2$$
$$= \begin{pmatrix} 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix} = \frac{1}{2} \hat{\sigma}_2$$

$\therefore$  The usual Pauli matrix (2D) representation of  $SU(2)$  comes from a spin  $\frac{1}{2}$  states system.

$\Rightarrow$  Fundamental representation of  $SU(2)$

Let's consider a system of spin-1 particle

$$\text{states} = \{ |1,1\rangle, |1,0\rangle, |1,-1\rangle \}$$

$\Rightarrow$  We can find  $\hat{L}_3$  in matrix form (3x3 matrix)

$$|1,1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1,0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1,-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Exercise Use the previous method to construct

$$\hat{L}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\hat{L}_+ = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}, \quad \hat{L}_- = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

$$\hat{L}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \hat{L}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

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## Roots and Weights

We want to describe a more general Lie group from the point of view of our simplest group ( $SU(2)$ )

## Roots and Weights

We want to describe a more general Lie group from the point of view of our simplest group ( $SU(2)$ )

Suppose that there are  $n$  generators in a Lie group.

$\Rightarrow$  There are  $l$  generators that are mutually commuting

$\hat{H}_i, i=1, \dots, l \Rightarrow$  mutually commuting generators  
\* which are also Hermitian.

$\hat{E}_j, j=1, \dots, n-l \Rightarrow$  The rest of generators

$$\hat{H}_i^\dagger = \hat{H}_i \quad \& \quad [\hat{H}_i, \hat{H}_j] = 0, \quad \forall i, j$$

Hermitian

mutually commuting

$\rightarrow$  Cartan generator / The group that generated by  $\{\hat{H}_i\} \Rightarrow$  Cartan subgroup.

\* The number of  $\hat{H}_i = n = \text{rank of Lie group}$

As a consequence  $\Rightarrow$  There are shared eigenstates of  $\hat{H}_i$ .

$|\vec{\mu}, \kappa, D\rangle$  such that

$$\hat{H}_i |\vec{\mu}, \kappa, D\rangle = \mu_i |\vec{\mu}, \kappa, D\rangle, \quad i=1, \dots, n$$

vector of eigenvalue

$$\vec{\mu} = (\mu_1, \mu_2, \dots, \mu_n)$$

eigenvalue of  $\hat{H}_1, \dots$

$D = \text{representation}, \quad \kappa = \text{remaining label} \quad \left. \vphantom{\begin{matrix} D \\ \kappa \end{matrix}} \right\} \Rightarrow \text{Casimir operators}$

$$\underline{SU(2)} = \{\hat{L}_1, \hat{L}_2, \hat{L}_3\}$$

Cartan generator  $\Rightarrow$  only 1 generator  $= \{\hat{L}_3\}$   
 choose  $\hat{L}_3$

$$\text{rank} = 1$$

$\Rightarrow$  The eigenvectors of Cartan generator

$$|j, m\rangle \Rightarrow \hat{L}_3 |j, m\rangle = m |j, m\rangle$$

$m$  is an eigenvalue of  $\hat{L}_3$  ( $\vec{\mu} = m$  &  $x, D = j$ )

More name to remember!

The eigenvalues  $\mu_i$  are called weights

&  $\vec{\mu}$  is called weight vector.

$\therefore$  In case of  $SU(2)$ ,  $m$  is the weight

We consider the adjoint representation.

$$[\hat{X}_a]_{bc} = -if_{abc}, \quad a, b, c = 1, \dots, m$$

#generators

$\therefore$  The dimension of the matrix = The number of generators

We can define the state associated with  $\hat{X}_a$  by

writing  $|\hat{X}_a\rangle \Rightarrow$  basis of vector space

$$\hat{X}_a = \left( \underbrace{\quad}_m \right) \Bigg\}_m, \quad |\hat{X}_a\rangle = \underbrace{\{|\hat{X}_1\rangle, \dots, |\hat{X}_m\rangle\}}_m$$

Since we have the linear operator

$$\alpha \hat{X}_a + \beta \hat{X}_b \Rightarrow \alpha |\hat{X}_a\rangle + \beta |\hat{X}_b\rangle = |\alpha \hat{X}_a + \beta \hat{X}_b\rangle$$

In the adjoint rep.  $\text{Tr}(\hat{X}_a \hat{X}_b) = \lambda \delta_{ab}$

$\therefore$  we define the dot (inner) product via

$$\langle \hat{X}_a | \hat{X}_b \rangle \equiv \frac{1}{\lambda} \text{Tr}(\hat{X}_a^\dagger \hat{X}_b) = \delta_{ab}$$

*complex*

we can calculate

$$\begin{aligned} \hat{X}_a | \hat{X}_b \rangle &= \sum_c | \hat{X}_c \rangle \langle \hat{X}_c | \hat{X}_a | \hat{X}_b \rangle \\ &= \sum_c i f_{abc} | \hat{X}_c \rangle \\ &= i \sum_c f_{abc} \hat{X}_c | \hat{X}_b \rangle \end{aligned}$$

$[\hat{X}_a]_{cb} = -i f_{acb} = i f_{abc}$

linear operator

$i f_{abc} \hat{X}_c = [\hat{X}_a, \hat{X}_b]$

$\hat{X}_a | \hat{X}_b \rangle = | [\hat{X}_a, \hat{X}_b] \rangle$

In adjoint rep.

For the Cartan subalgebra, in adjoint representation

we can calculate the state as follow

$$\Rightarrow \hat{H}_i | \hat{H}_j \rangle = | [\hat{H}_i, \hat{H}_j] \rangle = | 0 \rangle = 0$$

states that corresponds to  $\hat{H}_j$

More name to remember  $\Rightarrow$  The weight in adjoint representation is called root

$\therefore$  From  $\hat{H}_i | \hat{H}_j \rangle = 0$ ,  $| \hat{H}_j \rangle$  has zero weight (root)

eigenvalue = 0

How about the states correspond to the rest?

$$\hat{E}_i \rightarrow |\hat{E}_i\rangle$$

$\Rightarrow$  They will have non zero root (non-Cartan)

$\therefore$  We write

$$\hat{H}_i |\hat{E}_j\rangle = \alpha_i |\hat{E}_j\rangle$$

eigenvalue of  $|\hat{E}_j\rangle$  (weight vector)  $\Rightarrow$  root vector  $\rightarrow |\hat{E}_{\vec{\alpha}}\rangle$  eigenvalue as a label

$$\hat{H}_i |\hat{E}_{\vec{\alpha}}\rangle = |[\hat{H}_i, \hat{E}_{\vec{\alpha}}]\rangle = |\alpha_i \hat{E}_{\vec{\alpha}}\rangle$$

$$\therefore [\hat{H}_i, \hat{E}_{\vec{\alpha}}] = \alpha_i \hat{E}_{\vec{\alpha}} \quad + \text{ Hermitian}$$

$$[\hat{E}_{\vec{\alpha}}^+, \underbrace{\hat{H}_i}_{\hat{H}_i}] = \alpha_i \hat{E}_{\vec{\alpha}}^+$$

$$\hat{E}_{\vec{\alpha}}^+ = \hat{E}_{-\vec{\alpha}}$$

$$[\hat{H}_i, \hat{E}_{\vec{\alpha}}^+] = -\alpha_i \hat{E}_{\vec{\alpha}}^+$$

This is similar to  $[\hat{L}_3, \hat{L}_+] = \hat{L}_+$

$$\& [\hat{L}_3, \hat{L}_-] = -\hat{L}_-$$

Cartan  $\hat{H}$  generators

Linear combi of  $\hat{L}_1$  &  $\hat{L}_2$

$$\text{In } \text{SU}(2), \quad \hat{E}_{\vec{\alpha}} = \hat{L}_+, \quad \hat{E}_{\vec{\alpha}}^+ = \hat{E}_{-\vec{\alpha}} = \hat{L}_-, \quad \vec{\alpha} = 1$$

$|\hat{L}_3\rangle$  has zero root,  $|\hat{L}_+\rangle$  has root=1,  $|\hat{L}_-\rangle$  has root=-1

This is similar to  $[\hat{L}_3, \hat{L}_+] = \hat{L}_+$  &  $[\hat{L}_3, \hat{L}_-] = -\hat{L}_-$

Cartan  $\hat{H}$  generators

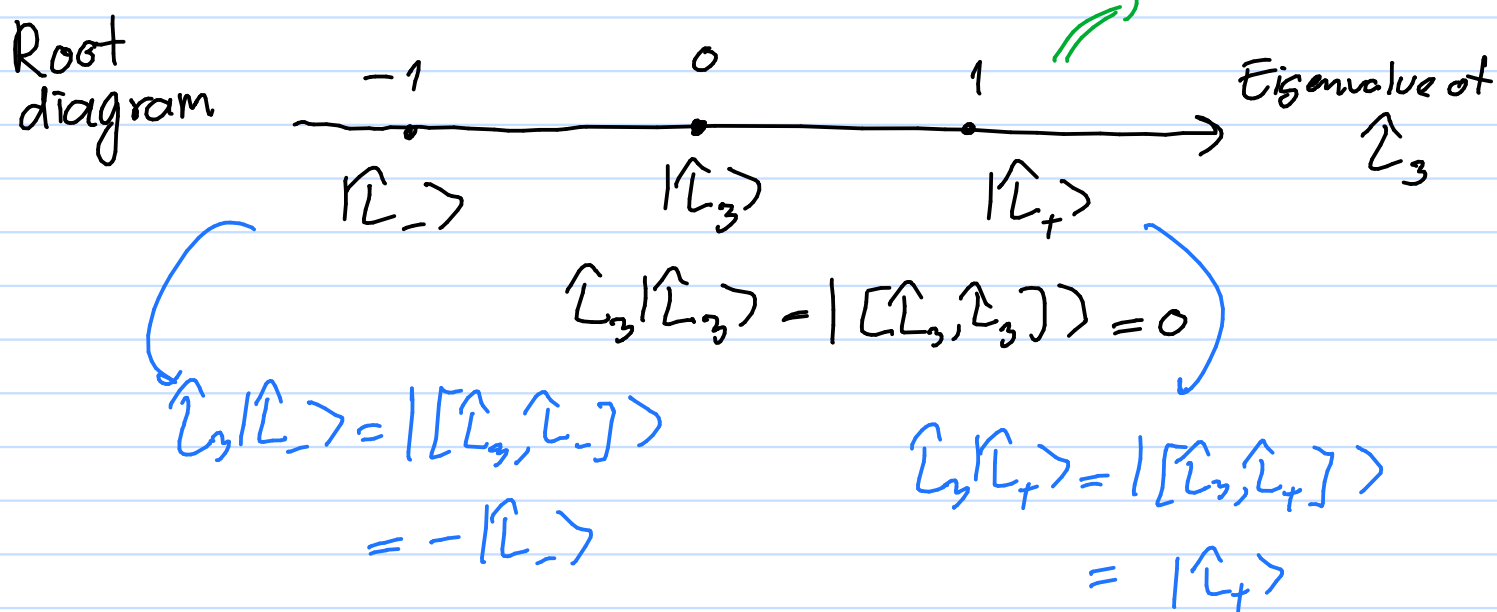
Linear combi of  $\hat{L}_1$  &  $\hat{L}_2$

In  $SU(2)$ ,  $\hat{E}_{\vec{\alpha}} = \hat{L}_+$ ,  $\hat{E}_{-\vec{\alpha}} = \hat{L}_-$ ,  $\vec{\alpha} = 1$

$|\hat{L}_3\rangle$  has zero root,  $|\hat{L}_+\rangle$  has root = 1

,  $|\hat{L}_-\rangle$  has root = -1

$\therefore$  We can also draw familiar picture Spin 1 system



Now, We have seen similarity between

$$\{\hat{H}_i, \hat{E}_{\vec{\alpha}}, \hat{E}_{-\vec{\alpha}}\} \quad \& \quad \{\hat{L}_3, \hat{L}_+, \hat{L}_-\}$$

We expect to see the raising/lowering duty for  $\hat{E}_{\vec{\alpha}}$

In a representation  $D$ , a state  $|\vec{\mu}, D\rangle$  can be raised/lowered by  $\hat{E}_{\vec{\alpha}}$

Now, We have seen similarity between

$$\{\hat{H}_i, \hat{E}_{\vec{\alpha}}, \hat{E}_{-\vec{\alpha}}\} \quad \& \quad \{\hat{L}_3, \hat{L}_+, \hat{L}_-\}$$

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In a representation  $D$ , a state  $|\vec{\mu}, D\rangle$  can be raised/lowered by  $\hat{E}_{\vec{\alpha}}$

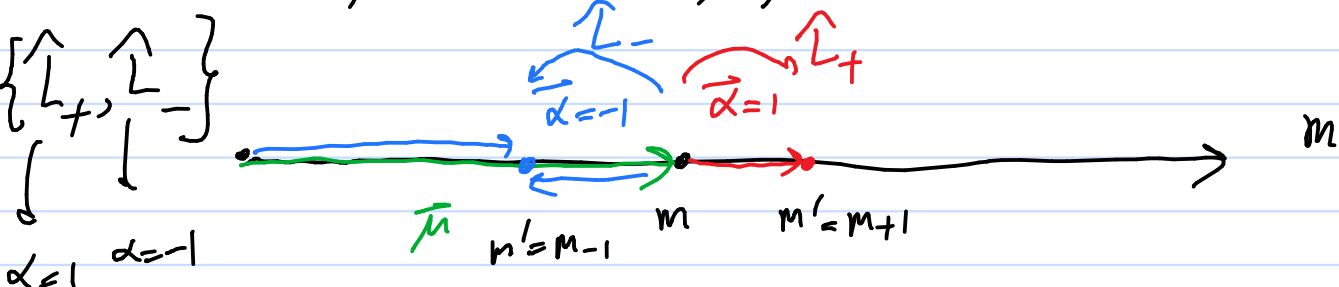
$$\begin{aligned} \hat{H}_i (\hat{E}_{\vec{\alpha}} |\vec{\mu}, D\rangle) &= \hat{H}_i \hat{E}_{\vec{\alpha}} |\vec{\mu}, D\rangle \\ &= (\underbrace{\hat{H}_i \hat{E}_{\vec{\alpha}} - \hat{E}_{\vec{\alpha}} \hat{H}_i}_{[\hat{H}_i, \hat{E}_{\vec{\alpha}}] = \alpha_i \hat{E}_{\vec{\alpha}}} + \hat{E}_{\vec{\alpha}} \hat{H}_i) |\vec{\mu}, D\rangle \\ &= (\alpha_i \hat{E}_{\vec{\alpha}} + \underbrace{\mu_i}_{\text{eigenvalue of } \hat{H}_i} \hat{E}_{\vec{\alpha}}) |\vec{\mu}, D\rangle \\ &= (\underbrace{\mu_i + \alpha_i}_{\text{eigenvalue of } \hat{H}_i}) \hat{E}_{\vec{\alpha}} |\vec{\mu}, D\rangle \end{aligned}$$

*Annotations:*  
 -  $\mu_i |\vec{\mu}, D\rangle$  (blue arrow from  $\mu_i$  to  $|\vec{\mu}, D\rangle$ )  
 -  $[\hat{H}_i, \hat{E}_{\vec{\alpha}}] = \alpha_i \hat{E}_{\vec{\alpha}}$  (blue bracket)  
 -  $\mu_i$  is the eigenvalue of  $\hat{H}_i$  (green arrow from  $\mu_i$  to  $\hat{H}_i$ )  
 -  $\mu_i + \alpha_i$  is the eigenvalue of  $\hat{H}_i$  for the new state (green arrow from  $\mu_i + \alpha_i$  to  $\hat{H}_i$ )

$$\begin{aligned} \hat{E}_{\vec{\alpha}} |\vec{\mu}, D\rangle &\propto |\vec{\mu} + \vec{\alpha}, D\rangle \\ \hat{E}_{-\vec{\alpha}} |\vec{\mu}, D\rangle &\propto |\vec{\mu} - \vec{\alpha}, D\rangle \end{aligned}$$

*Diagram:* A vector diagram showing  $\vec{\mu}$  (green arrow),  $\vec{\alpha}$  (blue arrow), and  $\vec{\mu} + \vec{\alpha}$  (blue arrow).  $\vec{\mu}$  is labeled "weight".  $\vec{\alpha}$  is labeled "root".

In  $SU(2)$ , rank = 1  $\Rightarrow \vec{\mu} \propto \vec{\alpha}$  are 1D vector (number)





In the  $SU(2)$  case, we have the commutator

$$[\hat{L}_+, \hat{L}_-] = \hat{L}_3$$

How about  $[\hat{E}_\alpha, \hat{E}_{-\alpha}] = ?$

We can guess that  $[\hat{E}_\alpha, \hat{E}_{-\alpha}]$  should be a linear combination of  $\hat{H}_i$

Suppose this is true

$$[\hat{E}_\alpha, \hat{E}_{-\alpha}] = \sum_{i=1}^n \beta_i \hat{H}_i \equiv \vec{\beta} \cdot \hat{H}$$

In adjoint representation

$$|[\hat{E}_\alpha, \hat{E}_{-\alpha}]\rangle = \hat{E}_\alpha |\hat{E}_{-\alpha}\rangle = \sum_{i=1}^n \beta_i |\hat{H}_i\rangle$$

$$\sum_{i=1}^n \langle \hat{H}_j | \beta_i | \hat{H}_i \rangle = \langle \hat{H}_j | \hat{E}_\alpha | \hat{E}_{-\alpha} \rangle$$

$$\langle \hat{H}_j | \hat{H}_i \rangle = \delta_{ij}$$

$$\langle \hat{H}_j | [\hat{E}_\alpha, \hat{E}_{-\alpha}] \rangle = \frac{1}{\lambda} \text{Tr}(\hat{H}_j^\dagger [\hat{E}_\alpha, \hat{E}_{-\alpha}])$$

$$\sum_{i=1}^n \beta_i \delta_{ij} = \beta_j = \frac{1}{\lambda} \text{Tr}(\hat{H}_j^\dagger [\hat{E}_\alpha, \hat{E}_{-\alpha}])$$

$$\hat{H}_j \hat{E}_\alpha \hat{E}_{-\alpha} - \hat{H}_j \hat{E}_{-\alpha} \hat{E}_\alpha$$

$$\hat{E}_{-\alpha} \hat{H}_j \hat{E}_\alpha - \hat{E}_{-\alpha} \hat{E}_\alpha \hat{H}_j$$

$$= \frac{1}{\lambda} \text{Tr}(\hat{E}_{-\alpha} [\hat{H}_j, \hat{E}_\alpha])$$

$$= \alpha_j \frac{1}{\lambda} \text{Tr}(\hat{E}_{-\alpha} \hat{E}_\alpha) = \alpha_j \frac{1}{\lambda} \text{Tr}(\hat{E}_\alpha^\dagger \hat{E}_\alpha)$$

$$\beta_j = \alpha_j$$

$$\therefore [\hat{E}_{\vec{\alpha}}, \hat{E}_{-\vec{\alpha}}] = \vec{\alpha} \cdot \hat{H}$$

In summary, Lie algebra in this basis

$$[\hat{H}_i, \hat{H}_j] = 0, \quad [\hat{H}_i, \hat{E}_{\vec{\alpha}}] = \alpha_i \hat{E}_{\vec{\alpha}}$$

$$[\hat{H}_i, \hat{E}_{-\vec{\alpha}}] = -\alpha_i \hat{E}_{-\vec{\alpha}}, \quad [\hat{E}_{\vec{\alpha}}, \hat{E}_{-\vec{\alpha}}] = \vec{\alpha} \cdot \hat{H}$$

(We call this basis - Cartan-Weyl basis)

The benefit is that we can use  $SU(2)$  language to describe the representation theory.

• Another point is that we might need to do linear combi. to bring them in this form. (Think about  $\{\hat{L}_1, \hat{L}_2, \hat{L}_3\} \Rightarrow \{\hat{L}_3, \hat{L}_+, \hat{L}_-\}$ )

The difference between  $\{\hat{H}_i, \hat{E}_{\vec{\alpha}}, \hat{E}_{-\vec{\alpha}}\}$  &  $\{\hat{L}_3, \hat{L}_+, \hat{L}_-\}$  is that root is not unity  $\Rightarrow$  We can also normalise the  $\hat{E}_{\vec{\alpha}}$

Consider 1 direction of  $\vec{\alpha}$  (choose 1 vector from root vectors)

$\Rightarrow$  We absorb the root into the operator.

$$\hat{E}_+ = \frac{1}{|\vec{\alpha}|} \hat{E}_{\vec{\alpha}}, \quad \hat{E}_- = \frac{1}{|\vec{\alpha}|} \hat{E}_{-\vec{\alpha}}$$

Consider 1 direction of  $\vec{\alpha}$  (choose 1 vector from root vectors)

$\Rightarrow$  We absorb the root into the operator.

$$\hat{E}_+ \equiv \frac{1}{|\vec{\alpha}|} \hat{E}_{\vec{\alpha}}, \quad \hat{E}_- \equiv \frac{1}{|\vec{\alpha}|} \hat{E}_{-\vec{\alpha}}$$

Also, we absorb a factor into  $\hat{H}$

$$\hat{E}_3 \equiv \frac{\vec{\alpha} \cdot \hat{H}}{|\vec{\alpha}|^2}$$

$$\begin{aligned} & \{ \hat{H}_i, \hat{E}_{\vec{\alpha}}, \hat{E}_{-\vec{\alpha}} \} \\ & \Downarrow \\ & \{ \hat{E}_3, \hat{E}_+, \hat{E}_- \} \end{aligned}$$

Consider the commutator

$$[\hat{E}_3, \hat{E}_+] = \left[ \frac{\vec{\alpha} \cdot \hat{H}}{|\vec{\alpha}|^2}, \frac{1}{|\vec{\alpha}|} \hat{E}_{\vec{\alpha}} \right]$$

$$= \frac{1}{|\vec{\alpha}|^3} \left[ \sum_i \alpha_i \hat{H}_i, \hat{E}_{\vec{\alpha}} \right]$$

$$= \frac{1}{|\vec{\alpha}|^3} \sum_i \alpha_i \underbrace{[\hat{H}_i, \hat{E}_{\vec{\alpha}}]}_{\alpha_i \hat{E}_{\vec{\alpha}}}$$

$$= \frac{1}{|\vec{\alpha}|^3} \left( \sum_i \alpha_i^2 \right) \hat{E}_{\vec{\alpha}} = \frac{1}{|\vec{\alpha}|} \hat{E}_{\vec{\alpha}} = \hat{E}_+$$

$\vec{\alpha} \cdot \vec{\alpha} = |\vec{\alpha}|^2$

$$\text{Also, } [\hat{E}_+, \hat{E}_-] = \left[ \frac{1}{|\vec{\alpha}|} \hat{E}_{\vec{\alpha}}, \frac{1}{|\vec{\alpha}|} \hat{E}_{-\vec{\alpha}} \right]$$

$$= \frac{1}{|\vec{\alpha}|^2} [\hat{E}_{\vec{\alpha}}, \hat{E}_{-\vec{\alpha}}] = \frac{1}{|\vec{\alpha}|^2} \vec{\alpha} \cdot \hat{H}$$

$$= \hat{E}_3$$

$\therefore$  We normally view a Lie group into a bunch of  $SU(2)$  in different direction.

∴ We normally view a Lie group into a bunch of  $SU(2)$  in different direction.

# directions = # of  $\vec{\alpha}$ .

Let's do the  $SU(3)$

$SU(3) \subset SL(3, \mathbb{C})$  with Unitarity.

$$\hat{g} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & i & h \end{pmatrix} \rightarrow \text{all complex} \Rightarrow \text{parameters} = 9 \times 2 = 18$$

$$\hat{g}^\dagger \hat{g} = \hat{1} \rightarrow 9 \text{ constraints}$$

$$\hookrightarrow \text{free parameter} = 18 - 9 = 9$$

$$\det \hat{g} = 1 \rightarrow 1 \text{ constraint}$$

$$\therefore \text{Free parameters} = 9 - 1 = 8$$

$$\# \text{ of generators} = 8$$

Generator of  $SU(n)$  must be traceless Hermitian matrix.

The obvious choice for these is Gell-Mann matrices

$$\hat{\lambda}_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \hat{\lambda}_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \hat{\lambda}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\hat{\lambda}_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \hat{\lambda}_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \hat{\lambda}_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{\lambda}_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \hat{\lambda}_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

We also write generators as

$$\hat{F}_i = \frac{1}{2} \hat{\Lambda}_i$$

( $\frac{1}{2}$  is chosen as a normalisation)

$$\text{Tr}(\hat{F}_i \hat{F}_j) = \frac{1}{2} \delta_{ij}$$

check

$$\begin{aligned} \text{Tr}\left(\frac{1}{2}\hat{\Lambda}_3 \cdot \frac{1}{2}\hat{\Lambda}_3\right) &= \text{Tr}\left(\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}\right) \\ &= \text{Tr}\left(\begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 \end{pmatrix}\right) = \frac{1}{2} \dots \end{aligned}$$

We need to bring these into Cartan-Weyl basis

What are Cartan generators?

Due to their diagonal forms, we choose  $\hat{F}_3, \hat{F}_8$

as Cartan generators.

$$[\hat{F}_3, \hat{F}_8] = 0$$

$$\hat{F}_3 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \hat{F}_8 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$\therefore$  The rank of  $SU(3) = 2$

$$\hat{H}_1 = \hat{F}_3, \quad \hat{H}_2 = \hat{F}_8$$

What their shared eigenvectors look like?

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\hat{F}_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\hat{F}_8 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$|\vec{\mu}, 0\rangle \Rightarrow$

$\hookrightarrow$  eigenvalues of  $\hat{H}_1, \hat{H}_2$

What their shared eigenvectors look like?

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad \hat{F}_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|\vec{\mu}, 0\rangle \Rightarrow$$

↳ eigenvalues of  $\hat{H}_1, \hat{H}_2$

$$\hat{F}_8 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\hat{F}_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{F}_8 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = |\vec{\mu} = (\frac{1}{2}, \frac{1}{2\sqrt{3}})\rangle$$

$$\hat{F}_3 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \hat{F}_8 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2\sqrt{3}} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

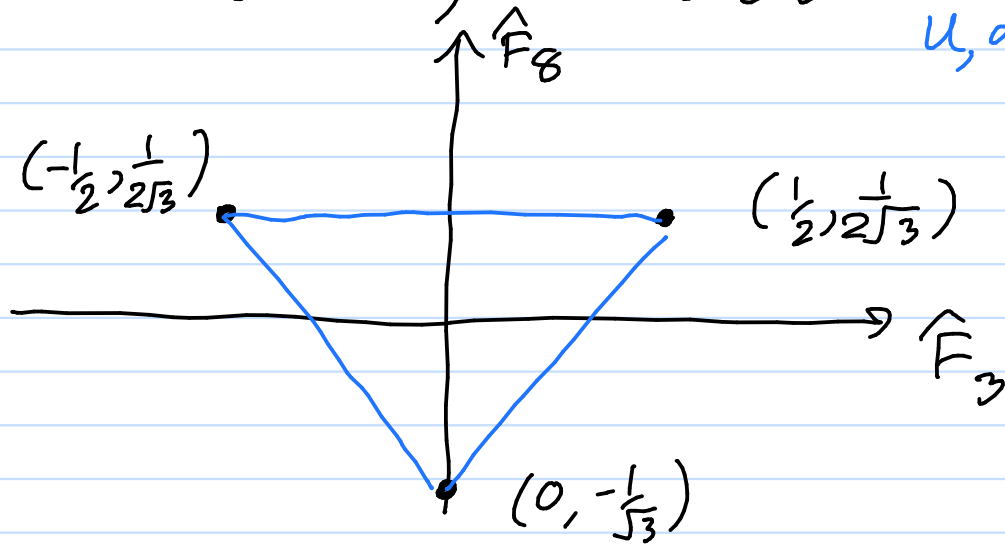
$$\therefore \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = |\vec{\mu} = (-\frac{1}{2}, \frac{1}{2\sqrt{3}})\rangle$$

$$\hat{F}_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \hat{F}_8 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \frac{-2}{2\sqrt{3}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = |\vec{\mu} = (0, -\frac{1}{\sqrt{3}})\rangle$$

$\therefore$  In this representation, we have 3 states and we can draw the  $\vec{\mu}$  vectors as

$\therefore$  In this representation, we have 3 states and we can draw the  $\vec{\mu}$  vectors as u, d, s quarks



### \* Fundamental representation of $SU(3)$

Next, we construct the adjoint representation of  $SU(3)$

$\Rightarrow$  Find the weight vectors (root vectors)  $\rightarrow \vec{\alpha}_i$

In adjoint rep.  $\Rightarrow$  States  $\Rightarrow$  corresponds with generators

$$\hat{H}_i \Rightarrow |\hat{H}_i\rangle, \quad \hat{E}_i \Rightarrow |\hat{E}_i\rangle$$

$$\hat{H}_i |\hat{H}_j\rangle = 0, \quad \forall i, j \rightarrow \text{The weight of } |\hat{H}_i\rangle = \vec{0} \quad (\text{root})$$

$$\hat{H}_i |\hat{E}_\alpha\rangle = \alpha_i |\hat{E}_\alpha\rangle, \quad \rightarrow \text{The weight of } |\hat{E}_\alpha\rangle = \alpha_i \quad (\text{root})$$

$$[\hat{H}_i, \hat{E}_\alpha] = \alpha_i \hat{E}_\alpha$$

$$(|\hat{E}_\alpha\rangle = \vec{\alpha})$$

Adjoint rep of  $SU(3) \Rightarrow \# \text{ of root vectors} = 8$   
(2 of them is zero)

$\Rightarrow$  Bring  $\lambda_1 \dots \lambda_8$  to the Cartan-Weyl form!

## Gell-Mann matrices

$$\hat{\lambda}_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \hat{\lambda}_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \hat{\lambda}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\hat{\lambda}_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \hat{\lambda}_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \hat{\lambda}_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{\lambda}_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \hat{\lambda}_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

## Exercise

The structure constant of  $\lambda_i$  are

$$\left[ \frac{\hat{\lambda}_i}{2}, \frac{\hat{\lambda}_j}{2} \right] = i f_{ijk} \frac{\hat{\lambda}_k}{2} \quad \left( [\hat{F}_i, \hat{F}_j] = i f_{ijk} \hat{F}_k \right)$$

$$f_{123} = 1, \quad f_{147} = \frac{1}{2}, \quad f_{156} = -\frac{1}{2}, \quad f_{246} = \frac{1}{2}, \quad f_{257} = \frac{1}{2}$$

$$f_{345} = \frac{1}{2}, \quad f_{367} = -\frac{1}{2}, \quad f_{458} = \frac{\sqrt{3}}{2}, \quad f_{678} = \frac{\sqrt{3}}{2}$$

& the rest vanish

We have  $\hat{F}_3$  &  $\hat{F}_8$  as the Cartan generator ( $\hat{H}_i$ )

$\Rightarrow$  we need  $\hat{E}_\alpha, \hat{E}_{-\alpha}$

$8 - 2 = 6 \Rightarrow$  we need 3 pairs of  $(\hat{E}_\alpha, \hat{E}_{-\alpha} = \hat{E}_\alpha^\dagger)$

(total) (cartan) We define

$$\hat{T}_\pm = \hat{F}_1 \pm i\hat{F}_2, \quad \hat{V}_\pm = \hat{F}_4 \pm i\hat{F}_5, \quad \hat{U}_\pm = \hat{F}_6 \pm i\hat{F}_7$$



$$\hat{T}_{\pm} = \hat{F}_1 \pm i\hat{F}_2, \quad \hat{V}_{\pm} = \hat{F}_4 \pm i\hat{F}_5, \quad \hat{U}_{\pm} = \hat{F}_6 \pm i\hat{F}_7$$

$$\hat{T}_{+}^{\dagger} = \hat{T}_{-} \quad \text{since } \hat{F}_i^{\dagger} = \hat{F}_i \quad (\text{also } \hat{V}_{+}^{\dagger} = \hat{V}_{-}, \hat{U}_{+}^{\dagger} = \hat{U}_{-})$$

We normally normalize

$$\hat{T}_3 = \hat{F}_3, \quad \hat{Y} = \frac{2}{\sqrt{3}} \hat{F}_8$$

Cartan generators

Let's find eigenvalue  $\alpha_i$  of  $\hat{T}_{\pm}$  ( $i=1,2$ )

$$\hat{T}_3 |\hat{T}_{+}\rangle = \alpha_1 |\hat{T}_{+}\rangle$$

$$[\hat{T}_3, \hat{T}_{+}] = \alpha_1 \hat{T}_{+}$$

$$\hat{T}_3 |\hat{T}_{+}\rangle = |[\hat{T}_3, \hat{T}_{+}]\rangle$$

$$[\hat{F}_3, \hat{F}_1 \pm i\hat{F}_2] = [\hat{F}_3, \hat{F}_1] \pm i[\hat{F}_3, \hat{F}_2]$$

$$= i f_{312} \hat{F}_2 \pm i(i f_{321} \hat{F}_1)$$

$$= i \cdot 1 \cdot \hat{F}_2 \pm (-1)(-1) \hat{F}_1$$

$$= i\hat{F}_2 \pm \hat{F}_1$$

$$= \pm (\hat{F}_1 \pm i\hat{F}_2)$$

$$[\hat{T}_3, \hat{T}_{\pm}] = \pm 1 \hat{T}_{\pm}$$

eigenvalue of  $|\hat{T}_{\pm}\rangle$   
of operator  $\hat{T}_3$

$$\begin{aligned} f_{123} &= 1 \\ f_{312} &= 1 \\ f_{321} &= -1 \end{aligned}$$

$$(\pm)^2 = 1$$

Another eigenvalue:

$$\hat{Y} |\hat{T}_{\pm}\rangle = \alpha_2 |\hat{T}_{\pm}\rangle$$

$$[\hat{Y}, \hat{T}_{\pm}] = \alpha_2 \hat{T}_{\pm}$$

$$\hat{Y} |\hat{T}_{\pm}\rangle = |[\hat{Y}, \hat{T}_{\pm}]\rangle$$

Another eigenvalue:

$$\hat{Y} |\hat{T}_{\pm}\rangle = \alpha_2 |\hat{T}_{\pm}\rangle, \quad \hat{Y} |\hat{T}_{\pm}\rangle = -|\hat{Y}, \hat{T}_{\pm}\rangle$$

$$[\hat{Y}, \hat{T}_{\pm}] = \alpha_2 \hat{T}_{\pm}$$

no  $f_{s2} \dots, f_{s1} \dots$

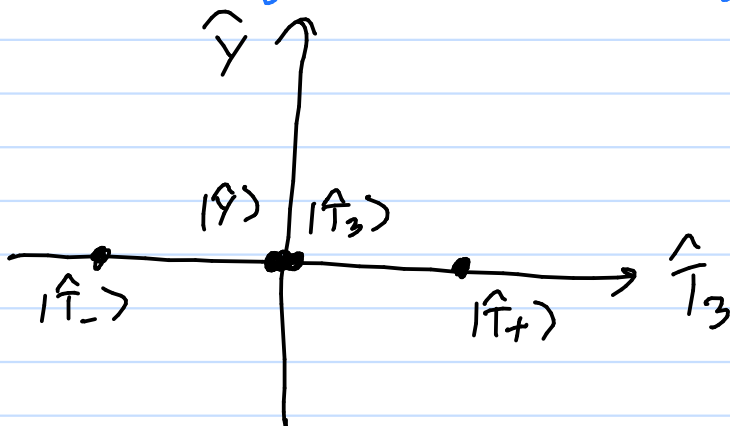
$$\left[ \frac{2}{\sqrt{3}} \hat{F}_8, \hat{F}_1 \pm i \hat{F}_2 \right] = \frac{2}{\sqrt{2}} [\hat{F}_8, \hat{F}_1] \pm i \frac{2}{\sqrt{3}} [\hat{F}_8, \hat{F}_2]$$

$$= 0 \rightarrow \text{eigenvalue of } |\hat{T}_{\pm}\rangle \text{ of } \hat{Y} \text{ is } 0$$

$\therefore$  The root (weight) vector of  $|\hat{T}_{\pm}\rangle$  is just

$$\vec{\alpha} = (1, 0), \quad -\vec{\alpha} = (-1, 0)$$

eigenvalue of  $\hat{T}_3$     eigenvalue of  $\hat{Y}$



Then  $|\hat{V}_{\pm}\rangle \rightarrow$  eigenvectors of  $\hat{T}_3, \hat{Y}$   
 $\hookrightarrow [\hat{T}_3, \hat{V}_{\pm}], [\hat{Y}, \hat{V}_{\pm}]$

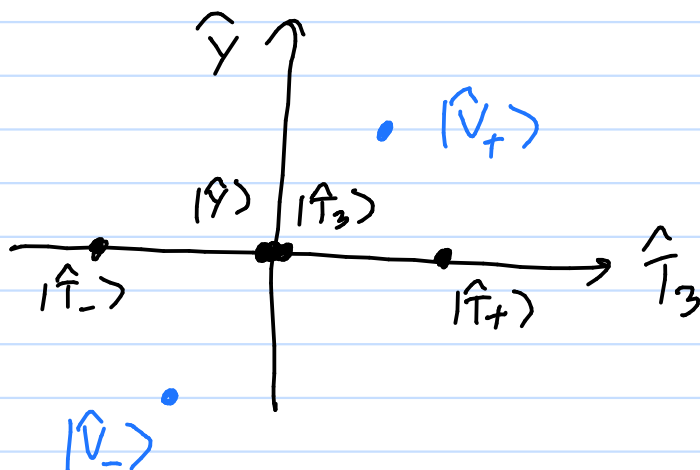
### Exercise

$$[\hat{T}_3, \hat{V}_{\pm}] = \pm \frac{1}{2} \hat{V}_{\pm}$$

$$\& [\hat{Y}, \hat{V}_{\pm}] = \pm 1 \hat{V}_{\pm}$$

$\therefore$  The root vector (weight vector) is just

$$|\hat{V}_{\pm}\rangle \Rightarrow \vec{\alpha} = \left(\frac{1}{2}, 1\right), -\vec{\alpha} = \left(-\frac{1}{2}, -1\right)$$



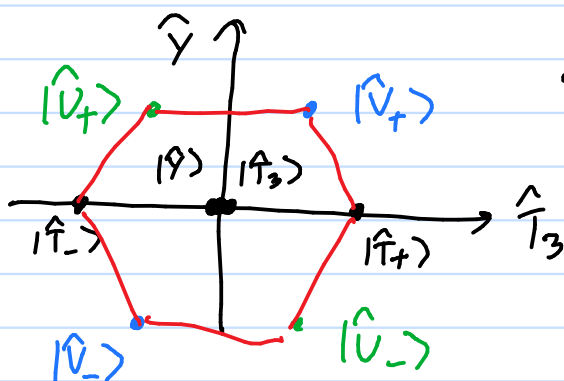
Then, finally, we need  $|\hat{U}_{\pm}\rangle$

### Exercise

$$[\hat{T}_3, \hat{U}_{\pm}] = \mp \frac{1}{2} \hat{U}_{\pm}, [\hat{Y}, \hat{U}_{\pm}] = \pm \hat{U}_{\pm}$$

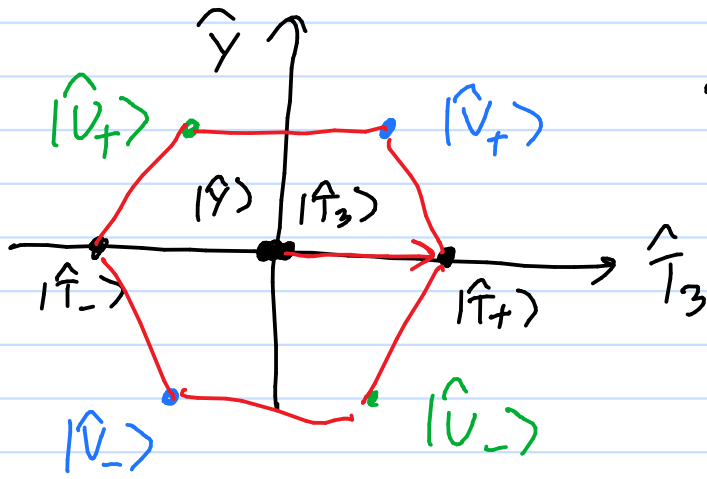
The root vectors are

$$|\hat{U}_{\pm}\rangle \rightarrow \vec{\alpha} = \left(-\frac{1}{2}, 1\right), -\vec{\alpha} = \left(\frac{1}{2}, -1\right)$$



The root diagram of  
adjoint rep. of  $SU(3)$

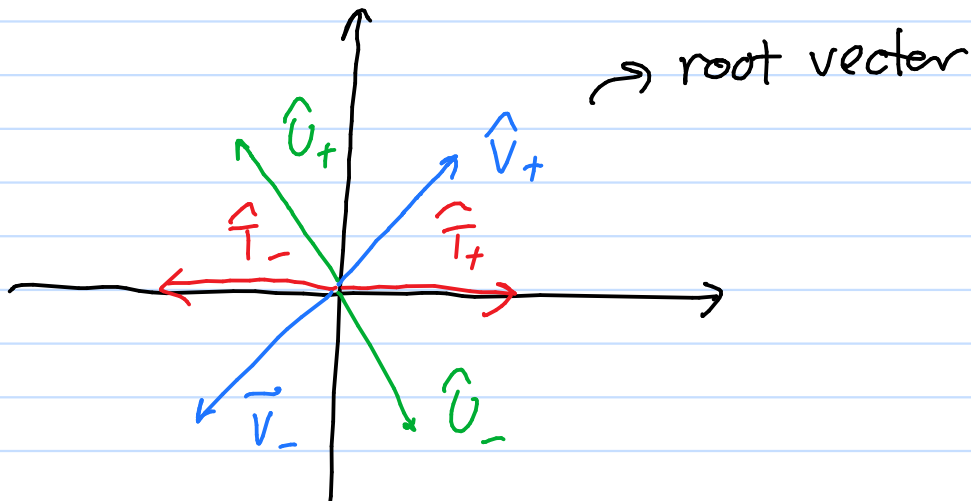
The root diagram of  
adjoint rep. of  $SU(3)$



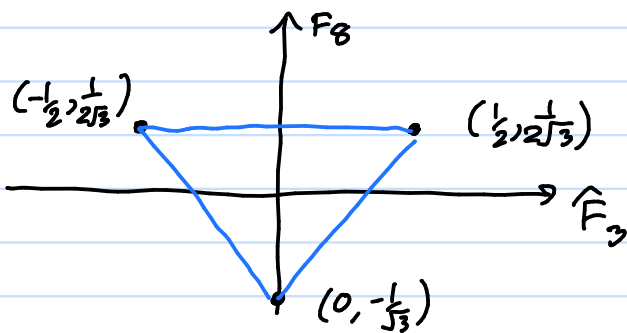
↓ This can also be interpreted as the root which is the increase/decrease of weight

$$\hat{T}_+ |\vec{\mu}, D\rangle \propto |\vec{\mu} + (1, 0), D\rangle$$

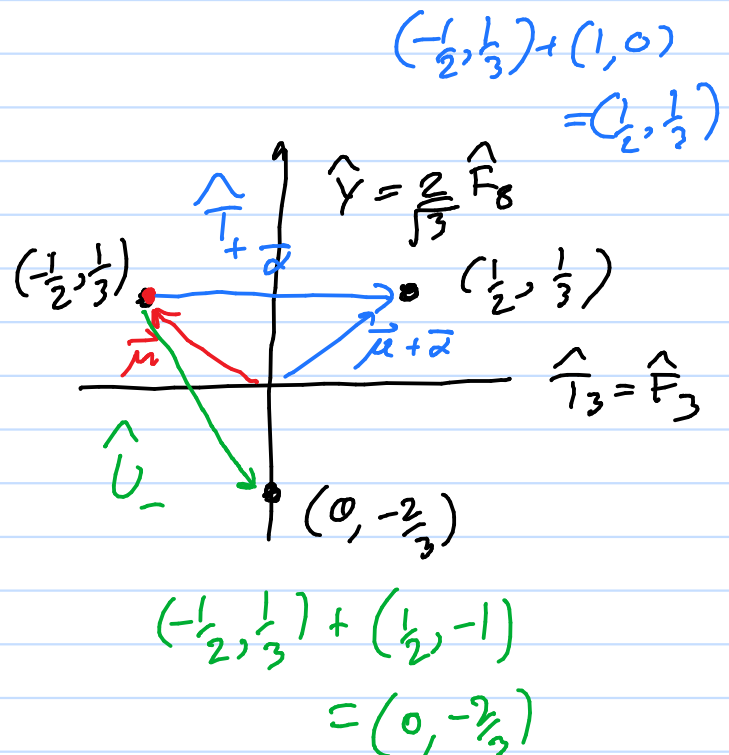
$$\hat{T}_- |\vec{\mu}, D\rangle \propto |\vec{\mu} + (-1, 0), D\rangle$$



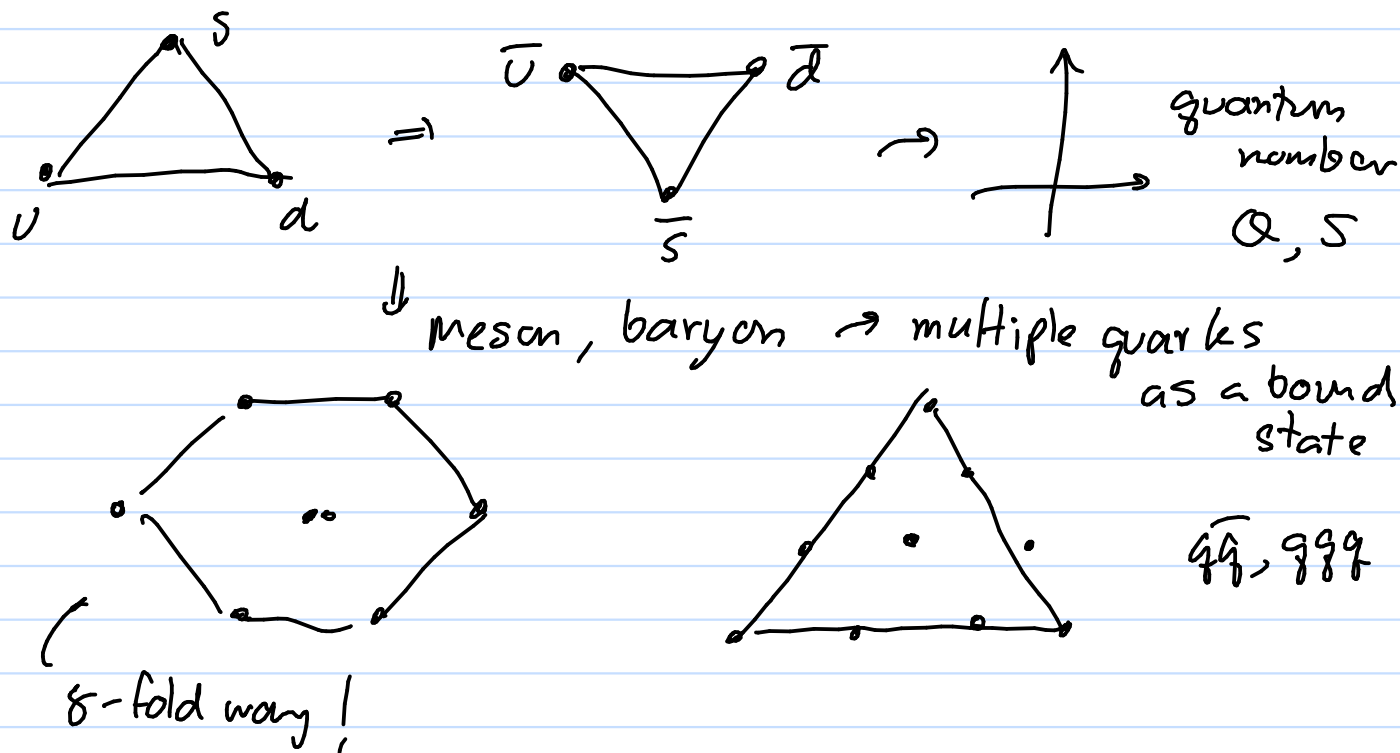
Ex Fundamental rep



=>



The irrep of  $SU(3)$  is useful to the theory of quarks



$\Rightarrow$  A composite state of multiple particles

$\Rightarrow$  tensor product of representation

Let's start with  $SU(2)$

The tensor product of operator / state

$\hat{T}, \hat{S} \Rightarrow \hat{T} \otimes \hat{S}$  tensor product

$\hat{T}|u\rangle, \hat{S}|u\rangle \quad (\hat{T} \otimes \hat{S})(|u\rangle \otimes |u\rangle)$

$\hookrightarrow$  tensor product of state.

The 2-particle state of spin- $\frac{1}{2}$

1 particle  $\left\{ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right\} \quad \uparrow \downarrow$

1 particle  $\left\{ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right\} \quad \uparrow \uparrow$

$\Rightarrow$  The tensor product state

1 particle  $\{ | \frac{1}{2}, -\frac{1}{2} \rangle, | \frac{1}{2}, \frac{1}{2} \rangle \}$   $\uparrow$  1

$$\left\{ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right.$$

$$\left\{ \left| \frac{1}{2}, \frac{1}{2} \right\rangle \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right\}$$

$$|\frac{1}{2}, -\frac{1}{2}\rangle \otimes |\frac{1}{2}, -\frac{1}{2}\rangle \Rightarrow \text{The total } \hat{L}_z = -\frac{1}{2} + -\frac{1}{2} = -1$$

$$|\frac{1}{2}, -\frac{1}{2}\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle \rightarrow \text{The total } \hat{L}_z = -\frac{1}{2} + \frac{1}{2} = 0$$

$$|\frac{1}{2}, \frac{1}{2}\rangle \otimes |\frac{1}{2}, -\frac{1}{2}\rangle \Rightarrow \text{---}, \frac{1}{2} - \frac{1}{2} = 0$$

$$|1/2, 1/2\rangle \otimes |1/2, 1/2\rangle \Rightarrow \hbar \longrightarrow \hbar \quad 1/2 + 1/2 = 1$$

$$\text{rep of } \text{SU}(2) = \left\{ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right\} \quad j = \frac{1}{2} \rightarrow |j, m\rangle$$

$$\dim = 2$$

$$\dim = 3$$

$$j = \frac{1}{2} \otimes j = \frac{1}{2} \Rightarrow j = 1 + j = 0$$

tensor product

$m = -1, 0, 1$  direct sum  $\dim = 1$

$$2 \otimes 2 = 3 \oplus 1$$

## The decomposition of tensor product of representation

The decomposition of tensor product of representation

$$2 \otimes 2 = 3 \oplus 1$$

$\otimes$  = tensor product

$\oplus$  = direct sum  $\Rightarrow$

$$\begin{pmatrix} (d_1) & & \\ & (d_2) & \\ & & (d_3) \end{pmatrix}$$

block diagonal form

$$d_1 \oplus d_2 \oplus d_3$$

representation with

$$j = \frac{1}{2} \left\{ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right\}$$

$$\rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

dim = 2

$$\text{rep with } j=0 \\ \left\{ |0, 0\rangle \right\}$$

dim = 1

rep with  $j=1$

$$\left\{ |1, -1\rangle, |1, 0\rangle, |1, 1\rangle \right\}$$

$$\rightarrow \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

dim = 3

The generalisation of tensor decomposition

For example,  $SU(3)$  reps  $\otimes$   $SU(3)$  reps.

$$\hat{L}_3 \Rightarrow \hat{T}_3, \hat{Y} \quad \text{nontrivial}$$

We need a general method!

- We need a way to label representation of  $SU(n)$  group. (or a Lie group)
- Note that in  $SU(2)$  we label them by

• We need a way to label representation of  $SU(n)$  group. (or a Lie group)

• Note that in  $SU(2)$  we label them by

1). The dimension (# of states)

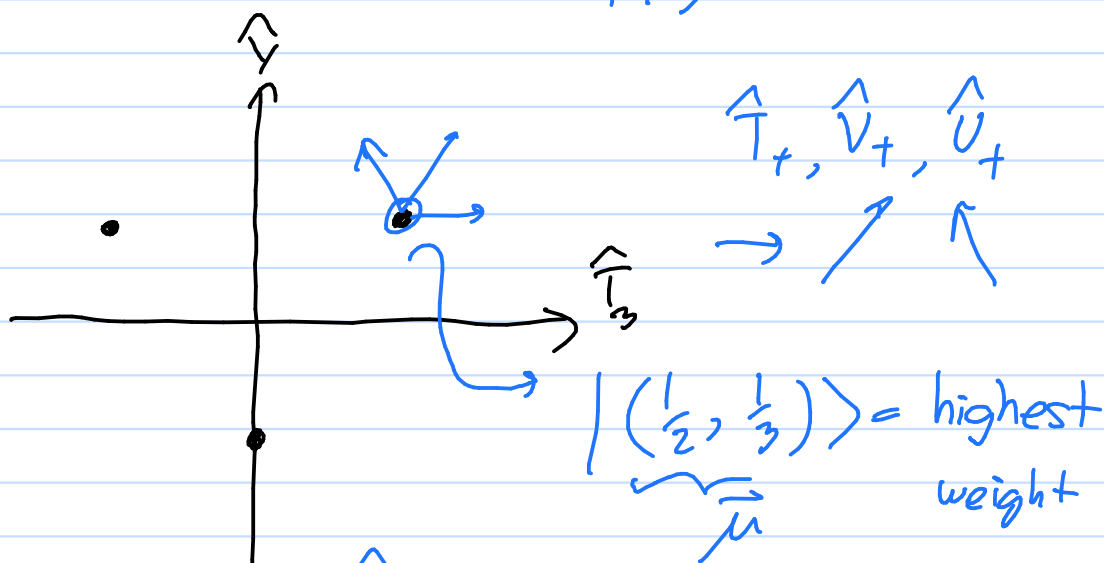
2).  $j$  value  $\Rightarrow$  highest weight

$$\{ | \frac{1}{2}, -\frac{1}{2} \rangle, | \frac{1}{2}, \frac{1}{2} \rangle \} \Rightarrow 2 \quad \text{or} \quad j = \frac{1}{2}$$

$$\{ | 1, -1 \rangle, | 1, 0 \rangle, | 1, 1 \rangle \} \Rightarrow 3 \quad \text{or} \quad j = 1$$

$\hat{L}_+ | 1, 1 \rangle = 0$  the highest

$SU(3)$

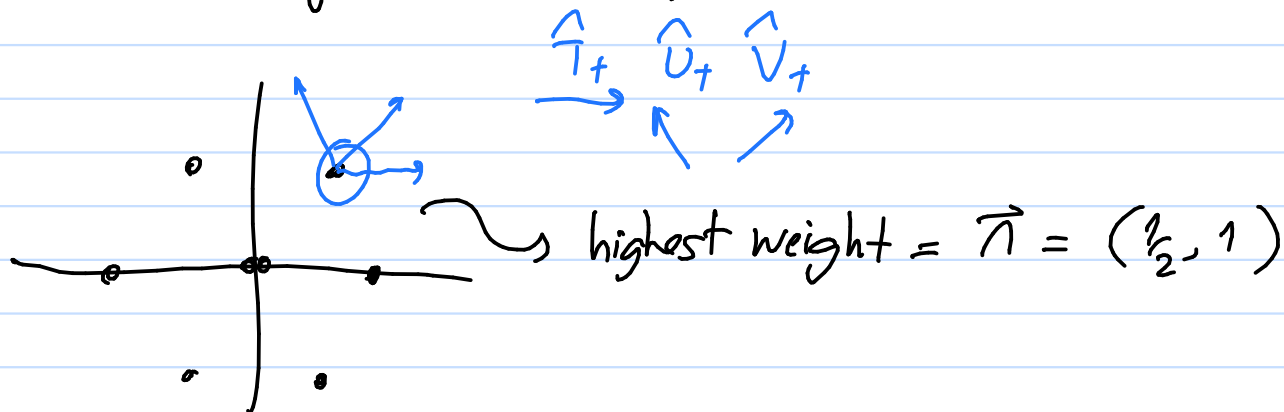


$$| \vec{\lambda}, \vec{\mu} \rangle \Leftrightarrow \begin{cases} \hat{T}_+ | \vec{\mu} = (\frac{1}{2}, \frac{1}{3}) \rangle = 0 \\ \hat{U}_+ | \vec{\mu} = (\frac{1}{2}, \frac{1}{3}) \rangle = 0 \\ \hat{V}_+ | \vec{\mu} = (\frac{1}{2}, \frac{1}{3}) \rangle = 0 \end{cases}$$

$\vec{\lambda} = \text{highest weight}$   
( $m = j$ )



How about adjoint of  $SU(3)$



In  $SU(2)$ ,  $|j, j\rangle \Rightarrow$  The rest are constructed from  
 $|j, j-n\rangle \propto (\hat{T}_-)^n |j, j\rangle$

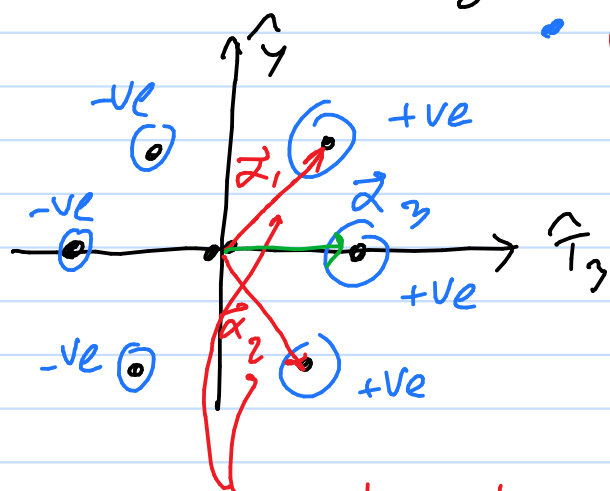
Similarly, the rest are constructed by  
 $(\hat{T}_-)^{n_1} (\hat{U}_-)^{n_2} (\hat{V}_-)^{n_3} |\vec{\lambda}, \vec{\lambda}\rangle = \dots$

Let's name more things

In adjoint rep., we have roots (weights) of  $|\hat{E}_a\rangle$

We have  $\{\vec{\alpha}, -\vec{\alpha}\}$

$\vec{\alpha}$  is called positive root when the first component is positive, negative if the first component is negative



Then we call the positive roots that cannot be written as a sum of other positive roots  $\Rightarrow$  simple roots

$$\left\{ \vec{\alpha}_1 = (\frac{1}{2}, 1), \vec{\alpha}_2 = (\frac{1}{2}, -1) \right\}$$

Positive root

$$\vec{\alpha}_3 = \vec{\alpha}_1 + \vec{\alpha}_2$$

Note that the number of simple roots  $\rightarrow$  basis vector  
 = number of Cartan generators

# of simple roots = rank of Lie group

$\rightarrow$  rank of Lie group

$\Rightarrow$  We can identify  $\vec{\Lambda}$  from Linear combination of simple roots

Suppose that there are  $r$  simple roots  $\vec{\alpha}_i, i=1, \dots, r$

$$\vec{\Lambda} = m_1 \vec{\alpha}_1 + m_2 \vec{\alpha}_2 + \dots + m_r \vec{\alpha}_r$$

identify rep

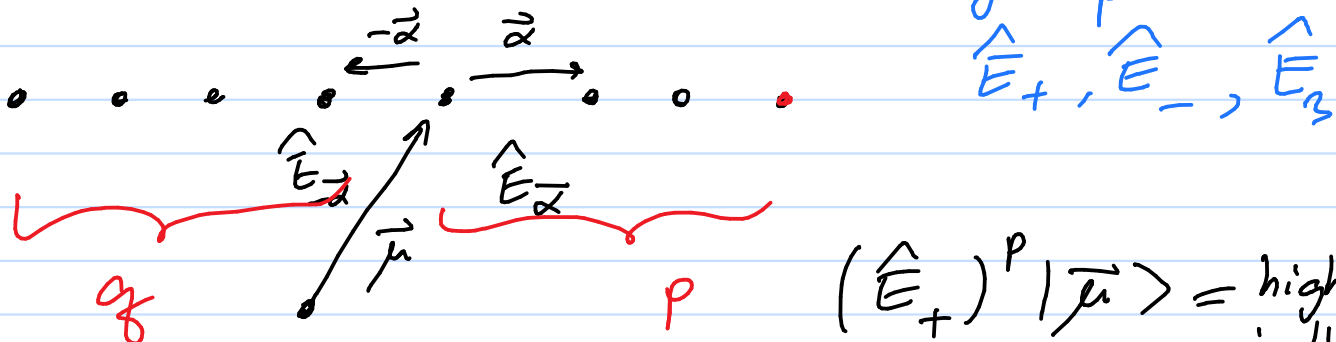
$(m_1, m_2, \dots, m_r) =$  identify rep.



★ Claim  $m_1, m_2, \dots, m_r$  are integer

## Master formula

Consider a root  $\vec{\alpha} \rightarrow$   $SU(2)$  subgroup



$(\hat{E}_+)^p |\vec{\mu}\rangle =$  highest in this direction

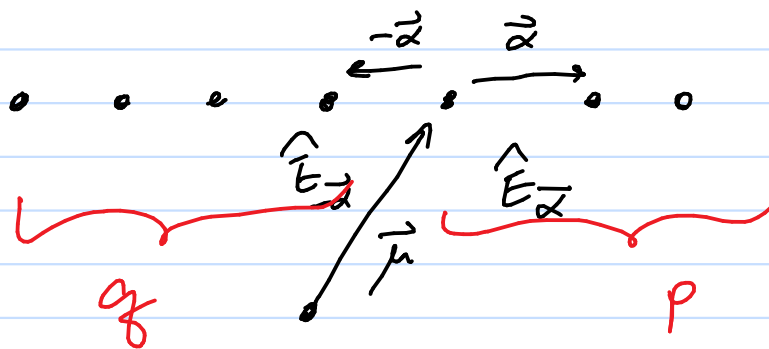
$$\begin{aligned} \hat{E}_3 |\vec{\mu}\rangle &= \frac{\vec{\alpha} \cdot \hat{H}}{|\vec{\alpha}|^2} |\vec{\mu}\rangle \\ &= \frac{\vec{\alpha} \cdot \vec{\mu}}{|\vec{\alpha}|^2} |\vec{\mu}\rangle \end{aligned}$$

$$(\hat{E}_+)^{p+1} |\vec{\mu}\rangle = 0$$

# Master formula

Consider a root  $\vec{\alpha}$   $\rightarrow$   $SU(2)$  subgroup

$$\hat{E}_+, \hat{E}_-, \hat{E}_3$$



$(\hat{E}_+)^p |\vec{\mu}\rangle = \text{highest in this direction}$

$$\begin{aligned} \hat{E}_3 |\vec{\mu}\rangle &= \frac{\vec{\alpha} \cdot \hat{H}}{|\vec{\alpha}|^2} |\vec{\mu}\rangle \\ &= \frac{\vec{\alpha} \cdot \vec{\mu}}{|\vec{\alpha}|^2} |\vec{\mu}\rangle \end{aligned}$$

$$(\hat{E}_+)^{p+1} |\vec{\mu}\rangle = 0$$

maximum weight

$\therefore$  Suppose we want  $\hat{E}_3 \left( (\hat{E}_+)^p |\vec{\mu}\rangle \right) = j \left( (\hat{E}_+)^p |\vec{\mu}\rangle \right)$

$\vec{\mu}' = \vec{\mu} + \vec{\alpha}p$

$$j = \frac{\vec{\alpha} \cdot \vec{\mu}'}{|\vec{\alpha}|^2} = \frac{\vec{\alpha} \cdot (\vec{\mu} + \vec{\alpha}p)}{|\vec{\alpha}|^2}$$

in  $SU(2)$

$j = \text{integer}$

or half-integer

$\therefore \boxed{\frac{\vec{\alpha} \cdot \vec{\mu}}{|\vec{\alpha}|^2} = j - p}$   $\rightarrow p = \text{integer}$

Similarly,  $(\hat{E}_-)^q |\vec{\mu}\rangle = \text{lowest}$ .

$$\hat{E}_3 \left( (\hat{E}_-)^q |\vec{\mu}\rangle \right) = (-j) \left( (\hat{E}_-)^q |\vec{\mu}\rangle \right)$$

$\vec{\mu}' = \vec{\mu} - \vec{\alpha}q$

$$\frac{\vec{\alpha} \cdot (\vec{\mu} - \vec{\alpha}q)}{|\vec{\alpha}|^2} = -j \Rightarrow \boxed{\frac{\vec{\alpha} \cdot \vec{\mu}}{|\vec{\alpha}|^2} = q - j}$$

$q = \text{int}$

① + ②

$$2 \frac{\vec{\alpha} \cdot \vec{\mu}}{|\vec{\alpha}|^2} = q - p \rightarrow \text{integer}$$

Master formula.

Implication of this formula is that, in the adjoint rep  $\Rightarrow$  consider  $\vec{\mu} = \vec{\beta} \Rightarrow$  another root

$$2 \frac{\vec{\alpha} \cdot \vec{\beta}}{|\vec{\alpha}|^2} = q - p \quad \text{--- ①}$$

$\Rightarrow$  If we consider  $SU(2)$  subgroup in  $\vec{\beta}$  direction

$$2 \frac{\vec{\beta} \cdot \vec{\mu}}{|\vec{\beta}|^2} = q' - p' \Rightarrow \vec{\mu} = \vec{\alpha}$$

$$2 \frac{\vec{\beta} \cdot \vec{\alpha}}{|\vec{\beta}|^2} = q' - p' \quad \text{--- ②}$$

$$\text{①} \times \text{②} = \frac{|\vec{\alpha} \cdot \vec{\beta}|^2}{|\vec{\alpha}|^2 |\vec{\beta}|^2} = \frac{(q-p)(q'-p')}{4} = \cos^2 \theta_{\alpha\beta}$$

integer

$$\vec{\alpha} \cdot \vec{\beta} = |\vec{\alpha}| |\vec{\beta}| \cos \theta_{\alpha\beta}$$

$$\cos^2 \theta_{\alpha\beta} = 0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}$$

$\therefore$  The angle between roots are fixed

$90^\circ, 60^\circ \text{ or } 120^\circ, 45^\circ \text{ or } 135^\circ, 30^\circ \text{ or } 150^\circ$

Let's get back to the highest weight.

$$\vec{\lambda} = \sum_{i=1} m_i \vec{\alpha}_i$$

Let's get back to the highest weight.

$$\vec{\lambda} = \sum_{i=1}^r m_i \vec{\alpha}_i$$

The properties of highest weight state

$$\hat{E}_{\vec{\alpha}} |\vec{\lambda}, \vec{\mu} = \vec{\lambda}\rangle = 0, \quad \forall \vec{\alpha}$$

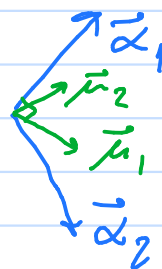
$\therefore$  Apply master formula

$$\frac{2 \vec{\alpha}_i \cdot \vec{\lambda}}{|\vec{\alpha}_i|^2} = \text{integer}$$

simple roots

Let's define fundamental weights  $\vec{\mu}_i$  such that

$$\frac{2 \vec{\alpha}_i \cdot \vec{\mu}_j}{|\vec{\alpha}_i|^2} = \delta_{ij}$$



$\vec{\mu}_i$  = fundamental weight also

span vector space

$$\Rightarrow \vec{\lambda} = \sum_{i=1}^r m_i \vec{\mu}_i$$

$$\therefore \frac{2 \vec{\alpha}_i \cdot \vec{\lambda}}{|\vec{\alpha}_i|^2} = \frac{2 \vec{\alpha}_i \cdot (\sum_{j=1}^r m_j \vec{\mu}_j)}{|\vec{\alpha}_i|^2} = \sum_j m_j \cdot \frac{2 \vec{\alpha}_i \cdot \vec{\mu}_j}{|\vec{\alpha}_i|^2} = \sum_j m_j \delta_{ij} = m_i$$

$$\therefore m_i = \text{integer.}$$

\*\*\*

integer

Highest weight  $\vec{\lambda}$  can be labeled by  $(m_1, \dots, m_r)$

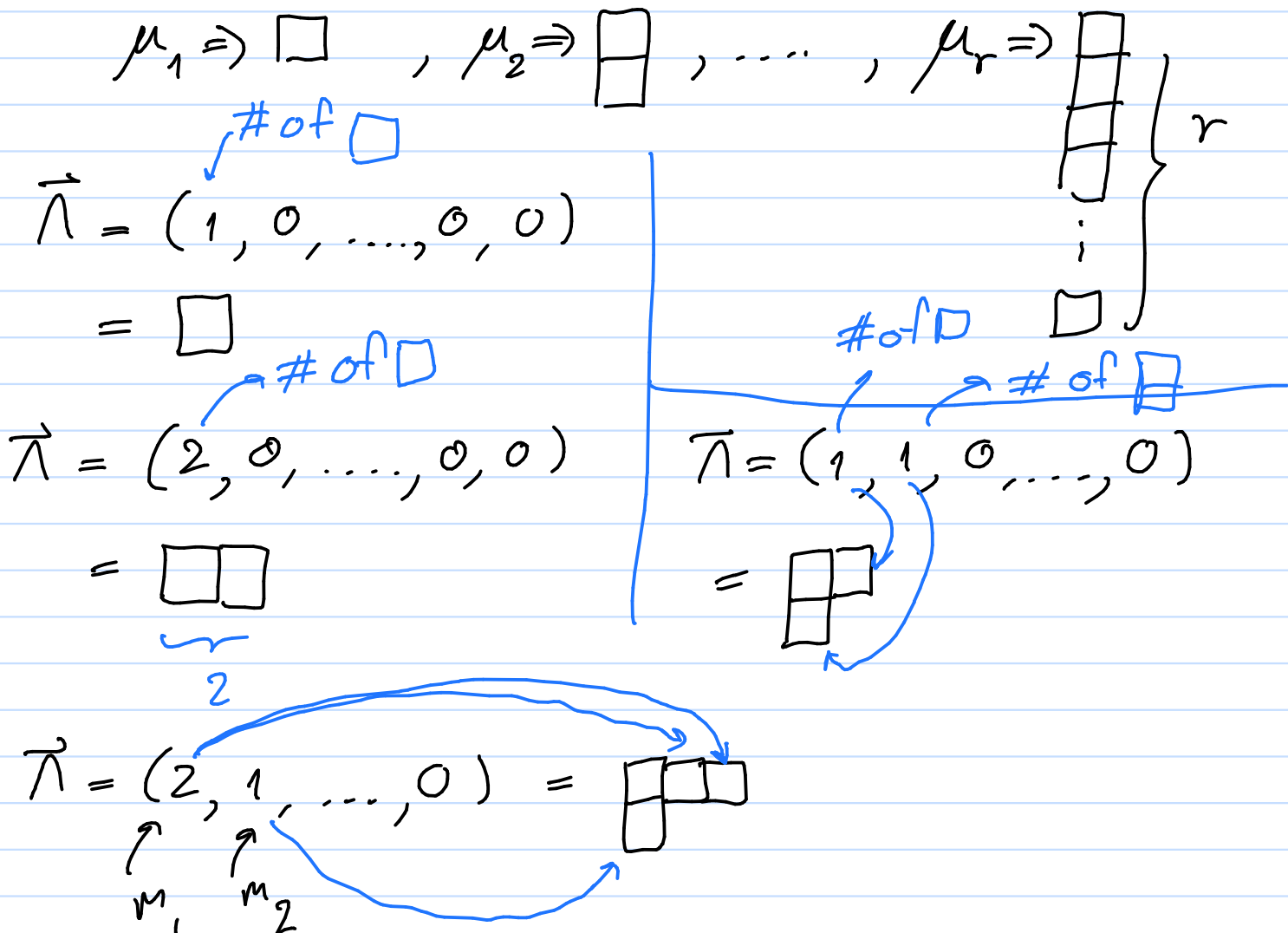
\*\* integer

Highest weight  $\vec{\lambda}$  can be labeled by  $(m_1, \dots, m_r)$

$m_1, \dots, m_r$  = Dynkin coefficients.

## Young Tableaux

We can translate  $m_1, \dots, m_r$  by using Young tableaux.



In the case of  $SU(n) \Rightarrow \text{rank} = n - 1$

$SU(2) \Rightarrow \text{rank} = 1$  ,  $SU(3) \Rightarrow \text{rank} = 2$

$SU(2)$ :  $\square$ ,  $\otimes$ ,  $\square\square$ ,  $\square\square\square$ , ...  
 $(1)$   $(2)$   $(3)$

$SU(3)$ :  $\square$ ,  $\begin{smallmatrix} \square \\ \square \end{smallmatrix}$ ,  $\begin{smallmatrix} \square & \square \end{smallmatrix}$ ,  $\begin{smallmatrix} \square & \square & \square \end{smallmatrix}$ ,  $\begin{smallmatrix} \square & \square \\ \square \end{smallmatrix}$ , ...  
 $(1,0)$   $(0,1)$   $(1,1)$   $(1,2)$   $(2,1)$

$SU(4)$ :  $\square$ ,  $\begin{smallmatrix} \square \\ \square \end{smallmatrix}$ ,  $\begin{smallmatrix} \square & \square \\ \square \end{smallmatrix}$ ,  $\begin{smallmatrix} \square & \square & \square \end{smallmatrix}$ ,  $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$ ,  $\begin{smallmatrix} \square & \square & \square \\ \square \end{smallmatrix}$ , ...  
 $(1,0,0)$   $(0,1,0)$   $(0,0,1)$   $(1,1,0)$   $(0,1,1)$   $(1,0,1)$

How do we calculate dimension?

$\Rightarrow$  In principle, we write  $\pi$  from Young Tableaux

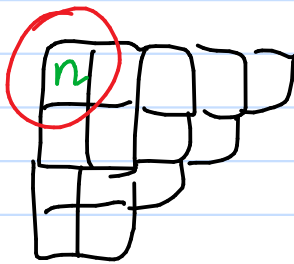
$\Rightarrow$  Find the rest of state from  $E_{-\alpha_i}$  

$\Rightarrow$  Then count the number

$\Rightarrow$  Very difficult.

There is another way to calculate the dimension from the Young tableaux directly.

# Dimension from a Young table



$\Rightarrow$

1). Write the distance from the most top left box

move to the right + 1

$SU(n) \Rightarrow$  put  $n$  in the top left box first  $\Rightarrow$

move down - 1

EX The distance in  $SU(3)$

is

|   |   |   |   |
|---|---|---|---|
| 3 | 4 | 5 | 6 |
| 2 | 3 |   |   |

The distance is  $SU(2)$

is

|   |   |   |   |
|---|---|---|---|
| 2 | 3 | 4 | 5 |
|---|---|---|---|

2). Write hook length for each box

= # of boxes on its right + # of boxes down + itself (+1)

EX 4 boxes

The dimension of irreducible representation

=  $\prod$  Distance  $\div \prod$  hook length

|   |   |   |   |
|---|---|---|---|
| 3 | 4 | 5 | 6 |
| 2 | 3 |   |   |

$\div$

|   |   |   |   |
|---|---|---|---|
| 5 | 4 | 2 | 1 |
| 2 | 1 |   |   |

$$= \frac{3 \times 4 \times 5 \times 6 \times 2 \times 3}{5 \times 4 \times 2 \times 1 \times 2 \times 1}$$

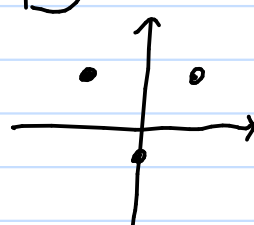


## Ex $SU(2)$

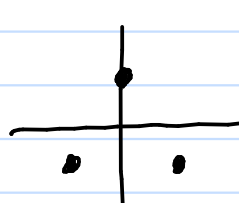
$$\begin{array}{c} \square \\ (1) \end{array} : \dim = \boxed{2} \div \boxed{1} = 2 \quad \rightarrow j = m/n = 1 \times \frac{1}{2}$$
$$\Rightarrow j = \frac{1}{2} \quad \left( \left\{ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right\} \right)$$

$$\begin{array}{c} \square \square \\ (2) \end{array} : \dim = \boxed{2 \mid 3} \div \boxed{2 \mid 1} = \frac{2 \times 3}{2 \times 1} = 3$$
$$\Rightarrow j = 1 \quad \left( \left\{ |1, -1\rangle, |1, 0\rangle, |1, 1\rangle \right\} \right) \quad \rightarrow j = m/n = 2 \cdot \frac{1}{2} = 1$$

## $SU(3)$ :

$$\begin{array}{c} \square \\ (1, 0) \end{array} : \dim = \boxed{3} \div \boxed{1} = 3 \Rightarrow \text{fundamental}$$
$$\vec{\lambda} = \left( \frac{1}{2}, \frac{1}{3} \right) \Rightarrow$$


... quarks

$$\begin{array}{c} \square \\ (0, 1) \end{array} : \dim = \boxed{\frac{3}{2}} \div \boxed{\frac{2}{1}} = \frac{3 \times 2}{2 \times 1} = 3$$
$$\vec{\lambda} \Rightarrow$$


Sometimes we call antiquarks

$$\begin{array}{c} \square \square \\ (2, 0) \end{array} : \dim \boxed{3 \mid 4} \div \boxed{2 \mid 1} = \frac{3 \times 4}{2} = 6$$

$$\begin{array}{c} \square \square \\ \square \square \\ (1, 1) \end{array} : \dim \begin{array}{|c|c|} \hline 3 & 4 \\ \hline 2 & \\ \hline \end{array} \div \begin{array}{|c|c|} \hline 3 & 1 \\ \hline 1 & \\ \hline \end{array} = \frac{3 \times 4 \times 2}{3 \times 1 \times 1} = 8 \quad (\text{adjoint rep})$$

Another great benefit is to find the tensor decomposition!

# Tensor Decomposition from Young Tableaux

Recipe for tensor decomposition.

$$\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array} = \dots \oplus \dots \oplus \dots$$

Ex  $\square \otimes \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline \end{array}$  in  $SU(3)$

Step 1 put  $a$  &  $b$  &  $c, \dots$  in the second diagram such that  $a$ 's are in the same row  
 $b$ 's are in the same row, ...

$$\square \otimes \begin{array}{|c|c|} \hline a & a \\ \hline b & \\ \hline \end{array} \quad \begin{array}{c} a \\ \downarrow \\ b \\ ; \\ c \end{array}$$

Step 2 Put each " $a$ " box from the second diagram in the first diagram (only allowed position)

- 1). The length of column  $< n$
- 2). no same  $a$  box in the same column

$$\begin{aligned} \square \otimes \begin{array}{|c|c|} \hline a & a \\ \hline b & \\ \hline \end{array} &= \left\{ \begin{array}{|c|c|} \hline & a \\ \hline & \\ \hline \end{array} + \begin{array}{|c|} \hline a \\ \hline \end{array} \right\} \otimes \begin{array}{|c|} \hline a \\ \hline b \\ \hline \end{array} \\ &= \left\{ \begin{array}{|c|c|c|} \hline & a & a \\ \hline & & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & a \\ \hline a & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & a \\ \hline a & \\ \hline \end{array} + \begin{array}{|c|} \hline a \\ \hline a \\ \hline \end{array} \right\} \begin{array}{|c|} \hline b \\ \hline \end{array} \end{aligned}$$

pattern of a's is equivalent
disallowed

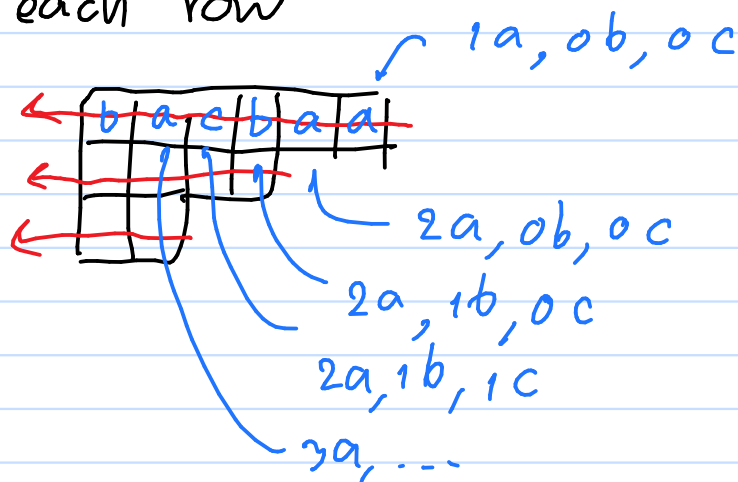
$$\begin{aligned}
 \square \otimes \begin{array}{|c|c|} \hline a & a \\ \hline b & \\ \hline \end{array} &= \left\{ \begin{array}{|c|} \hline a \\ \hline \end{array} + \begin{array}{|c|} \hline a \\ \hline \end{array} \right\} \otimes \begin{array}{|c|} \hline a \\ \hline b \\ \hline \end{array} \\
 &= \left\{ \begin{array}{|c|c|} \hline & a \\ \hline & a \\ \hline \end{array} + \begin{array}{|c|c|} \hline & a \\ \hline a & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & a \\ \hline a & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & a \\ \hline a & \\ \hline \end{array} \right\} \otimes \begin{array}{|c|} \hline b \\ \hline \end{array} \\
 &\quad \text{pattern of a's is equivalent} \quad \text{disallowed (a's are in the same column)} \\
 &= \left\{ \begin{array}{|c|c|} \hline a & a \\ \hline & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & a \\ \hline a & \\ \hline \end{array} \right\} \otimes \begin{array}{|c|} \hline b \\ \hline \end{array} \\
 &= \left\{ \begin{array}{|c|c|c|} \hline a & a & \\ \hline b & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & a & a \\ \hline & & b \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & a & b \\ \hline a & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & a & b \\ \hline a & & \\ \hline \end{array} + \begin{array}{|c|c|} \hline a & \\ \hline b & \\ \hline \end{array} \right\}
 \end{aligned}$$

### Step 3 Cleaning by counting

Make a count of a, b, c, ...

at each box from right to left

for each row



We disallow the diagram that at any point  
the count of b > count of a or count of c > count of b

$$= \left\{ \begin{array}{|c|c|c|} \hline & a & a \\ \hline b & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & a & a \\ \hline & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & a & b \\ \hline a & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline & a & \\ \hline a & b & \\ \hline \end{array} \right. + \left\{ \begin{array}{|c|c|} \hline & a \\ \hline a & \\ \hline b & \\ \hline \end{array} \right\}$$

$$= \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline & \\ \hline \end{array}$$

dim  $\sim$  dim  $\square$

Step 4 Delete  $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline & \\ \hline \end{array}$  since the dimension of invariant = singlet

$$\square \otimes \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline & \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline & \\ \hline \end{array} \oplus \square$$

$\sim \epsilon_{ijk}$   
invariant  
under  $SU(3)$

$$3 \otimes 8 = 15 \oplus 6 \oplus 3 \quad SU(3)$$

Ex  $\square \otimes [a] \quad SU(3)$

$$3 \otimes 3 \Rightarrow \begin{array}{|c|c|} \hline & a \\ \hline & \\ \hline & \\ \hline & \\ \hline \end{array} + \begin{array}{|c|} \hline a \\ \hline \end{array}$$

$$\therefore 3 \otimes 3 = 6 \oplus \bar{3} \quad 6 \quad \bar{3}$$

$$\square \otimes \begin{array}{|c|c|} \hline a & \\ \hline b & \\ \hline \end{array} = \left\{ \begin{array}{|c|c|} \hline & a \\ \hline & \\ \hline \end{array} + \begin{array}{|c|} \hline a \\ \hline \end{array} \right\} \otimes \begin{array}{|c|} \hline b \\ \hline \end{array} \quad SU(3)$$

$$= \begin{array}{|c|c|} \hline & a \\ \hline b & \\ \hline \end{array} + \begin{array}{|c|c|} \hline & a \\ \hline & \\ \hline & \\ \hline & \\ \hline \end{array}$$

$q\bar{q} = \text{meson} + \text{singlet}$

$$3 \otimes \bar{3} = 8 + 1$$

$$SU(2) : \begin{array}{c} \square \otimes \square \\ 2 \otimes 2 \end{array} = \begin{array}{c} \square \\ 3 \end{array} + \begin{array}{c} \uparrow \\ \square \\ \downarrow \\ 1 \end{array} \quad \left. \vphantom{\begin{array}{c} \square \\ 3 \end{array}} \right\} 2$$

$$= 3 \oplus 1$$

Final for Undergrad

$$\begin{array}{c} \uparrow \\ \square \\ \downarrow \\ 1 \end{array} = 1, \quad \begin{array}{c} \uparrow \\ \square \\ \downarrow \\ 2 \end{array} = 2$$

$$2 \rightarrow \psi_i, i=1, 2$$

$$\psi_i \otimes \psi_j$$